02 Elasticity, Heat and Thermodynamics

ELASTICITY

A body is said to be rigid if the relative positions of its constituent particles remains unchanged when external deforming forces are applied to it. The nearest approach to a rigid body is diamond or silicon carbide (carborundum).

Actually, nobody is perfectly rigid and everybody can be deformed more or less by the application of suitable forces. All these deformed bodies however regain their original shape or size, when the deforming forces are removed.

Deforming Force:

An external force applied to a body which changes its size or shape or both is called deforming force.

Elasticity:

The property of body of virtue by which it tends to regain its original size and shape when the applied forces are removed, is known as **elasticity** and the deformation caused is known as **elastic deformation**. Quartz is a very nearly perfectly elastic body.

However, if we apply a force to a lump or mud or putty, they have no cross tendency to regain their previous shape and they get permanently deformed. Such substances are called **plastic** and the property is called **plasticity**.

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Examples:

- Fluids, including liquids and gases, possess no definite shape, and therefore cannot possess elasticity.
- The volume of matter, whether this matter be solid or fluid, can be altered by the application of suitable forces, and in most cases the original volume is regained when the forces cease to act ; hence, all matter possesses volume elasticity.
- Any alteration produced in the shape or volume of matter leads to what is called a "strain".

Cause of Elasticity

In a solid, atoms and molecules are arranged in such a way that each molecule is acted upon by the forces due to the neighbouring molecules. These forces are known as intermolecular forces. When no deforming force is applied on the body, each molecule of the solid (i.e. body) is in its equilibrium position and the inter molecular forces between the molecules of the solid are minimum.

On applying the deforming force on the body, the molecules either come closer or go far apart from each other. As a result of this, the molecules are displaced from their equilibrium position. In other words, intermolecular forces get changed and restoring forces are developed on the molecules. When the deforming force is removed, these restoring forces bring the molecules of the solid to their respective equilibrium positions and hence the solid (or the body) regains its original form.

Internal Restoring Force

When an external force acts at any substance then due to the intermolecular force there is an internal resistance produced into the substance called internal restoring force.

At equilibrium the numerical value of internal restoring force is equal to the external force.

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Restoring Mechanism:

The restoring mechanism can be visualised by taking a model of spring-ball system shown in the Figure. Here the balls represent atoms and springs represent interatomic forces.

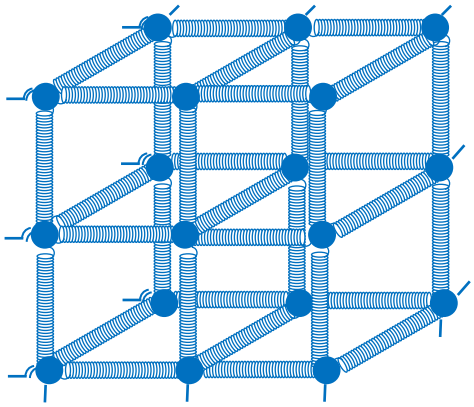


Figure : Spring-ball model for the illustration of elastic behaviour of solids

If you try to displace any ball from its equilibrium position, the spring system tries to restore the ball back to its original position. Thus, elastic behaviour of solids can be explained in terms of microscopic nature of the solid. When a body is subjected to a deforming force, a restoring force is developed in the body. This restoring force is equal in magnitude but opposite in direction to the applied force.

Stress:

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The internal restoring force acting per unit area of cross–section of the deformed body is called stress.

$$\text{Stress} = \frac{\text{Internal restoring force}}{\text{Area of cross section}} = \frac{F_{\text{Internal}}}{A} = \frac{F_{\text{External}}}{A}$$

Stress depends on direction of force as well as direction of area of application so it is tensor.

SI Unit : $N–m^{-2}$

Dimensions : $M^1\ L^{-1}\ T^{-2}$

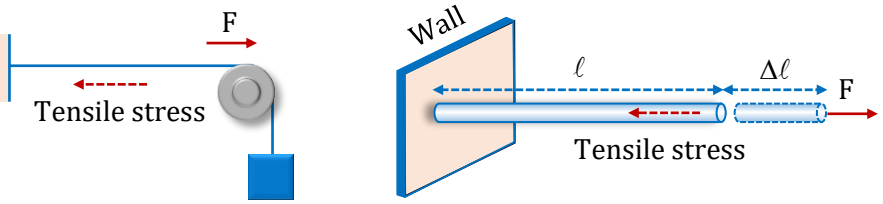
There are three types of stress:

(a) Longitudinal Stress:

When the stress is normal to the surface of body, then it is known as longitudinal stress. There are two types of longitudinal stress:

(i) Tensile Stress:

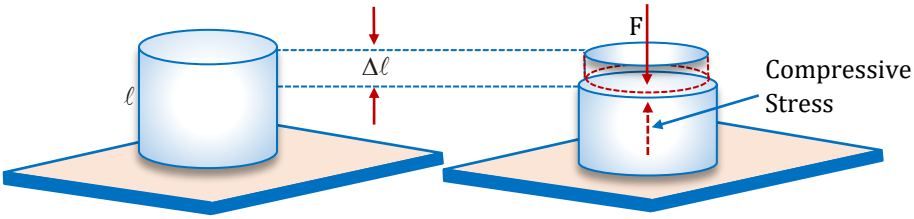
The longitudinal stress, produced due to increase in length of a body, is defined as tensile stress.



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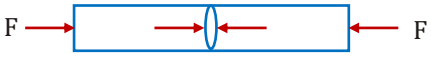
(ii) Compressive Stress:

The longitudinal stress, produced due to decrease in length of a body, is defined as compressive stress.



Tensile 

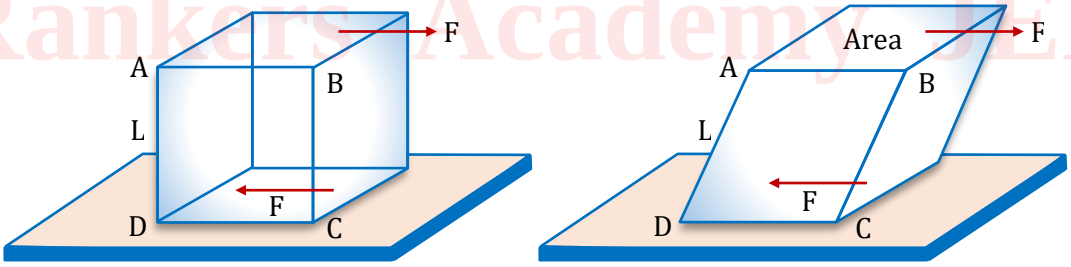
$\sigma = \frac{F}{A}$ as $F \propto A$

Compressive 

(b) Tangential Stress or Shear Stress:

When the applied external force is tangential or parallel to the surface of a body then the stress developed is known as shear stress. Due to this stress, the shape of the body changes or it gets twisted.

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$$\text{Shear stress} = \frac{F}{\text{Area}}$$

(c) Volume Stress or Bulk Stress or Hydraulic Stress:

Since the fluid presses inward on all sides of the object (figure), the solid is compressed-its volume is reduced. The fluid pressure P is the force per unit surface area ; it can be through of as the volume stress on the solid object. Pressure has the same units as the other kinds of stress: N/m^2 or Pa.

$$\text{Volume stress} = \text{Pressure} = \frac{F}{A} = P$$

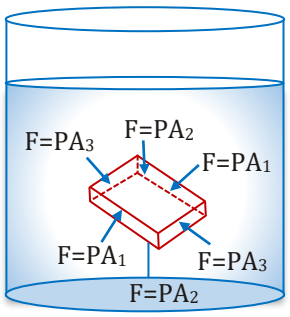
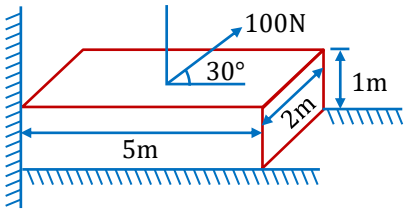


Figure : Forces on an object when submerged in a fluid

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Illustration 1



Find out longitudinal stress and tangential stress on a fixed block. Force is uniformly applied on the upper surface of the block.

Solution:

Longitudinal or normal stress,

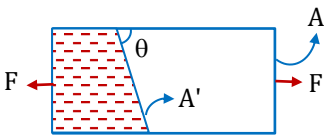
$$\sigma_1 = \frac{100\sin 30^0}{5\times 2} = 5\ N/m^2$$

Tangential stress,

$$\sigma_2 = \frac{100\cos 30^0}{5\times 2} = 5\sqrt{3} N/m^2$$

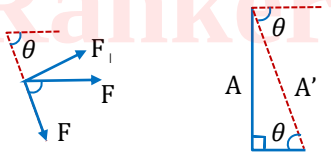
Illustration 2

Find longitudinal stress and shear stress on A' ?



Solution:

$$F_\perp = F\sin\theta, F_t = F\cos\theta$$



$$\therefore \quad \sin\theta = \frac{A}{A'}$$

$$A' = \frac{A}{\sin\theta}$$

$$\therefore \quad \text{Longitudinal stress} = \frac{F\sin\theta}{A/\sin\theta} = \frac{F\sin^2\theta}{A}$$

$$\therefore \quad \text{Shear stress} = \frac{F\cos\theta}{A/\sin\theta} = \frac{F\sin\theta\cos\theta}{A} = \frac{F\sin 2\theta}{2A}$$

Strain:

The ratio of the change in configuration (i.e. shape, length or volume) to the original configuration of the body is called strain,

$$\text{i.e.,} \quad \text{Strain, } \in = \frac{\text{Change in configuration}}{\text{Original configuration}}$$

It has no unit. It is dimensionless.

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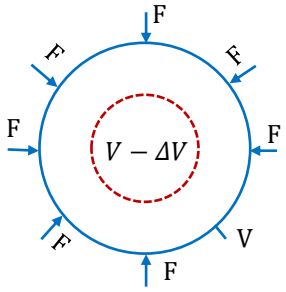
Types of Strain:

There are three types of strain:

(i) **Longitudinal Strain :** This type of strain is produced when the deforming force causes a change in length of the body. It is defined as the ratio of the change in length to the original length of the body. Consider a wire of length L : When the wire is stretched by a force F, then let the change in length of the wire is ΔL.

$$\therefore \quad \text{Longitudinal strain, } \in_\ell = \frac{\text{Change in length}}{\text{Original length}} \quad \text{or} \quad \text{Longitudinal strain} = \frac{\Delta L}{L}$$

(ii) **Volume Strain or Hydraulic Strain :** This type of strain is produced when the deforming force produces a change in volume of the body as shown in the figure. It is defined as the ratio of the change in volume to the original volume of the body.

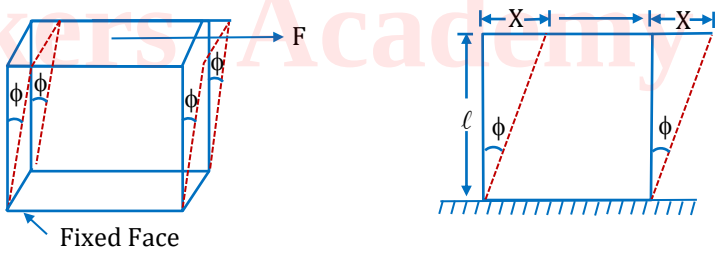


If ΔV = Change in volume ; V = Original volume

$$\in_v = \text{Volume strain} = \frac{\Delta V}{V}$$

(iii) **Shear Strain :** This type of strain is produced when the deforming force causes a change in the shape of the body. It is defined as the angle ϕ through which a face originally perpendicular to the fixed face is turned as shown in the figure.

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$$\tan\phi \text{ or } \phi = \frac{x}{\ell} \quad (\because \tan\phi \approx \phi \text{ for small deformation})$$

Hooke's Law and Modulus of Elasticity:

According to this law, within the elastic limit, stress is proportional to the strain.

i.e., Stress $\propto$ Strain

or Stress = Constant × Strain or $\frac{\text{Stress}}{\text{Strain}}$ = Modulus of Elasticity

This constant is called modulus of elasticity.

Thus, modulus of elasticity is defined as the ratio of the stress to the strain.

Modulus of elasticity depends on the nature of the material of the body and is independent of its dimensions (i.e. length, volume etc.).

Unit : The Sl unit of modulus of elasticity is Nm^{-2} or Pascal (Pa).

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Types of Modulus of Elasticity:

Corresponding to the three types of strain there are three types of modulus of elasticity:

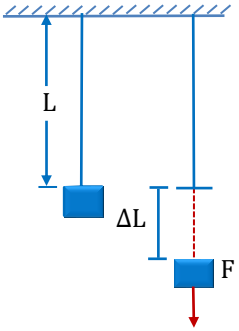
(a) Young's modulus of elasticity (Y)

(b) Bulk modulus of elasticity (K)

(c) Modulus of rigidity or shear modulus (η).

(a) Young's Modulus of Elasticity:

It is defined as the ratio of the normal stress to the longitudinal strain.



i.e., Young's modulus (Y) = $\frac{\text{Longitudinal stress}}{\text{Longitudinal strain}}$

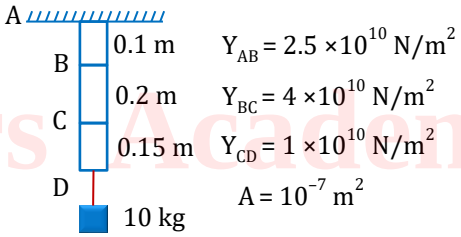
Normal stress = F/A

Longitudinal strain = $\Delta L/L$

$$Y = \frac{F/A}{\Delta L/L} = \frac{FL}{A\Delta L}$$

Illustration 3

Find out the shift in point B, C and D. Take $g = 10\ m/s^2$.



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Solution:

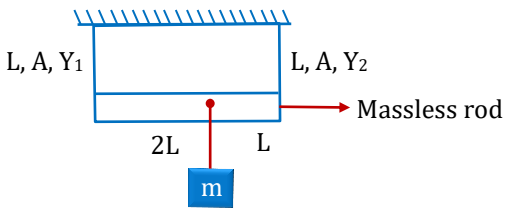
$$\Delta L_B = \Delta L_{AB} = \frac{FL}{AY} = \frac{MgL}{AY} = \frac{10\times10\times0.1}{10^{-7}\times2.5\times10^{10}} = 4 \times 10^{-3}\ m = 4mm$$

$$\Delta L_C = \Delta L_B + \Delta L_{BC} = 4 \times 10^{-3} + \frac{100\times0.2}{10^{-7}\times4\times10^{10}} = 4 \times 10^{-3} + 5 \times 10^{-3} = 9mm$$

$$\Delta L_D = \Delta L_C + \Delta L_{CD} = 9 \times 10^{-3} + \frac{100\times0.15}{10^{-7}\times1\times10^{10}} = 9 \times 10^{-3} + 15 \times 10^{-3} = 24\ mm$$

Illustration 4

Rod is horizontal. Find $\frac{Y_1}{Y_2}$ if rod remained horizontal after hanging mass m ?



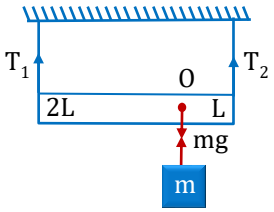
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Solution:

For rod remains horizontal, $\Delta L_1 = \Delta L_2$

$$\frac{T_1 L}{AY_1} = \frac{T_2 L}{AY_2}$$

$$\frac{T_1}{Y_1} = \frac{T_2}{Y_2} \quad \Rightarrow \quad \frac{Y_1}{Y_2} = \frac{T_1}{T_2} \qquad ...(1)$$



Also, for rotational equilibrium about O,

$$\tau_{net} = 0 \quad \Rightarrow \quad T_1 2L - T_2 L = 0$$

$$\Rightarrow \quad \frac{T_1}{T_2} = \frac{1}{2} \qquad ...(2)$$

From (1) and (2) $\frac{Y_1}{T_2} = \frac{1}{2}$

Increment of Length Due to Own Weight:

Let a rope of mass M and length L is hung vertically. As the tension of different point on the rope is different. Stress as well as strain will be different at different point.

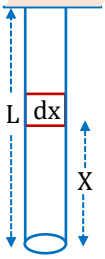
(i) Maximum stress at hanging point

(ii) Minimum stress at lower point

Consider a dx element of rope at x distance from lower end then tension, $T = \left(\frac{M}{L}\right) x\, g$

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So, $\text{Stress} = \frac{T}{A} = \left(\frac{M}{L}\right)\frac{xg}{A}$



Let increase in length of dx is dy then strain = $\frac{dy}{dx}$

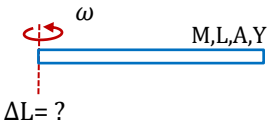
So, Young modulus of elasticity $Y = \frac{\text{Stress}}{\text{Strain}} = \frac{\frac{M}{L}\frac{xg}{A}}{dy/dx} \Rightarrow \left(\frac{M}{L}\right)\frac{xg}{A}dx = Y\,dy$

For full length of rope $\frac{Mg}{LA}\int_0^L x dx = Y\int_0^{\Delta\ell} dy \quad \Rightarrow \frac{Mg}{LA}\frac{L^2}{2} = Y\Delta\ell \quad \Rightarrow \Delta\ell = \frac{MgL}{2AY}$

[Since the stress is varying linearly we may apply average method to evaluate strain.]

Illustration 5

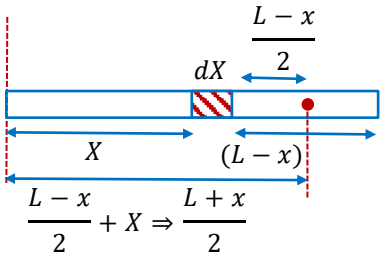
Consider a rod of length L and mass M and young's modulus of elasticity, Y and cross-sectional area A is hanged horizontal and rotated with angular velocity ω. Calculate change in length.



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Solution:

Considering the part having $(L-x)$ length an applying, $\vec{F}_{ext} = m\vec{a}_{com}$



$$T = m\omega^2 R = \frac{M}{L}(L-x)\omega^2\left(\frac{L+x}{2}\right) = \frac{M\omega^2}{2L}(L^2-x^2)$$

$$\int dL = \int \frac{M\omega^2(L^2-x^2)dx}{2LAY}$$

$$\Delta L = \frac{M\omega^2}{2AY}\int\left(\frac{L^2-x^2}{L}\right)dx \quad \Rightarrow \quad \Delta L = \frac{M\omega^2}{2AY}\left(\int_0^L L dx - \int_0^L \frac{x^2}{L}dx\right)$$

$$\Delta L = \frac{M\omega^2}{2AY}\left(L\int_0^L 1\,dx - \int_0^L \frac{x^2}{L}dx\right) \quad \Rightarrow \quad \Delta L = \frac{M\omega^2}{2AY}\left(L^2 - \frac{L^2}{3}\right)$$

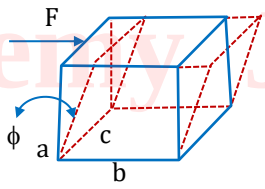
$$\Delta L = \frac{M\omega^2}{2AY}\times\frac{2L^2}{3} \quad \Rightarrow \quad \Delta L = \frac{M\omega^2 L^2}{3AY}$$

Modulus of Rigidity 'η':

Within elastic limit the ratio of shearing stress and shearing strain is called modulus of rigidity of a material.

$$\eta = \frac{\text{Shearing stress}}{\text{Shearing strain}} = \left(\frac{\frac{F_{\text{tangential}}}{A}}{\phi}\right) = \frac{F_{\text{tangential}}}{A\phi}$$

$$\eta = \frac{F}{bc.\phi}$$



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Note:

Angle of shear 'ϕ' always take in radian.

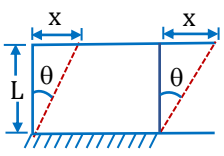
Illustration 6

A rubber cube of side 5 *cm* has one side fixed while a tangential force equal to 1800 *N* is applied to opposite face find the shearing strain and the lateral displacement of the strained face. Modulus of rigidity for rubber is $2.4 \times 10^6\ N/m^2$.

Solution:

$$\frac{F}{A} = \eta\ \frac{x}{L} = \eta\theta$$

$$\text{Strain } \theta = \frac{F}{A\eta} = \frac{1800}{25\times10^{-4}\times2.4\times10^6} = \frac{180}{25\times24} = \frac{3}{10} = 0.3 \text{ radian}$$



$$\frac{x}{L} = 0.3 \Rightarrow x = 0.3 \times 5 \times 10-2 = 1.5 \times 10-2\ m = 1.5\ cm$$ **Ans.**

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(b) Bulk Modulus:

It is defined as the ratio of the volume stress to the volume strain

i.e., $B = \dfrac{\text{Pressure}}{\text{Volumetric strain}}$

The stress being the normal force applied per unit area and is equal to the pressure applied (p).

$$B = \frac{p}{\frac{-\Delta V}{V}} = -\frac{pV}{\Delta V}$$

Negative sign shows that increase in pressure (p) causes decrease in volume (ΔV).

Quite often, the change in volume is measured corresponding to a change in pressure. The bulk modulus is then defined as

$$B = \frac{-\Delta P}{\Delta V / V} = -V\frac{\Delta P}{\Delta V} = -V\frac{dP}{dV}$$

Compressibility : The reciprocal of bulk modulus of elasticity is called compressibility. Unit of compressibility in Sl is $N^{-1}\,m^2$ or pascal -1 (Pa^{-1}).

Bulk modulus of solids is about fifty times that of liquids, and for gases it is 10^{-8} times of solids.

$$B_{\text{solids}} > B_{\text{liquids}} > B_{\text{gases}}$$

In thermodynamics we will study:

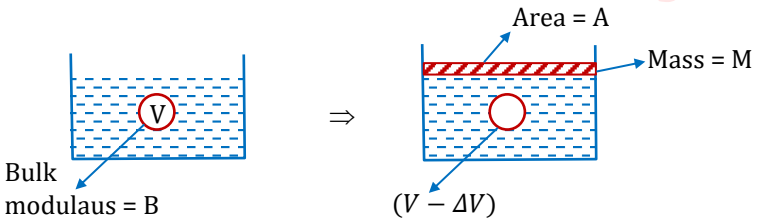
Isothermal bulk modulus of elasticity of gas $B = P$ (pressure of gas)

Adiabatic bulk modulus of elasticity of gas $B = \gamma \times P$ where $\gamma = \dfrac{C_p}{C_v}$.

Illustration 7

Find $\left|\dfrac{\Delta V}{V}\right| = ?$

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Solution:

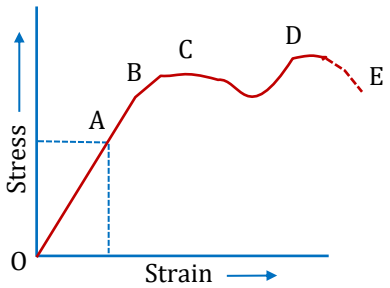
$$\Delta P = \frac{Mg}{A}$$

$$\therefore \quad \left|\frac{\Delta V}{V}\right| = \frac{\Delta P}{B} = \frac{Mg}{AB}$$

Variation of Strain with Stress:

When a wire is stretched by a load, it is seen that for small value of load, the extension produced in the wire is proportional to the load. On removing the load, the wire returns to its original length. The wire regains its original dimensions only when load applied is less or equal to a certain limit. This limit is called elastic limit. Thus, elastic limit is the maximum stress, where if external deforming force is removed, the bodies regain their original dimensions. In shown figure, this type of behaviour is represented by *OB* portion of the graph. Till *A* the stress is proportional to strain and from *A* to *B* if deforming forces are removed then the wire comes to its original length but here stress is not proportional to strain. (Graph is non-linear)

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OA → Limit of Proportionality
OB → Elastic Limit
C → Yield Point
CD → Plastic Behaviour
D → Ultimate point
DE → Fracture

As we go beyond the point *B*, then even for a very small increase in stress, the strain produced is very large. This type of behaviour is observed around point *C* and at this stage the wire begins to flow like a viscous fluid. The point *C* is called yield point. If the stress is further increased, then the wire breaks off at a point *D* called the breaking point. The stress corresponding to this point is called breaking stress or tensile strength of the material of the wire. A material for which the plastic range *CD* is relatively high is called ductile material. These materials get permanently deformed before breaking. The materials for which plastic range is relatively small are called brittle materials. These materials break as soon as elastic limit is crossed.

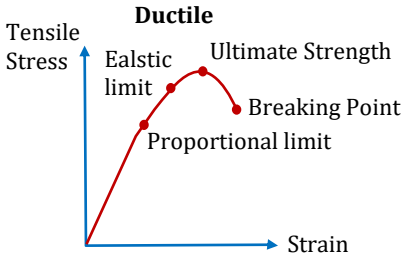
Important Points:

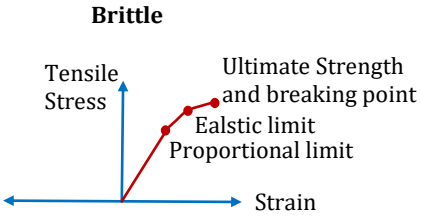
- Breaking stress = Breaking force/area of cross section.
- Breaking stress is constant for a material.
- Breaking force depends upon the area of the section of the wire of a given material.
- The working stress is always kept lower than that of a breaking stress so that safety factor = breaking stress/working stress may have a large value.
- Breaking strain = Elongation or compression/original dimension.

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Elastic Fatigue

The, the loss of strength of the material due to repeated strains on the material is called elastic fatigue. That is why bridges are declared unsafe after a long time of their use.

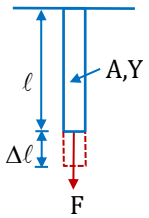


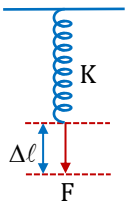


Analogy of Rod as a Spring

$$Y = \frac{\text{Stress}}{\text{Strain}} \quad \Rightarrow Y = \frac{F\ell}{A\Delta\ell}$$

or $$F = \frac{AY}{\ell}\Delta\ell$$

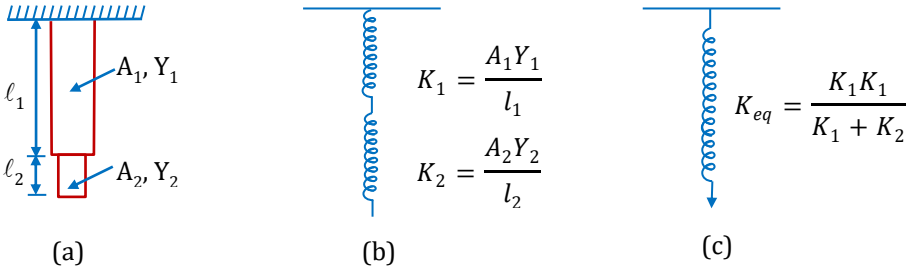




$\frac{AY}{\ell}$ = Constant, depends on type of material and geometry of rod.

$F = k\Delta\ell$, where $k = \frac{AY}{\ell}$ = Equivalent spring constant.

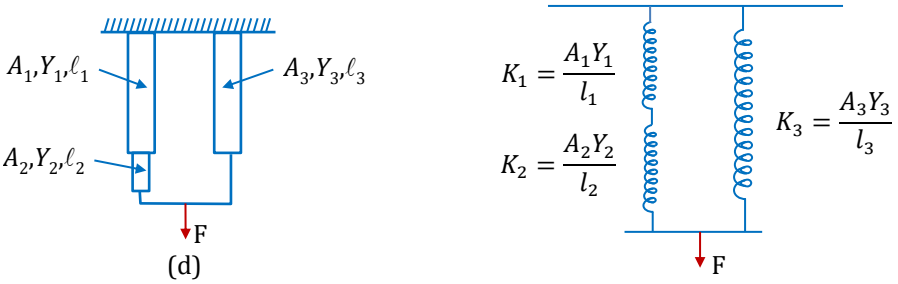
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(a) (b) (c)

For the system of rods shown in figure (a), the replaced spring system is shown in figure (b) two spring in series]. Figure (c) represents equivalent spring system.

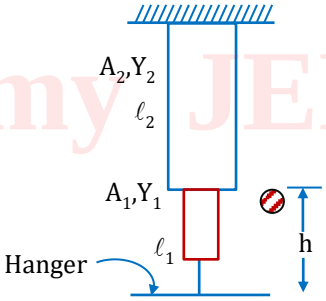
Figure (d) represents another combination of rods and their replaced spring system.



(d)

Illustration 8

A ball of mass *'m'* drops from a height *'h'*, which sticks to massless hanger after striking. Neglect over turning. Find K_{eq}. Also find out the maximum extension in rod. Assuming rod is massless, express answer in terms of K_{eq}.



Solution:

Applying energy conservation $mg\,(h + x) = \frac{1}{2}\frac{k_1k_2}{k_1+k_2}x^2$

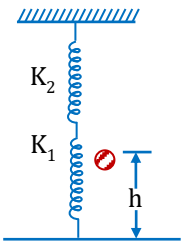
where $k_1 = \frac{A_1Y_1}{\ell_1}$; $k_2 = \frac{A_2Y_2}{\ell_2}$

$\Rightarrow \quad 2mgh + 2mgx = K_{eq}x^2$

$\Rightarrow \quad k_{eq}x^2 - 2mgx - 2mgh = 0$

$$x = \frac{2mg \pm \sqrt{4m^2g^2 + 8mghk_{eq}}}{2k_{eq}}$$

$$x_{max} = \frac{mg}{k_{eq}} + \sqrt{\frac{m^2g^2}{k_{eq}^2} + \frac{2mgh}{k_{eq}}}$$



and $k_{eq} = \frac{A_1A_2Y_1Y_2}{A_1Y_1\ell_2 + A_2Y_2\ell_1}$

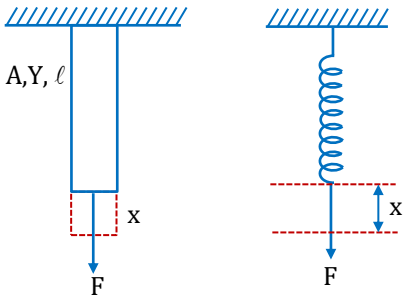
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Elastic Potential Energy Stored in a Stretched Wire or in a Rod:

Strain energy stored in equivalent spring

$$U = \frac{1}{2}kx^2$$

Where $x = \frac{F\ell}{AY}$, $k = \frac{AY}{\ell}$



$$U = \frac{1}{2}\frac{AY}{\ell}\frac{F^2\ell^2}{A^2Y^2} = \frac{1}{2}\frac{F^2\ell}{AY}$$

Equation can be re-arranged

$$U = \frac{1}{2}\frac{F^2}{A^2} \times \frac{\ell A}{Y} \qquad [\ell A = \text{Volume of rod}, F/A = \text{Stress}]$$

$$U = \frac{1}{2}\frac{(\text{Stress})^2}{Y} \times \text{Volume}$$

Again, $U = \frac{1}{2}\frac{F}{A} \times \frac{F}{AY} \times A\,\ell \qquad \left[\text{Strain} = \frac{F}{AY}\right]$

$$U = \frac{1}{2}\text{Stress} \times \text{Strain} \times \text{Volume}$$

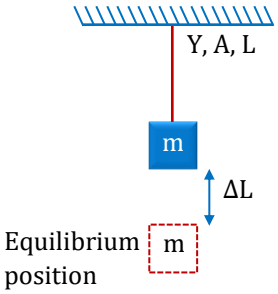
Again, $U = \frac{1}{2}\frac{F^2}{A^2Y^2}A\ell Y \qquad \Rightarrow U = \frac{1}{2}Y\,(\text{Strain})^2 \times \text{Volume}$

Strain energy density $= \frac{\text{Strain energy}}{\text{Volume}} = \frac{1}{2}\frac{(\text{Stress})^2}{Y} = \frac{1}{2}Y(\text{Strain})^2 = \frac{1}{2}\text{Stress} \times \text{Strain}$

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Illustration 9

Block stops at equilibrium. (a) Find U stored in wire in equilibrium position. (b) Find work done by gravity.

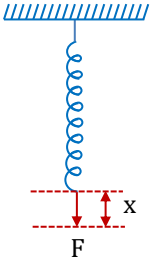


Solution:

(a) $U = \frac{1}{2} \times \text{Stress} \times \text{Strain} \times \text{Volume}$

$$U = \frac{1}{2} \times \frac{mg}{A} \times \frac{mg}{AY} \times AL \qquad \Rightarrow \quad U = \frac{1}{2} \times \frac{m^2g^2L}{AY}$$

(b) $W_g = mg \times (\Delta L) = mg \times \frac{mgL}{AY} \quad \Rightarrow \quad \frac{m^2g^2L}{AY}$



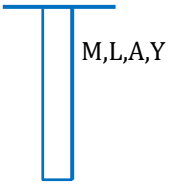
Note:

Rest half energy is lost in **Heat**.

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Illustration 10

A steel wire of mass M length L cross-sectional A and young's modulus of elasticity Y hanged vertically. Calculate U stored in equilibrium position.



Solution:

$$\text{Energy density} = \frac{dU}{dV} = \frac{dU}{Adx} = \frac{1}{2} \times \text{Stress} \times \text{Strain}$$

$$dU = \frac{1}{2} \times \frac{T}{A} \times \frac{T}{Ay} \times Adx$$

$$\Rightarrow \quad \int dU = \frac{1}{2} \frac{m^2 g^2}{L^2 Ay} \int_0^L x^2 dx$$

$$\Rightarrow \quad U = \frac{m^2 g^2}{6 L^2 Ay} (L)^3$$

$$U = \frac{m^2 g^2 L}{6 Ay}$$

Applications of Elasticity:

Some of the important applications of the elasticity of the materials are discussed as follows:

1. The material used in bridges lose its elastic strength with time bridges are declared unsafe after long use.

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2. **To estimate the maximum height of a mountain:**

The pressure at the base of the mountain = $h\rho g$ = stress. The elastic limit of a typical rock is 3×10^8 $N\,m^{-2}$. The stress must be less than the elastic limits, otherwise the rock begins to flow.

$$h < \frac{3 \times 10^8}{\rho g} \qquad \Rightarrow h < 10^4\ m\ (\because \rho = 3 \times 10^3\ kg\ m^{-3}\ ;\ g = 10\ ms^{-2}) \quad \text{or} \quad h = 10\ km$$

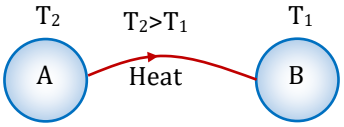
It may be noted that the height of Mount Everest is nearly 9 *km.*

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CALORIMETRY

Heat:

The energy that is being transferred between two bodies or between adjacent parts of a body as a result of temperature difference is called heat. Thus, heat is a form of energy. It is energy in transit whenever temperature differences exist. Once it is transferred, it becomes the internal energy of the receiving body. It should be clearly understood that the word "heat" is meaningful only as long as the energy is being transferred. Thus, expressions like "heat in a body" or "heat of a body" are meaningless.



When we say that a body is heated it means that its molecules begin to move with greater kinetic energy.

S.I. unit of heat energy is joule (J). Another common unit of heat energy is calorie (cal).

1 calorie = 4.18 joules

1 calorie : The amount of heat needed to increase the temperature of 1 gm of water from 14.5°C to 15.5 °C at one atmospheric pressure is 1 calorie.

Mechanical Equivalent of Heat:

In early days heat was not recognized as a form of energy. Heat was supposed to be something needed to raise the temperature of a body or to change its phase. Calorie was defined as the unit of heat. A number of experiments were performed to show that the temperature may also be increased by doing mechanical work on the system. These experiments established that heat is equivalent to mechanical energy and measured how much mechanical energy is equivalent to a calorie. If mechanical work W produces the same temperature change as heat H, we write, $\boldsymbol{W = JH}$, where J is called mechanical equivalent of heat. J is expressed in joule/calorie. The value of J gives how many joules of mechanical work is needed to raise the temperature of 1g of water by 1°C. Value of J is 4.18 J/cal.

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Specific Heat:

Specific heat of substance is equal to heat gain or released by that substance to raise or fall its temperature by 1°C for a unit mass of substance.

When a body is heated, it gains heat. On the other hand, heat is lost when the body is cooled. The gain or loss of heat is directly proportional to:

(a) the mass of the body $\Delta Q \propto m$

(b) rise or fall of temperature of the body $\Delta Q \propto \Delta T$

$\Delta Q \propto m\,\Delta T$ or $\Delta Q = m\,s\,\Delta T$ or $dQ = msdT$ or $Q = \int msdT$

where s is a known as the specific heat of the body.

If s is independent of body temperature then $\Delta Q = ms\,\Delta T$ is used. If s is temperature dependent $Q = \int msdT$ is used.

$s = \frac{Q}{m\Delta T}$. S.I. unit of s is joule/kg-Kelvin and C.G.S. unit is cal./gm °C.

Specific heat of water : S = 4200 J/kg°C = 1000 $cal/kg°C$ = 1 $Kcal/kg°C$ = 1 $cal/gm°C$

Specific heat of steam = Half of specific heat of water = Specific heat of ice

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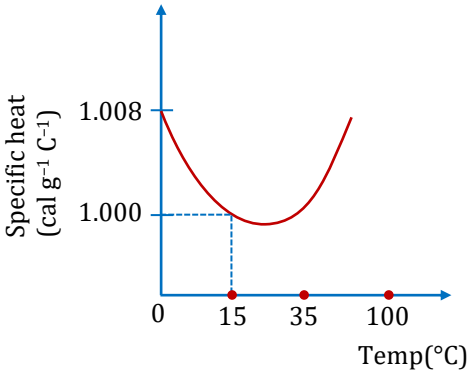
Heat Capacity or Thermal Capacity:

Heat capacity of a body is defined as the amount of heat required to raise the temperature of that body by 1°*C*. If *'m'* is the mass and *'s'* the specific heat of the body, then **Heat capacity = *ms***.

Units of heat capacity in: *CGS* system is, ***cal °C*$^{-1}$**; *SI* unit is, ***JK*$^{-1}$**

Important Points:

(a) We know, $s = \frac{Q}{m\Delta T}$, if the substance undergoes the change of state which occurs at constant temperature ($\Delta T = 0$), then $s = Q/0 = \infty$. Thus, the specific heat of a substance when it melts or boils at constant temperature is infinite. More about this to be studied in thermodynamics.



(b) If the temperature of the substance changes without the transfer of heat ($Q = 0$) then $s = 0$. Thus, when liquid in the thermos flask is shaken, its temperature increases without the transfer of heat and hence the specific heat of liquid in the thermos flask is zero. More about this to be studied in thermodynamics.

(c) To raise the temperature of saturated water vapours, heat (Q) is withdrawn. Hence, specific heat of saturated water vapours is negative. (This is for your information only and not in the course)

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(d) The slight variation of specific heat of water with temperature is shown in the graph at 1 atmospheric pressure. Its variation is less than 1% over the interval form 0 to 100°*C*. Hence it can be taken as a constant.

Illustration 11

If 15 *gm* liquid with $s = 3$ *cal/g-k* with 25°*C* is supplied 450 cal. then find its final temperature.

Solution:

$\Delta Q = ms\Delta T$

$450 = 15 \times 3 \times (T_f - 25)$

$\frac{30}{3} = T_f - 25$

$T_f = 35°C$

Illustration 12

A geyser of power 2.09 *k* watt has capacity of 5*kg* water. Just after starting it, find rate of change of temperature of water. $s_{water} = 1\frac{cal}{g-k}$

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Solution:

$$\left(\frac{\Delta Q}{\Delta t}\right)=ms\left(\frac{\Delta T}{\Delta t}\right)$$

$$2.09\times10^{3}\frac{J}{\text{sec.}}=5kg\times4.18\times10^{3}\frac{J}{kg-K}\times\left(\frac{\Delta T}{\Delta t}\right)$$

$$\left(\frac{\Delta T}{\Delta t}\right)=\frac{1}{10}K/\text{sec.}=0.1K/\text{sec}$$

Illustration 13

Specific heat of a substance depends on temperature as $s = 3T^2$, where temperature is in kelvin and the constant 3 is in SI units. Find heat required to raise temperature of unit mass from $2K$ to $5K$?

Solution:

$$\int dQ=\int msdT$$

$$\Delta Q=1\int_{2}^{5}3T^{2}dT$$

$\Delta Q = [5^3 - 2^3] \quad \Rightarrow \quad 117\,J$

Relation between Specific Heat and Water Equivalent:

It is the amount of water which requires the same amount of heat for the same temperature rise as that of the object (mass m and specific heat s).

$$ms\,\Delta T = m_W s_W \Delta T \quad \Rightarrow \quad m_W = \frac{ms}{s_W}$$

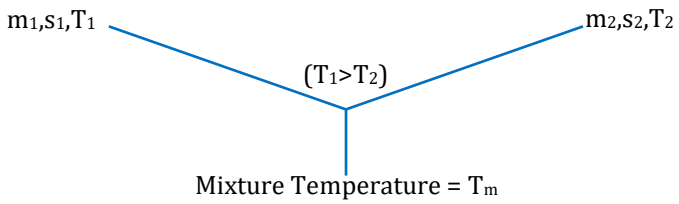
In calorie $s_W = 1$ $\quad \therefore \quad m_W = ms$

m_w is also represent by W

So, $\quad W = ms$

Law of Mixture:

When two substances at different temperatures are mixed together, then exchange of heat continues to take place till their temperatures become equal. This temperature is then called final temperature of mixture. Here, **Heat taken by one substance = Heat given by another substance**



$\Rightarrow \quad m_1 s_1 (T_1 - T_m) = m_2 s_2 (T_m - T_2)$

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Illustration 14

The temperature of equal masses of three different liquids *A, B,* and *C* are 10°*C*, 15°*C* and 20°*C* respectively. The temperature when *A* and *B* are mixed is 13°*C* and when *B* and *C* are mixed, it is 16°*C*. What will be the temperature when *A* and *C* are mixed?

Solution:

When *A* and *B* are mixed $mS_1 \times (13 - 10) = m \times S_2 \times (15 - 13)$

$$3S_1 = 2S_2 \quad ...(1)$$

When *B* and *C* are mixed

$$S_2 \times 1 = S_3 \times 4 \quad ...(2)$$

When *C* and *A* are mixed

$$S_1(\theta - 10) = S_3 \times (20 - \theta) \quad ...(3)$$

By using equation (1), (2) and (3)

We get $\theta = \frac{140}{11}$ °C

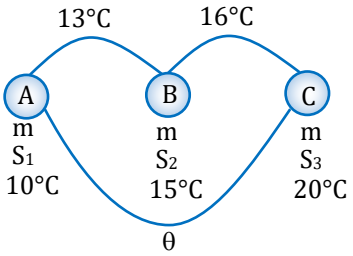


Illustration 15

If three different liquid of different masses, specific heats and temperature are mixed with each other and then what is the temperature of mixture at thermal equilibrium.

$m_1, s_1, T_1 \rightarrow$ Specification for liquid 1

$m_2, s_2, T_2 \rightarrow$ Specification for liquid 2

$m_3, s_3, T_3 \rightarrow$ Specification for liquid 3

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Solution:

Total heat lost or gain by all substance is equal to zero,

$$\Delta Q = 0$$

$$m_1s_1(T - T_1) + m_2s_2(T - T_2) + m_3s_3(T - T_3) = 0$$

So, $$T = \frac{m_1s_1T_1 + m_2s_2T_2 + m_3s_3T_3}{m_1s_1 + m_2s_2 + m_3s_3}$$

Phase Change

Heat required for the change of phase or state,

$\boldsymbol{Q = mL}$, L = Latent heat.

Latent Heat (L):

The heat supplied to a substance of unit mass which changes its state at constant temperature is called latent heat of the body.

Latent Heat of Fusion (L_f):

The heat supplied to a substance which changes it from solid to liquid state at its melting point and 1 *atm*. pressure is called latent heat of fusion. Latent heat of fusion of ice is 80 $kcal/kg$

Latent Heat of Vaporization (L_v):

The heat supplied to a substance which changes it from liquid to vapour state at its boiling point and 1 *atm*. pressure is called latent heat of vaporization. Latent heat of vaporization of water is 540 $kcal\ kg^{-1}$.

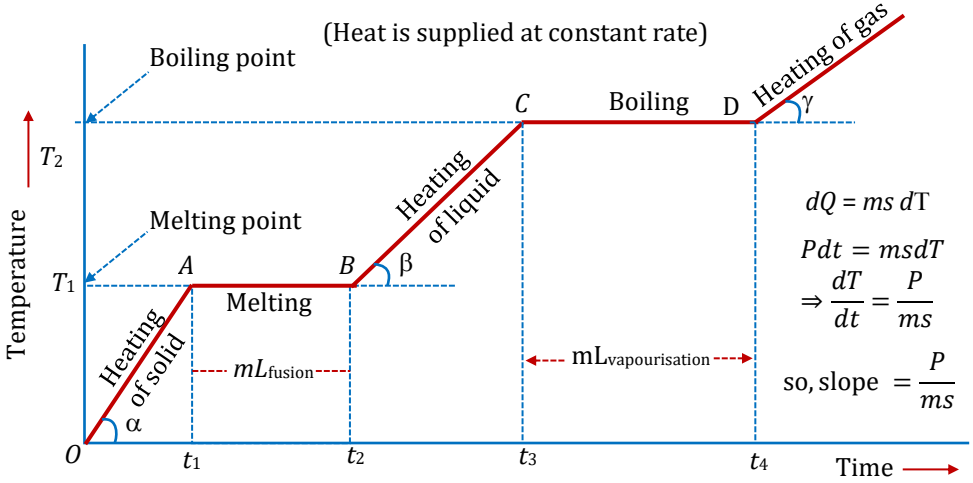
Latent Heat of Ice : $L = 80\ cal/gm = 80\ Kcal/kg = 4200 \times 80\ J/kg$

Latent Heat of Steam : $L = 540\ cal/gm = 540\ Kcal/kg = 4200 \times 540\ J/kg$

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Heating Curve:

If to a given mass (m) of a solid, heat is supplied at constant rate and a graph is plotted between temperature and time as shown in figure, then the graph is called heating curve.



- **In the Region *OA*:**

  Rate of heat supply P is constant and temperature of solid is changing with time

  So, $Q = mc_S\Delta T \Rightarrow P\,\Delta t = mc_S\Delta T$ $[\because Q = P\,\Delta t]$

  $\because$ $\frac{\Delta T}{\Delta t}$ = The slope of temperature–time curve so specific heat of solid $c_S \propto \frac{1}{\text{Slope of line } OA}$

  specific heat (or thermal capacity) is inversely proportional to the slope of temperature–time curve.

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- **In the Region *AB*:**

  Temperature is constant, so it represents change of state, i.e., melting of solid with melting point T_1. At point A melting starts and at point B all solid is converted into liquid. So, between A and B substance is partly solid and partly liquid. If L_F is the latent heat of fusion then

  $$Q = mL_F \Rightarrow L_F = \frac{P(t_2 - t_1)}{m} \quad [\text{as } Q = P(t_2 - t_1)] \Rightarrow L_F \propto \text{length of line } AB$$

  i.e., Latent heat of fusion is proportional to the length of line of zero slope.

  [In this region specific heat $\propto \frac{1}{\tan 0^\circ} = \infty$]

- **In the Region *BC*:**

  Temperature of liquid increases so specific heat (or thermal capacity) of liquid will be inversely proportional to the slope of line BC, $c_L \propto \frac{1}{\text{Slope of line BC}}$

- **In the Region *CD*:**

  Temperature is constant, so it represents change of state, i.e., liquid is boiling with boiling point T_2. At C all substance is in liquid state while at D is vapour state and between C and D partly liquid and partly gas. The length of line CD is proportional to latent heat of vaporisation, i.e., $L_V \propto$ Length of line CD.

  [In this region specific heat $\propto \frac{1}{\tan 0^\circ} = \infty$]

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- **In the Region *DE*:**
  The line *DE* represents gaseous state of substance with its temperature increasing linearly with time. The reciprocal of slope of line will be proportional to specific heat or thermal capacity of substance in vapour state.

Illustration 16

Heat is supplied at constant rate to ice and its temperature remain constant for 5 *min* After that temperature started increasing at a rate of $2°C/min$. Find $\frac{L_{\text{fusion}}}{s_{\text{water}}}$?

Solution:

Let rate of heat supply = P

$P \times 5 = mL$ ...(1)

$$\frac{\Delta Q}{\Delta t} = ms_{water}\frac{\Delta T}{\Delta t}$$

$P = ms_{water} \times 2$ ...(2)

Solving equation (1) and (2)

$$\boxed{\frac{L_{\text{fusion}}}{s_{\text{water}}} = 10}$$

Calorimeter:

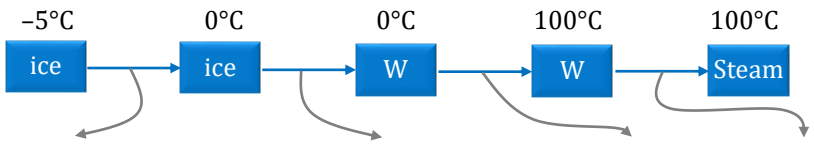
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A device in which heat measurement can be made is called a **calorimeter**. It consists a metallic vessel and stirrer of the same material like copper or aluminium. The vessel is kept inside a wooden jacket which contains heat insulating materials like glass, wool etc. The outer jacket acts as a heat shield and reduces the heat loss from the inner vessel. There is an opening in the outer jacket through which a mercury thermometer can be inserted into the calorimeter. The specific heat capacity of a given solid can be determinated by using the principle, heat gained is equal to the heat lost. Water equivalent of a calorimeter means amount of heat required for the equivalent amount of water to raise temperature by $1°C$.

Illustration 17

Calculate heat required to convert 10 *gm* ice at $-5°C$ to steam at $100°C$. (specific heat of water is 1 cal/*g*-°*c*, specific heat of water is 0.5 cal/*g*-°C)

Solution:



$\Delta Q = (10g \times 0.5\text{cal./g-°c} \times 5°c) + (10g \times 80\ \text{cal/g}) + (10g \times 1\ \text{cal/g-°c} \times (100)°c) + (10g \times 540\ \text{cal/g})$

$\Delta Q = 25 + 800 + 1000 + 5400 = 7225$ cal

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Illustration 18

A calorimeter of heat capacity 100 J/K is at room temperature of 30°C. 100 g of water at 40°C of specific heat 4200 $J/kg–K$ is poured into the calorimeter. What is the temperature of water in calorimeter?

Solution:

Let the temperature of water in calorimeter is t. Then heat lost by water = Heat gained by calorimeter

$$(0.1) \times 4200 \times (40 - t) = 100\,(t - 30)$$

$$\Rightarrow \quad 420 \times 40 - 420t = 100t - 3000$$

$$\Rightarrow \quad t = 38.07°C$$

Illustration 19

Steam at 100°C is passed into 1.1 kg of water contained in a calorimeter of water equivalent 0.02 kg at 15°C till the temperature of the calorimeter and its contents rises to 80°C. What is the mass of steam condensed? Latent heat of steam = 536 cal/g.

Solution:

Heat required by (calorimeter + water)

$$Q = (m_1c_1 + m_2c_2)\,\Delta\theta = (0.02 + 1.1 \times 1)\,(80 - 15) = 72.8\ kcal$$

If m is mass of steam condensed, then heat given by steam

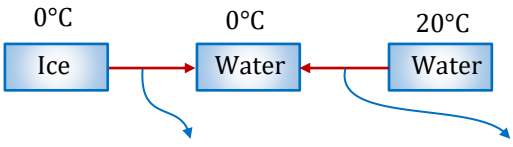
$$Q = mL + mc\,\Delta\theta = m \times 536 + m \times 1 \times (100 - 80) = 556\,m \quad \therefore\ 556\,m = 72.8$$

$$\therefore \quad \text{Mass of steam condensed } m = \frac{72.8}{556} = 0.130\ kg$$

Illustration 20

50 gm ice at 0°C + 100 g water at 20°C. Find final temperature and composition of mixture.

Solution:



If all ice to water = 50 × 80 = 4000 cal. 100 × 1 × 20 = 2000 cal.

Hence entire 100g ice will not melt.

Let suppose x gm ice melts

$$\therefore \quad x \times 80 = 100 \times 20$$

$$x = 25\ gm$$

$$\therefore \quad T_f = 0°C \text{ and it has } 25\ g \text{ ice} + 125\ gm \text{ water.}$$

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THERMAL EXPANSION

Potential Energy V/s Distance Curve:

When matter is heated without change in state, it usually expands. According to atomic theory of matter, asymmetry in potential energy curve is responsible for thermal expansion as with rise in temperature say from T_1 to T_2 the amplitude of vibration and hence energy of atoms increases from E_1 to E_2 and hence the average distance between the from r_1 to r_2.

$T_2 > T_1$

$E_2 > E_1$

$r_{2\ avg.} > r_{1\ avg.}$

(Due to asymmetric nature of curve)

Due to this increase in distance between atoms, the matter as a whole expands which is known as thermal expansion. Had the potential energy curve been symmetrical, no thermal expansion would have taken place in spite of heating.

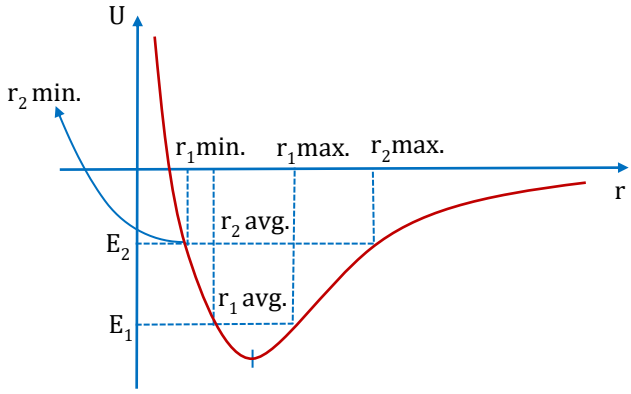
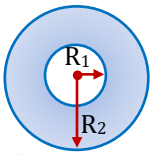


Illustration 21

In the given figure, when temperature is increased then which of the following increases:

(A) R_1 (B) R_2 (C) $R_2 - R_1$

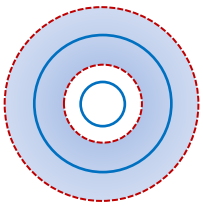


Solution:

All of the above

--------- Represents expanded Boundary

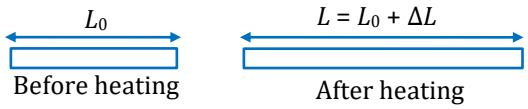
——— Represents original Boundary



As the intermolecular distance between atoms increases on heating hence the inner and outer perimeter increases. Also, if the atomic arrangement in radial direction is observed then we can say that it also increases hence all *A, B, C* are true.

Linear Expansion:

When the rod is heated, its increase in length ΔL is proportional to its original length L_0 and change in temperature ΔT where ΔT is in $^\circ C$ or K.



$$dL = \alpha L_0 dT \quad \Rightarrow \quad \Delta L = \alpha L_0 \Delta T$$

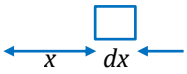
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If, $\alpha\Delta T << 1$

$\alpha = \dfrac{\Delta L}{L_0 \Delta T}$ where α is called the coefficient of linear expansion whose unit is $°C^{-1}$ or K^{-1}.

$L = L_0(1+\alpha\Delta T)$ where L is the length after heating the rod.

Variation of α with Temperature and Distance:



(a) If α varies with distance, for example $\alpha = ax + b$.
Then total expansion = $\int (ax + b)\Delta T dx$.

(b) If α varies with temperature, $\alpha = f(T)$. Then $\Delta L = \int \alpha L_0 dT$

Note:

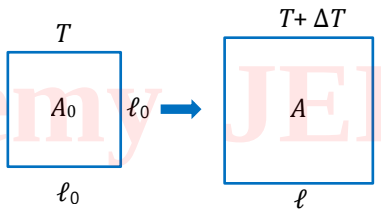
Actually, thermal expansion is always 3-*D* expansion. When other two dimensions of object are negligible with respect to one, then observations are significant only in one dimension and it is known as linear expansion.

Superficial (Areal) Expansion:

$A = A_0(1+\beta\Delta T)$

Also, $A_0 = \ell_0^2$ and $A = \ell^2$

So, $\ell^2 = \ell_0^2(1+\beta\Delta T) = \left[\ell_0(1+\alpha\Delta T)\right]^2 \Rightarrow \beta = 2\alpha$



Volume Expansion:

$V = V_0(1+\gamma\Delta T)$

Also, $V = \ell^3$ and $V_0 = \ell_0^3$ so $\gamma = 3\alpha$

$\Rightarrow$ $\alpha : \beta : \gamma = 1 : 2 : 3$

Unit of $\alpha : \beta : \gamma$ is $1/°C$ or $1/K$ or K^{-1}.

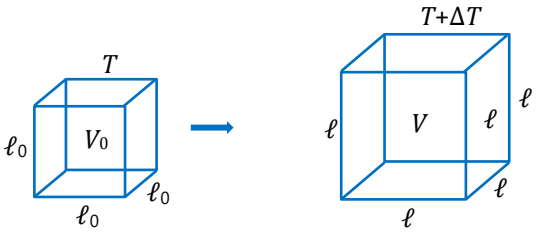
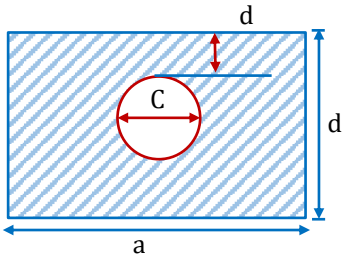


Illustration 22

A rectangular plate has a circular cavity as shown in the figure. If we increase its temperature then which dimension will increase in following figure.



Solution:

Distance between any two point on an object increases with increase in temperature. So, all dimension *a, b, c* and *d* will increase.

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Illustration 23

Iron rod of 50 *cm* is heated from 20°*C* to 100°*C*. Find length of rod at 100°*C* if $\alpha_{iron} = 12 \times 10^{-6}\ °C^{-1}$.

Solution:

$\ell_2 = \ell_1[1 + \alpha(T_2 - T_1)]$

$\ell_2 = 50[1 + 12 \times 10^{-6} \times 80]$

$\ell_2 = 50[1 + 96 \times 10^{-5}] = 50 + 48 \times 10^{-3}$

$\ell_2 = 50.048\ cm$

Illustration 24

The difference between lengths of a certain brass rod and of a steel rod is claimed to be constant at all temperatures. Is this possible ?

Solution:

If L_B and L_S are the lengths of brass and steel rods respectively at a given temperature, then the lengths of the rods when temperature is changed by θ°*C*.

$$L_B' = L_B(1+\alpha_B \Delta\theta) \text{ and } L_S' = L_S(1+\alpha_B \Delta\theta)$$

So, that $L_B' - L_S' = (L_B - L_S) + (L_B \alpha_B - L_S \alpha_S)\Delta\theta$

So, $(L_B' - L_S')$ will be equal to $(L_B - L_S)$ at all temperatures if , $L_B\alpha_B - L_S\alpha_S = 0$ [as $\Delta\theta \neq 0$] or $\frac{L_B}{L_S} = \frac{\alpha_S}{\alpha_B}$

i.e., the difference in the lengths of the two rods will be independent of temperature if the lengths are in the inverse ratio of their coefficients of linear expansion.

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Variation of Time Period of Pendulum Clocks:

The time represented by the clock hands of a pendulum clock depends on the number of oscillation performed by pendulum every time it reaches to its extreme position the second hand of the clock advances by one second that means second hand moves by two seconds when one oscillation in complete. Such a pendulum whose time period is 2 sec is called a "seconds pendulum".

Let $T = 2\pi\sqrt{\frac{L_0}{g}}$ at temperature θ_0 and $T' = 2\pi\sqrt{\frac{L}{g}}$ at temperature θ.

$$\frac{T'}{T} = \sqrt{\frac{L'}{L}} = \sqrt{\frac{L[1+\alpha\Delta\theta]}{L}} = 1 + \frac{1}{2}\alpha\,\Delta\,\theta$$

Therefore change (loss or gain) in time per unit time lapsed is $\frac{T'-T}{T} = \frac{1}{2}\alpha\,\Delta\theta$

Gain or loss in time in duration of 't' in

$\Delta T = \frac{1}{2}\alpha\,\Delta\theta T$, if T is the correct time then

(a) $\theta < \theta_0$, $T' < T$ clock becomes fast and gain time

(b) $\theta > \theta_0$, $T' > T$ clock becomes slow and loose time

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Illustration 25

A pendulum clock consists of an iron rod connected to a small, heavy bob. If it is designed to keep correct time at 20°C, how fast or slow will it go in 24 hours at 40°C? Coefficient of linear expansion of iron = 1.2×10^{-6} /°C.

Solution:

The time difference occurred in 24 hours (86400 seconds) is given by

$$\Delta t = \frac{1}{2}\,\alpha\,\Delta\theta\,t = \frac{1}{2} \times 1.2 \times 10^{-6} \times 20 \times 86400 = 1.04 \text{ sec.}$$

This is loss of time as θ is greater than θ_0. As the temperature increases, the time period also increases. Thus, the clock goes slow.

Measurement of Length by Metallic Scale:

Case (i): When object is expanded only;

No expansion of scale

$\ell_2 = \ell_1\{1 + \alpha_0(\theta_2 - \theta_1)$

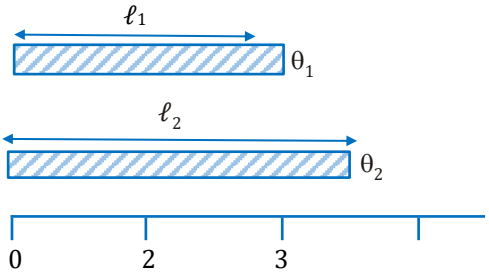
ℓ_1 = Actual length of object at θ_1°C

= Measured length of object at θ_1°C.

ℓ_2 = Actual length of object at θ_2°C

= Measured length of object at θ_2°C.

α_0 = Linear expansion coefficient of object.



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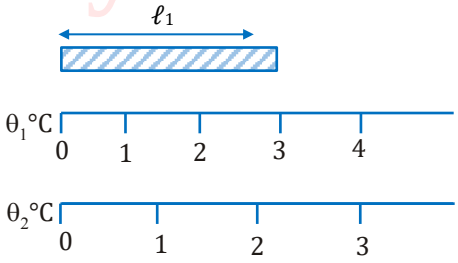
Case (ii): When only measurement instrument is expanded actual length of object will not change but measured value (MV) decreases.

$MV = \ell_1\{1 - \alpha_S(\theta_2 - \theta_1)\}$

α_S = Linear expansion coefficient of measuring instrument.

at θ_1 °C $MV = 3$

at θ_2 °C $MV = 2.2$



Case (iii): If both expanded simultaneously

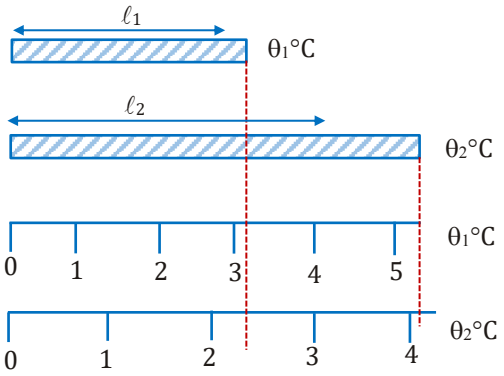
$MV = \ell_1\{1 + (\alpha_0 - \alpha_s)(\theta_2 - \theta_1)\}$

(i) If $\alpha_0 > \alpha_s$, then measured value is more then the actual value at θ_1°C

(ii) If $\alpha_0 < \alpha_s$, then measured value is less then the actual value at θ_1°C

at θ_1°C $MV = 3.4$

θ_2°C $MV = 4.1$



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Measured value = Calibrated value × {1 + α Δ θ}

where $\alpha = \alpha_0 - \alpha_s$

α_o = Coefficient of linear expansion of object material,

α_s = Coefficient of linear expansion of scale material

$\Delta\theta = \theta - \theta_C$

θ = Temperature at the time of measurement

θ_C = Temperature at the time of calibration.

For scale, true measurement = Scale reading $[1 + \alpha (\theta - \theta_0)]$

If $\theta > \theta_0$ true measurement > Scale reading

$\theta < \theta_0$ true measurement < Scale reading

Illustration 26

A bar measured with a Vernier caliper is found to be 180mm long. The temperature during the measurement is 10°C. The measurement error will be if the scale of the Vernier calliper has been graduated at a temperature of 20°C : ($\alpha = 1.1 \times 10^{-5}\ °C^{-1}$). Assume that the length of the bar does not change.)

(A) $1.98 \times 10^{-1}\ mm$ (B) $1.98 \times 10^{-2}\ mm$ (C) $1.98 \times 10^{-3}\ mm$ (D) $1.98 \times 10^{-4}\ mm$

Ans. (B)

Solution:

True measurement = Scale reading $[1 + \alpha (\theta - \theta_0)] = 180\ [1 - 10 \times 1.1 \times 10^{-5}]$

Error = $180 - 180\ [1 - 1.1 \times 10^{-4}] = 1.98 \times 10^{-2}\ mm$

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Illustration 27

A brass scale correctly calibrated at 15°C is employed to measure a length at a temperature of 35°C. If the scale gives a reading of 75cm, find the true length.

(Linear expansion coefficient of brass = $2.0 \times 10^{-5}\ °C^{-1}$)

Solution:

Let the distance between two fixed divisions on the scale at 15°C be L_1 and that at 35°C be L_2.

Clearly, $(L_2 - L_1) = \alpha L_1\ (35 - 15)$

or $L_2 = L_1\ (1 + 20 \times 2.0 \times 10 - 5) = L_1\ (1.0004)$

i.e., at 35°C, an actual length of L_2 will be read as L_1 (note), due to the increased separation of the divisions of the scale. In other words, the observed length will be less than the actual length.

Given, $L_1 = 75\ cm$

$\therefore$ $L_2 = 75\ (1.0004)\ cm = 75.03\ cm$

Illustration 28

A glass rod when measured with a zinc scale, both being at 30°C, appears to be of length 100cm. If the scale shows correct reading at 0°C, determine the true length of the glass rod at (a) 30°C and (b) 0°C.

('α' for glass = $8 \times 10^{-6}/°C$ and for zinc $26 \times 10^{-6}/°C$)

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Solution:

At 30°*C*, although the reading shown by the zinc scale corresponding to the length of the glass rod is 100*cm*, but the actual length would be more than 100*cm*, the reason being the increased separation between the markings, owing to a rise in temperature (from 0°*C* to 30°*C*).

Now, an actual (original at 0°*C*) length of 100*cm* on the zinc scale (or more precisely, two markings or divisions on the scale, separated by a distance of 100*cm*) would, at a temperature of 30°*C*, correspond to a length given by

$l = 100\ (1 + 26 \times 10^{-6} \times 30)\ cm$

$= 100.078\ cm$

The true length of the glass rod at 30°*C* is 100.078 *cm.*

Now, at 0°*C*, the length of glass rod would be lesser than that at 30°*C*,

$\therefore$ Using $l_t = l_0\ (1 + \alpha\Delta t)$, $l_0 = \dfrac{l_t}{1+\alpha\Delta t}$

$\therefore$ The length of the rod at 0°*C*, will be $l_0 = \dfrac{100.078 cm}{(1+8\times10^{-6}\times30)} = 100.054\ cm$

Thermal Stress of a Material:

If a rod is free to expand then there will be no stress and no strain. Stress and strain is produced only when an object is restricted to expand or contract according to change in temperature. When the temperature of the rod is decreased or increased under constrained condition, compressive or tensile stresses are developed in the rod. These stresses are known as thermal stresses.

$$\text{Strain} = \frac{\Delta L}{L_0} = \frac{\text{Final length} - \text{Original length}}{\text{Original length}} = \alpha\,\Delta T$$

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Here original length refers to natural length of rod at new temperature.

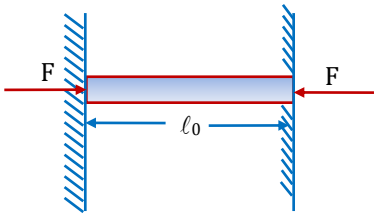
Note:

Original and final length should be at same temperature.

Consider a rod of length ℓ_0 which is fixed between two rigid end separated at a distance ℓ_0 now if the temperature of the rod is increased by $\Delta\theta$ then the strain produced in the rod will be:

$$\text{Strain} = \frac{\text{Length of the rod at new temperature-natural length of the rod at new temperature}}{\text{Natural length of the rod at new temperature}}$$

$$= \frac{\ell_0 - \ell_0(1+\alpha\Delta\theta)}{\ell_0(1+\alpha\Delta\theta)} = \frac{-\ell_0\alpha\Delta\theta}{\ell_0(1+\alpha\Delta\theta)}$$



$\because$ α is very small so, strain = $-\alpha\Delta\theta$ (negative sign in the answer represents that the length of the rod is less than the natural length that means is compressed by the ends.)

Illustration 29

A steel rod of length 1*m* rests on a smooth horizontal base. If it is heated from 0°*C* to 100°*C*, what is the longitudinal strain developed?

Solution:

In absence of external force, no strain or stress will be created hear rod is free to move.

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Illustration 30



In the given figure a rod is free at one end and other end is fixed. When we change the temperature of rod by $\Delta\theta$, then strain produced in the rod will be

(A) $\alpha\Delta\theta$ (B) $\frac{1}{2}\alpha\Delta\theta$ (C) zero (D) information incomplete

Solution:

Here rod is free to expand from one side by so by changing temperature no strain will be produced in the rod. Hence answer is (C).

Illustration 31

An iron ring measuring 15.00 *cm* in diameter is to be shrunk on a pulley which is 15.05 *cm* in diameter. All measurements refer to the room temperature 20°*C*. To what minimum temperature should the ring be heated to make the job possible? Calculate the strain developed in the ring when it comes to the room temperature. Coefficient of linear expansion of iron = $12 \times 10^{-6}/°C$.

Solution:

The ring should be heated to increase its diameter from 15.00 *cm* to 15.05 *cm*.

Using, $\ell_2 = \ell_1 (1 + \alpha\,\Delta\theta) = \dfrac{0.05cm}{15.00cm \times 12 \times 10^{-6}/°C} = 278°C$

The temperature = $20°C + 278°C = 298°C$.

The strain developed = $\dfrac{\ell_2 - \ell_1}{\ell_1} = 3.33 \times 10^{-3}$.

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Illustration 32

A steel rod is clamped at its two ends and rests on a fixed horizontal base. The rod is in natural length at 20°*C*. Find the longitudinal strain developed in the rod if the temperature rises to 50°*C*. Coefficient of linear expansion of steel = $1.2 \times 10^{-5}/°C$.

Solution:

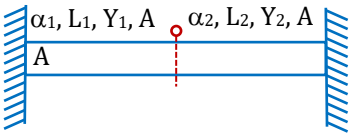
As we known that strain

$$\text{Strain} = \frac{\text{Change in length}}{\text{Original length}} = \frac{\Delta\ell}{\ell_0}$$

$\therefore$ Strain = $\alpha\Delta\theta = 1.2 \times 10^{-5} \times (50 - 20) = 3.6 \times 10^{-4}$

Here strain is compressive strain because final length is smaller than initial length.

Illustration 33



In the given figure, rods are fixed between two rigid end, on increasing temperature by ΔT. Find the shift in the interface of rods.

Solution:

Compression,

$$\Delta L_1 = L_1\alpha_1\Delta T - x \quad ; \quad \Delta L_2 = L_2\alpha_2\Delta T + x$$

$$F = \frac{YA\Delta L}{L}$$

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$$F_1 = \frac{Y_1 A\left(L_1\alpha_1\Delta T - x\right)}{L_1}$$

$$F_2 = \frac{Y_2 A\left(L_2\alpha_2\Delta T + x\right)}{L_2}$$

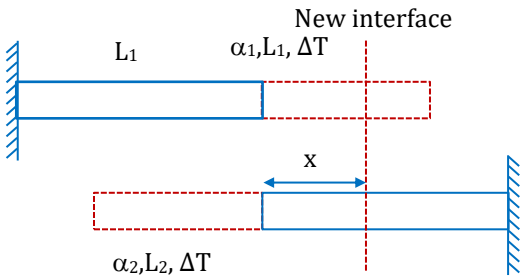
$F_1 = F_2$ (By Newton third law)

$$\frac{Y_1 A\left(L_1\alpha_1\Delta T - x\right)}{L_1} = \frac{Y_2 A\left(L_2\alpha_2\Delta T + x\right)}{L_2}$$

$$L_2\left(Y_1 L_1\alpha_1\Delta T - Y_1 x\right) = L_1\left(Y_2 L_2\alpha_2\Delta T + Y_2 x\right)$$

$$Y_1 L_1 L_2\alpha_1\Delta T - Y_2 L_2 L_1\alpha_2\Delta T = \left(Y_1 L_2 + Y_2 L_1\right)x$$

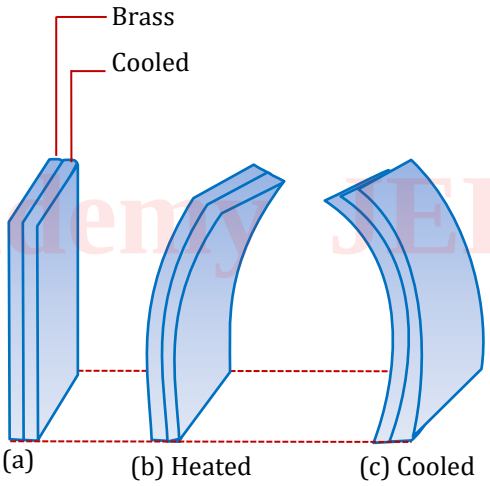
$$x = \frac{Y_1 L_1 L_2\alpha_1\Delta T - Y_2 L_2 L_1\alpha_2\Delta T}{\left(Y_1 L_2 + Y_2 L_1\right)} = \frac{\left(Y_1\alpha_1 - Y_2\alpha_2\right)\Delta T}{\left(\frac{Y_1}{L_1} + \frac{Y_2}{L_2}\right)}$$



The Bimetallic Strip:

A bimetallic strip is made from two thin strips of metal that have different coefficients of linear expansion, as figure shows.

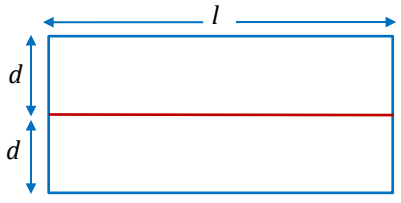
Often brass [$\alpha = 19 \times 10^{-6}\ (C°)^{-1}$] and steel [$\alpha = 12 \times 10^{-6}\ (C°)^{-1}$] are selected. The two pieces are welded or riveted together. When the bimetallic strip is heated, the brass, having the larger value of α, expands more than the steel. Since the two metals are bonded together, the bimetallic strip bends into an arc as in part *b*, with the longer brass piece having a larger radius than the steel piece. When the strip is cooled, the bimetallic strip bends in the opposite direction, as in part *c*.



(a) A bimetallic strip and how it behaves when (b) heated and (c) cooled

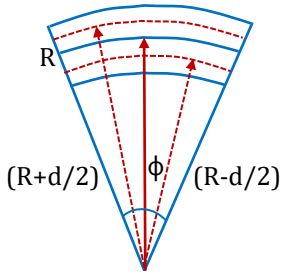
Mathematical Solution:

$$\left(R + \frac{d}{2}\right)\phi = L_0\ (1 + \alpha_1\Delta\theta)$$



$$\left(R - \frac{d}{2}\right)\phi = L_0\ (1 + \alpha_2\Delta\theta)$$

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$$\frac{R+\frac{d}{2}}{R-\frac{d}{2}}=\frac{1+\alpha_1\Delta\theta}{1+\alpha_2\Delta\theta}$$

Illustration 34

A metal sheet 50 *cm* long and 8 *cm* wide at 0°*C* has surface area 400.1 cm^2 at 100°*C*. Find α ?

Solution:

M-1 : $50(1+\alpha\Delta T)\times 8(1+\alpha\Delta T)=400.1$

$400(1+2\alpha\Delta T)=400.1$

$400\times 2\alpha\times 100=0.1$

$\alpha=1.25\times10^{-6}$

M-2 : $\beta=\frac{\Delta A}{A\Delta T}\Rightarrow\frac{0.1}{400\times100}=2\alpha=\beta=\frac{0.1}{4\times10^4}=\frac{10^{-5}}{4}=2.5\times10^{-6}$

Illustration 35

If volume of metal increases by 0.12% on increasing temperature by 20°*C*. Find α ?

Solution:

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$$\gamma=\frac{\Delta V}{V\Delta T}$$

$$\frac{\Delta V}{V}=\frac{0.12}{100}=12\times10^{-4}$$

$$3\alpha=\frac{12\times10^{-4}}{20}$$

$$\alpha=\frac{1}{5}\times10^{-4}$$

$\alpha=0.2\times10^{-4}$

Illustration 36

The radius of a metal sphere at room temperature *T* is *R*, and the coefficient of linear expansion of the metal is α. The sphere is heated a little by a temperature ΔT so that its new temperature is $T+\Delta T$. The increase in the volume of the sphere is approximately

(A) $2\pi R\,\alpha\,\Delta T$ (B) $\pi R^2\,\alpha\,\Delta T$ (C) $4\pi R^3\,\alpha\,\Delta T/3$ (D) $4\pi R^3\,\alpha\,\Delta T$

Ans. (D)

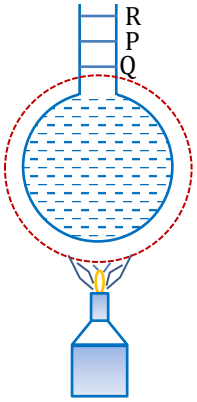
Solution:

$$\frac{\Delta V}{V}=3\alpha\Delta T$$

$$\Delta V=\left(\frac{4}{3}\pi R^3\right)(3\alpha\Delta T)=4\pi R^3\alpha\Delta T$$

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Thermal Expansion in Liquids:



- Liquids do not have linear and superficial expansion but these only have volume expansion.
- Since liquids are always to be heated along with a vessel which contains them so initially on heating the system (liquid + vessel), the level of liquid in vessel falls (as vessel expands more since it absorbs heat and liquid expands less) but later on, it starts rising due to faster expansion of the liquid.

  $PQ \rightarrow$ Represents expansion of vessel

  $QR \rightarrow$ Represents the real expansion of liquid.
- The actual increase in the volume of the liquid = The apparent increase in the volume of liquid + the increase in the volume of the vessel.

Co–Efficient of Real Expansion (γ_r):

- It is due to the actual increase in volume of liquid due to heating.

$$\gamma_r = \frac{\text{Real increase in volume}}{\text{Initial volume} \times \Delta\theta} = \frac{(\Delta V)}{V \times \Delta\theta}$$

- Also, coefficient of expansion of flask $\gamma_{\text{Vessel}} = \frac{(\Delta V)_{\text{Vessel}}}{V \times \Delta\theta}$
- $\gamma_{\text{Real}} = \gamma_{\text{Apparent}} + \gamma_{\text{Vessel}}$
- Change (apparent change) in volume in liquid relative to vessel is

$$\Delta V_{\text{app}} = V(\gamma_{\text{Real}} - \gamma_{\text{Vessel}})\Delta\theta = V(\gamma_r - 3\alpha)\Delta\theta$$

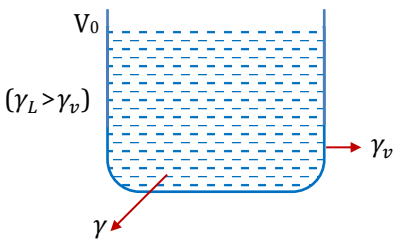
  α = Coefficient of linear expansion of the vessel.
- **Different level of liquid in vessel:**

γ	ΔV	Level
$\gamma_{Real} > \gamma_{Vessel}(= 3\alpha) \Rightarrow \gamma_{app} > 0$	ΔV_{app} is positive	Level of liquid in Vessle will rise on heating
$\gamma_{Real} < \gamma_{Vessel}(= 3\alpha) \Rightarrow \gamma_{app} > 0$	ΔV_{app} is negative	Level of liquid in Vessle will fall on heating
$\gamma_{Real} = \gamma_{Vessel}(= 3\alpha) \Rightarrow \gamma_{app} = 0$	$\Delta V_{app} = 0$	Level of liquid in Vessle will remain same

Illustration 37

On increasing temperature by ΔT. How much volume of liquid will overflow ?

(If container is initially filled up to brim.)



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Solution:

$$V_{\text{over flow}} = \Delta V_{\text{liq.}} - \Delta V_{\text{vessel}}$$

$$= V_0\gamma_L\Delta T - V_0\gamma_V\Delta T$$

$$= V_0(\gamma_L - \gamma_B)\Delta T$$

Illustration 38

The volume of a glass vessel filled with mercury is 500 *cc*, at 25°*C*. What volume of mercury will overflow at 45°*C*? the coefficients of volume expansion of mercury and glass are 1.8×10^{-4} / °*C* and 9.0×10^{-6} / °*C* respectively.

Solution:

The volume of mercury overflowing will be the expansion of mercury relative to the glass vessel (i.e., apparent expansion).

Now, since, $(\Delta V)_a = (\Delta V)_r - (\Delta V)_c$

Apparent expansion $(\Delta V)_a$ will be

$$(\Delta V)_a = V_1\gamma_1\Delta T - V_C\gamma_C\Delta T$$

$$= 500\ cc\ (180 - 9) \times \frac{10^{-6}}{°C}\ (45 - 25)°C = 1.71\ cc$$

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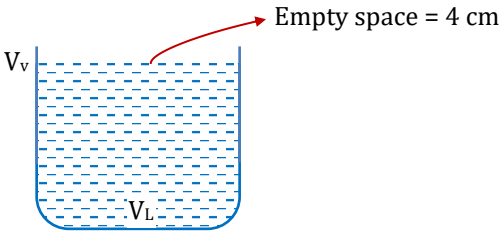
Thus, 1.71 *cc* of mercury overflows.

Illustration 39

Find V_0 of vessel and V of liquid if even on increasing T empty space remains constant ?

$\gamma_V = 2 \times 10^{-6}/°C$

$\gamma_L = 3 \times 10^{-6}/°C$



Solution:

$$\Delta V_{\text{liq}} = V_L \times \gamma_L \times \Delta T \qquad \Delta V_{\text{vessel}} = V_V \times \gamma_V \times \Delta T$$

$$= V_L \times 3 \times 10^{-6} \times \Delta T \qquad = V_V \times 2 \times 10^{-6} \times \Delta T$$

$$\Delta V_{\text{vessel}} = \Delta V_{\text{liq}}$$

$$\frac{V_V}{V_L} = \frac{3}{2}$$

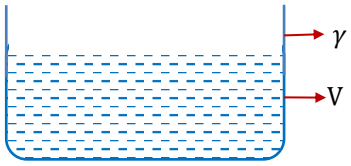
Initially $V_V - V_L = 4 \Rightarrow \frac{3}{2}V_L - V_L = 4$

$\frac{V_L}{2} = 4 \Rightarrow V_L = 8 \qquad \therefore V_V = 12$

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Illustration 40

If coefficient of volume expansion is same for liquid and vessel then,



(a) On increasing T empty space increases, decreases or remains same ?

(b) If vessel expanded by x%, empty space will expand by $>x$%, x%, $<x$%

Solution:

(a) $\Delta V_V = V_V \times \gamma \times \Delta T$

$\Delta V_L = V_L \times \gamma \times \Delta T$

$V_V > V_L$

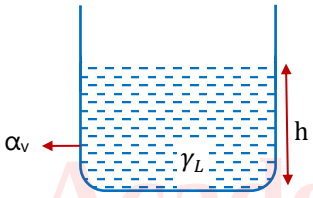
$\therefore \quad \Delta V_V > \Delta V_L$

(b) x%

In isotropic expansion everything will increased.

Illustration 41

Find Δh on increasing by ΔT ?



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Solution:

$$V_{\text{Liq.}} = hA_{\text{vessel}}$$

$$\frac{\Delta V_L}{V_L} = \frac{\Delta h}{h} + \frac{\Delta A_V}{A_V}$$

$$\gamma_L \Delta T = \frac{\Delta h}{h} + 2\alpha\Delta T$$

$$\boxed{\Delta h = h(\gamma_L - 2\alpha)\Delta T}$$

Variation of Density with Temperature:

As we known that mass = Volume × Density.

Mass of substance does not change with change in temperature so with increase of temperature, volume increases so density decreases and vice-versa.

$$d = \frac{d_0}{(1+\gamma\Delta T)}$$

For solids values of γ are generally small so we can write $d = d_0(1-\gamma\Delta T)$ (using binomial expansion).

Note:

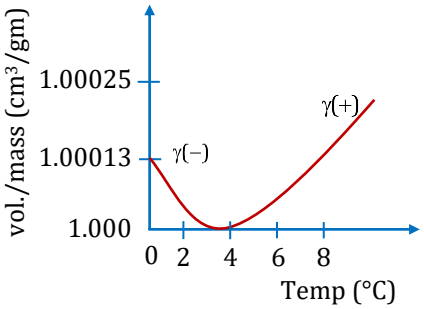
(i) γ for liquids are in order of 10^{-3}.

(ii) Anamolous expansion of water:

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For water density increases from 0 °C to 4 °C so γ is negative and for 4 °C to higher temperature γ is positive. At 4 °C density is maximum.

This anomalous behaviour of water causes ice to form first at the surface of a lake in cold weather. As winter approaches, the water temperature decreases initially at the surface. The water there sinks because of its increase density. Consequently, the surface reaches 0°C first and the lake becomes covered with ice. Due to ice at top surface further freezing is slowed. Ice is bad conductor of heat. Aquatic life is able to survive the cold winter as the lake bottom remains unfrozen at a temperature of about 4°C.



Variation of Force of Buoyancy with Temperature:

If body is submerged completely inside the liquid.

For solid, Buoyancy force,

$F_B = V_0\, d_L\, g$

V_0 = Volume of the solid inside liquid,

d_L = Density of liquid

Volume of body after increase its temperature $V = V_0\left[1+\gamma_S\Delta\theta\right]$,

Density of body after increase its temperature $d_L^{'} = \dfrac{d_L}{\left[1+\gamma_L\Delta\theta\right]}$.

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Buoyancy force of body after increase its temperature, $F_B^{'} = Vd_L^{'}g$, $\dfrac{F_B^{'}}{F_B} = \dfrac{\left[1+\gamma_S\Delta\theta\right]}{\left[1+\gamma_L\Delta\theta\right]}$

$F_B^{'} \approx F_B\left[1+\left(\gamma_S-\gamma_L\right)\Delta\theta\right]$

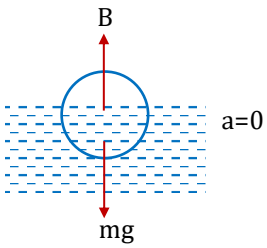
If $\gamma_S < \gamma_1$ $F_B^{'} < F_B$

(Buoyant force decreases) or apparent weight of body in liquid gets increased $\left[W-F_B^{'} > W-F_B\right]$.

Illustration 42

A body is floating over liquid if we increases temperature then what changes occur in Buoyancy force.

(Assume body is always in floating condition)



Solution:

Body is in equilibrium

So, $mg = B$

and gravitational force does not change with change in temperature. So, Buoyancy force remains constant.

By increasing temperature density of liquid decreases so volume of body inside the liquid increases to kept the Buoyance force constant for equal to gravitational force)

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Illustration 43

In previous question discuss the case when body move downward, upwards and remains at same position when we increase temperature.

Solution:

Let, f = Fraction of volume of body submerged in liquid

$$f = \frac{\text{Volume of body submerged in liquid}}{\text{Total volume of body}}$$

$f_1 = \frac{v_1}{v_0}$ at $\theta_1°C$

$f_2 = \frac{v_2}{v_0(1+3\alpha_s\Delta\theta)}$ at $\theta_2°C$

For equilibrium $mg = B = v_1d_1g = v_2d_2g$.

So, $v_2 = \frac{v_1d_1}{d_2}$

$\because$ $d_2 = \frac{d_1}{1+\gamma_L\Delta\theta} = v_1(1+\gamma_L\Delta\theta)$

$\therefore$ $f_2 = \frac{v_1(1+\gamma_L\Delta\theta)}{v_0(1+3\alpha_s\Delta\theta)}$

where $\Delta\theta = \theta_2 - \theta_1$

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Case I: Body move downward if $f_2 > f_1$

Means $\gamma_L > 3\alpha_S$

Case II: Body move upwards if $f_2 < f_1$

Means $\gamma_L < 3\alpha_S$

Case III: Body remains at same position

If $f_2 = f_1$

Means $\gamma_L = 3\alpha_S$

Measurement of Temperature:

The branch of thermodynamics which deals with the measurement of temperature is called thermometry. A thermometer is a device used to measure the temperature of a body. The substances like liquids and gases which are used in the thermometer are called thermometric substances.

Different Scales of Temperature:

A thermometer can be graduated into following scales:

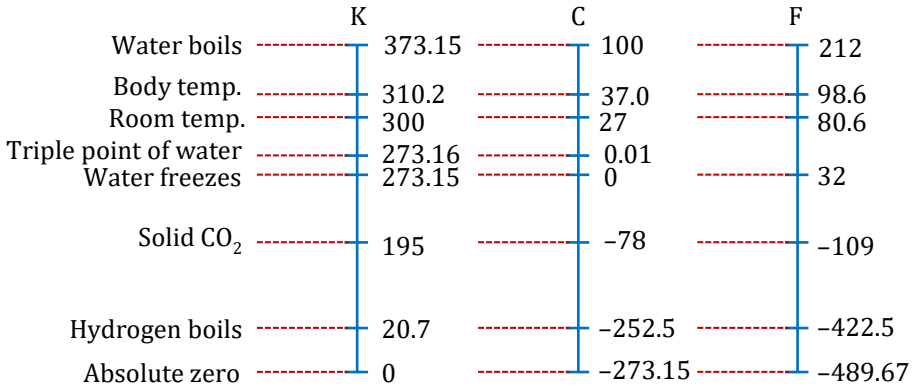
(a) The Centigrade or Celsius scale ($°C$)

(b) The Fahrenheit scale ($°F$)

(c) Kelvin scale of temperature (K)

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Comparison between Different Temperature Scales:



The formula for the conversion between different temperature scales is:

$$\frac{K-273}{100} = \frac{C}{100} = \frac{F-32}{180}$$

General formula for the conversion of temperature from one scale to another:

$$\frac{\text{Temp on one scale}(S_1)\text{-Lower fixed point}(S_1)}{\text{Upper fixed point}(S_2)\text{-Lower fixed point}(S_1)} = \frac{\text{Temp. on other scale}(S_2)\text{-Lower fixed point}(S_2)}{\text{Upper fixed point}(S_2)\text{-Lower fixed point}(S_2)}$$

Thermometers:

Thermometers are device that are used to measure temperatures. All thermometers are based on the principle that some physical property of a system changes as the system temperature changes.

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Required properties of good thermometric substance.

(1) Non-sticky (absence of adhesive force)

(2) Low melting point (in comparison with room temperature)

(3) High boiling temperature

(4) Coefficient of volumetric expansion should be high (to increase accuracy in measurement).

(5) Heat capacity should be low.

(6) Conductivity should be high

Mercury (Hg) suitably exhibits above properties.

Illustration 44

The readings of a thermometer at $0°C$ and $100°C$ are 50 cm and 75 cm of mercury column respectively. Find the temperature at which its reading is 80 cm of mercury column?

Solution:

Let reference temperature 0°C,

$75 = 50(1 + \alpha \times 100)$ ...(i)

$80 = 50(1 + \alpha \times T)$ ...(ii)

From equation (i) and (ii),

$$\frac{80-50}{75-50} = \frac{T-0}{100-0} \Rightarrow T = 120°C$$

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HEAT TRANSFER

Heat is a form of energy which transfers from a body at higher temperature to a body at lower temperature. The transfer of heat from one body to another may take place by any of the following modes:

- **Conduction:**
  The process in which the material takes an active part by molecular action and energy is passed from one particle to another is called conduction. It is predominant in solids.
- **Convection:**
  The transfer of energy by actual motion of particle of **medium** from one place to another is called convection. It is predominant is fluids (liquids and gases).
- **Radiation:**
  Quickest way of transmission of heat is known as radiation. In this mode of energy transmission, heat is transferred from one place to another without heating the inter–vening medium.

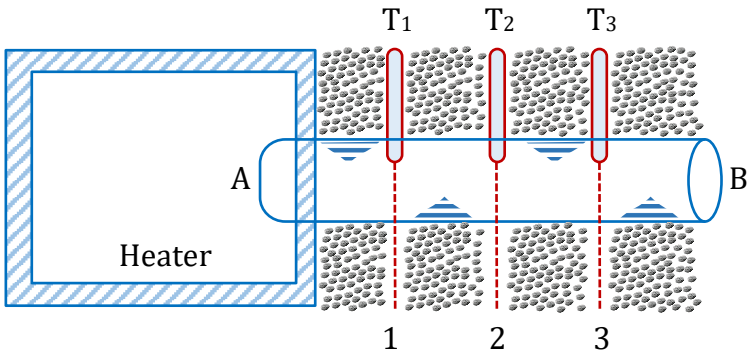
Conduction	Convection	Radiation
Due to free electron or vibration motion of molecules	Actual motion of particles	Electromagnetic radiation
Heat transfer in solid body (in mercury also)	Heat transfer in fluids (Liquid + gas)	All
Slow process	Slow process	Fast process (3×10^8 m/sec)

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	Medium
(1) Conduction	Required
(2) Convection	Required
(3) Radiation	No required

Steady and Variable State:

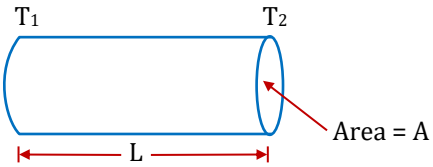
consider a metal rod AB, with one end A inserted into a chamber containing a heater with other end B left free and exposed to the surrounding as shown fig. The rod is thermally insulated sideways with some bad conductor of heat say cotton or felt fabric. Three thermometers are installed in the rod at three distinct sections numbered (1), (2) and (3). Initially, the entire system is at the room temperature and the three thermometers show the same room temperature. The heater is then switched on. The end A first gets heated up and simultaneously heat is conducted to the adjacent sections towards end B.



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Due to heat absorption at each section. The corresponding temperatures start rising with $T_1>T_2>T_3$. Such a state, encountered initially, is known as a variable state. It is in this state, the heat coming through end A, is continuously absorbed at each section with a temperature rise as time elapses.

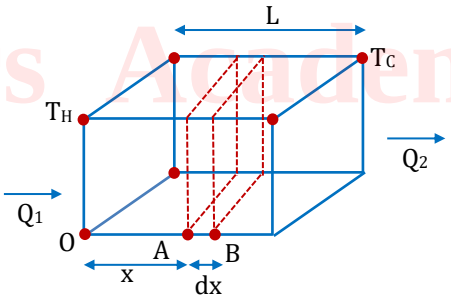
After some time when the temperature of end B becomes equal to that of surrounding and thus becomes constant. Consequently, the heat absorption at different sections of the rod ceases. So, there is no more rise in temperatures of any sections. Such a state is known as a steady state. It is in the steady state that temperatures of each section attain steady (constant) value with time.



Conduction:

The process of transmission of heat energy in which heat is transferred from one particle of the medium to the other, but each particle of the medium stays at its own position is called conduction, for example if you hold an iron rod with one of its end on a fire for some time, the handle will get hot. The heat is transferred from the fire to the handle by conduction along the length of iron rod. The vibrational amplitude of atoms and electrons of the iron rod at the hot end takes on relatively higher values due to the higher temperature of their environment. These increased vibrational amplitude are transferred along the rod, from atom to atom during collision between adjacent atoms. In this way a region of rising temperature extends itself along the rod to your hand.

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Consider a slab of face area A, Lateral thickness L, whose faces have temperatures T_H and T_C ($T_H > T_C$).

Now consider two cross sections in the slab at positions A and B separated by a lateral distance of dx. Let temperature of face A be T and that of face B be $T + dT$. Then experiments show that Q, the amount of heat crossing the area A of the slab at position x in time t is given by

$$\boxed{\frac{Q}{t} = -KA\frac{dT}{dx}} \qquad ...(1)$$

Here K is a constant depending on the material of the slab and is named thermal conductivity of the material, and the quantity $\left(\frac{dT}{dx}\right)$ is called temperature gradient. The (–) sign in equation (1) shows heat flows from high to low temperature (dT is a $-ve$ quantity).

- SI Unit : $J\,s^{-1}\,m^{-1}\,K^{-1}$ Dimensions : $M^1\,L^1\,T^{-3}\,\theta^{-1}$
- For an ideal or perfect conductor of heat the value of $K = \infty$
- For an ideal or perfect bad conductor or insulator the value of $K = 0$
- For cooking the food, low specific heat and high conductivity utensils are most suitable.

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Steady State:

If the temperature of a cross-section at any position x in the above slab remains constant with time (remember, it does vary with position x), the slab is said to be in steady state.

Remember steady-state is distinct from thermal equilibrium for which temperature at any position (x) in the slab must be same.

For a conductor in steady state there is no absorption or emission of heat at any cross-section. (As temperature at each point remains constant with time). The left and right face are maintained at constant temperatures T_H and T_C respectively, and all other faces must be covered with adiabatic walls so that no heat escapes through them and same amount of heat flows through each cross-section in a given Interval of time. Hence $Q_1 = Q = Q_2$. Consequently, the temperature gradient is constant throughout the slab.

Hence, $\frac{dT}{dx} = \frac{\Delta T}{L} = \frac{T_f - T_i}{L} = \frac{T_C - T_H}{L}$

and $\frac{Q}{t} = -KA\ \frac{\Delta T}{L}$

$\Rightarrow \quad \frac{Q}{t} = KA\left(\frac{T_H - T_C}{L}\right)$ ...(1)

Here Q is the amount of heat flowing through a cross-section of slab at any position in a time interval of t.

Illustration 45

One face of an aluminium cube of edge 2 metre is maintained at 100°C and the other end is maintained at 0°C. All other surfaces are covered by adiabatic walls. Find the amount of heat flowing through the cube in 5 seconds. (Thermal conductivity of aluminium is 209 W/m–°C)

Solution:

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Heat will flow from the end at 100°C to the end at 0°C. Area of cross-section perpendicular to direction of heat flow,

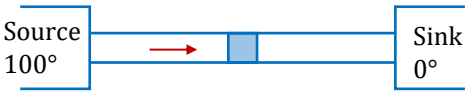
$A = 4m^2$

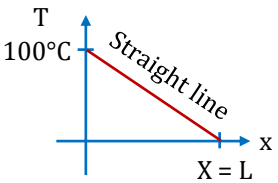
then $\frac{Q}{t} = KA\frac{(T_H - T_C)}{L}$

$Q = \frac{(209W/m°C)(4m^2)(100°C - 0°C)(5\text{sec})}{2m} = 209\ KJ$

Heat Flow from a Uniform Rod at Steady State:

In steady state temperature of each element of the rod becomes constant w.r.t. to time i.e. rate of heat flow at energy cross-section of the rod will be same.





In steady state $\frac{dQ}{dt}$ is same for each element

$\frac{dQ}{dt} = KA\frac{dT}{dx}$

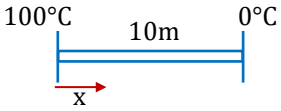
$\therefore$ for uniform rod (K and A same for each element)

$\Rightarrow \quad \frac{dT}{dx}$ is same $\Rightarrow \quad \frac{dT}{dx} = \frac{\Delta T}{\Delta x} \quad \Rightarrow \quad \frac{dQ}{dt} = kA\left(\frac{\Delta T}{\Delta x}\right)$

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Illustration 46

$k = 0.2\frac{W}{mK}$, $A = 2cm^2$ in steady state.



(a) Find x where $T = 60°$

(b) Find $\left(\frac{dQ}{dt}\right)$

Solution:

(a) Steady state

$\therefore$ Change same, $kA\left(\frac{100-60}{x}\right) = kA\left(\frac{100-0}{10}\right)$

$\Rightarrow$ $400 = x \times 100$ $\Rightarrow$ $x = 4m$

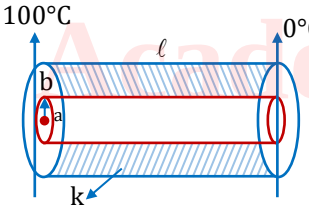
(b) $\frac{dQ}{dt} = kA\frac{\Delta T}{\Delta x} = \frac{0.2\times 2\times 10^{-4}(100-0)}{10}$

$\Rightarrow$ 4×10^{-4} watt

Illustration 47

For a thick cylinder find $\frac{dQ}{dt}$ = ?

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Solution:

$$\frac{dQ}{dt} = k(\pi b^2 - \pi a^2)\frac{(100-0)}{\ell}$$

Illustration 48

One end of a brass rod $2m$ long and having 1 cm radius is maintained at 250°C. When a steady state is reached, the rate of heat flow across any cross–section is 0.5 $cal\ s^{-1}$.

What is the temperature of the other end $K = 0.26\ cal\ s^{-1}\ cm^{-1}\ °C^{-1}$?

Solution:

$\frac{Q}{t}$ = 0.5 $cal\ s^{-1}$; $r = 1\ cm$

$\therefore$ Area $A = \pi r^2 = 3.142 \times 1\ cm^2 = 3.142\ cm^2$

L = Length of rod = $2m = 200\ cm$, $T_1 = 250°C$, $T_2 = ?$

We know $\frac{Q}{t} = \frac{KA(T_1 - T_2)}{L}$ or $(T_1 - T_2) = \frac{Q}{t} \times \frac{L}{kA} = \frac{0.5\times 200}{0.26C^{-1}\times 3.142} = 122.4°C$

$\therefore$ $T_2 = 250°C - 122.4°C = 127.6°C$

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Illustration 49

Steam at $373K$ is passed through a tube of radius 10 cm and length $2m$. The thickness of the tube is $5mm$ and thermal conductivity of the material is 390 $W\,m^{-1}\,K^{-1}$, calculate the heat lost per second. The outside temp. is $0°C$.

Solution:

Using the relation $Q = \dfrac{KA(T_1 - T_2)t}{L}$

Here, heat is lost through the cylindrical surface of the tube.

$A = 2\pi$ (Radius of the tube) (Length of the tube) $= 2\pi \times 0.1 \times 2 = 0.4\ \pi m^2$

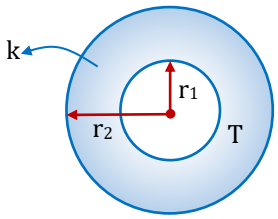
$K = 390\ W\,m^{-1}\,K^{-1}$

$T_1 = 373\ K, \quad T_2 = 0°C = 273\ K, \quad L = 5\ mm = 0.005\ m \quad$ and $\quad t = 1\ s$

$$\therefore \quad Q = \frac{390\times 0.4\pi\times(373-273)\times 1}{0.005} = \frac{390\times 0.4\pi\times 100}{0.005} = 98\times 10^5\ J.$$

Illustration 50

Thick spherical shell. Find rate of heat flow ?



Solution:

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$$\frac{dQ}{dt} = P = -K\frac{4\pi x^2}{dx}dT$$

$$\int_{r_1}^{r_2}\frac{dx}{x^2} = -\frac{K4\pi}{P}\int_{T_1}^{T_2}dT$$

$$\Rightarrow \quad \left[\frac{1}{r_1}-\frac{1}{r_2}\right] = -\frac{K4\pi}{P}(T_1 - T_2)$$

$$\boxed{\therefore \quad P = K4\pi\frac{r_1 r_2}{r_2 - r_1}(T_1 - T_2)}$$

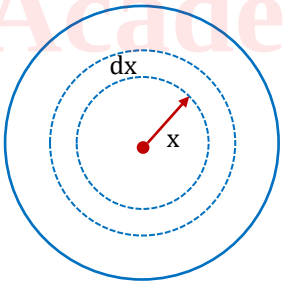
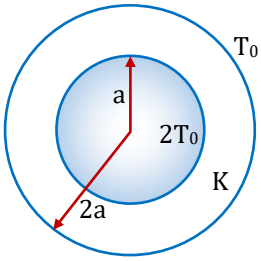


Illustration 51

A hollow spherical ball of inner radius a and outer radius $2a$ is made of a uniform material of constant thermal conductivity K. The temperature within the ball is maintained at $2T_0$ and outside the ball it is T_0. Find:

(a) the rate at which heat flows out of the ball in the steady state,

(b) the temperature at $r = 3a/2$, where r is radial distance from the centre of shell. Assume steady state condition.



Solution:

In the steady state, the net outward thermal current is constant, and does not depend on the radial position.

Thermal current, $C_1 = \left(\dfrac{dQ}{dt}\right) = -K.(4\pi r^2)\dfrac{dT}{dr} \Rightarrow \dfrac{dT}{dr} = -\dfrac{C_1}{4\pi K}\dfrac{1}{r^2} + C_2$

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At $r = a$, $T = 2T_0$ and $r = 2a$, $T = T_0 \Rightarrow T = \frac{2a}{r}T_0$

(a) $\frac{dQ}{dt} = 8\pi aKT_0$

(b) $T(r = 3a/2) = 4T_0/3$

Thermal Resistance to Conduction:

If you are interested in insulating your house from cold weather or for that matter keeping the meal hot in your tiffin-box, you are more interested in poor heat conductors, rather than good conductors. For this reason, the concept of thermal resistance R has been introduced.

For a slab of cross-section A, Lateral thickness L and thermal conductivity K,

$$R = \frac{L}{KA} \qquad ...(1)$$

In terms of R, the amount of heat flowing though a slab in steady-state (in time t)

$$\frac{Q}{t} = \frac{(T_H - T_C)}{R}$$

If we name $\frac{Q}{t}$ as thermal current i_T

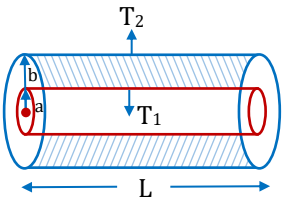
then, $i_T = \frac{T_H - T_C}{R}$ ...(2)

This is mathematically equivalent to OHM's law, with temperature playing the role of electric potential. Hence results derived from OHM's law are also valid for thermal conduction.

More over, for a slab in steady state we have seen earlier that the thermal current i_L remains same at each cross-section. This is analogous to Kirchhoff's current law in electricity, which can now be very conveniently applied to thermal conduction.

Illustration 52

Find $\frac{dQ}{dt}$?



Solution:

Method-1: $R_{eq.} = \int dR = \int_a^b \frac{dx}{k2\pi xL} = \frac{\ell n}{2\pi kL}\left(\frac{b}{a}\right)$

$$\frac{dQ}{dt} = \frac{(T_1 - T_2)2\pi kL}{\ell n\left(\frac{b}{a}\right)}$$

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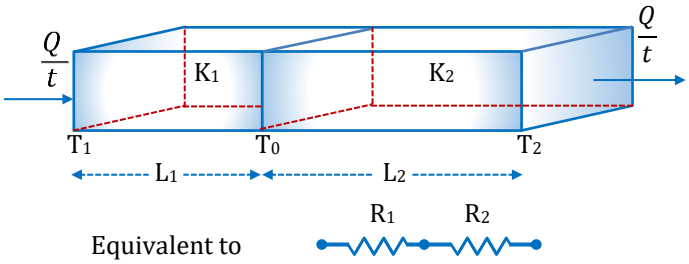
Method-2: $\frac{dQ}{dt} = P = -k\frac{2\pi kL}{dx}dT$

Solve and get $\frac{d\theta}{dt}$

Heat Flow Through Slabs in Series:

$R_{eq} = R_1 + R_2$

$\frac{L_1 + L_2}{K_{eq}A} = \frac{L_1}{K_1A} + \frac{L_2}{K_2A}$



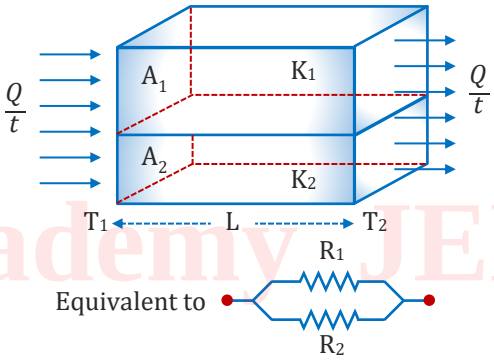
Equivalent thermal conductivity of the system is

$K_{eq} = \frac{L_1 + L_2}{\frac{L_1}{K_1} + \frac{L_2}{K_2}} = \frac{\Sigma L_i}{\Sigma \frac{L_i}{K_i}}$

Heat Flow Through Slabs in Parallel:

$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}, R = \frac{L}{KA}$

$\frac{K_{eq}}{L}(A_1 + A_2) = \frac{K_1A_1}{L} + \frac{K_2A_2}{L}$



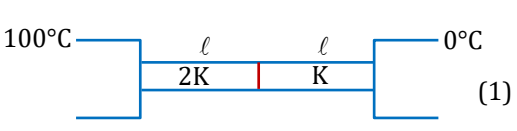
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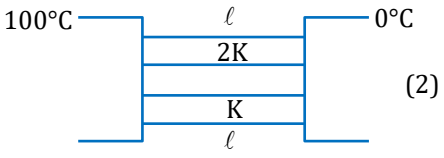
Equivalent thermal conductivity

$K_{eq} = \frac{K_1A_1 + K_2A_2}{A_1 + A_2} = \frac{\Sigma K_iA_i}{\Sigma A_i}$

Illustration 53

Ratio of rate of ice melting in $\frac{(1)}{(2)}$?





Solution:

$(R_{eq})_1 = R_1 + R_2 \Rightarrow \frac{\ell}{2KA} + \frac{\ell}{KA} \Rightarrow \frac{3\ell}{2KA}$

$\frac{1}{(R_{eq})_2} = \frac{1}{R_1} + \frac{1}{R_2}$

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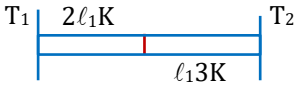
$$\Rightarrow \quad \frac{1}{(R_{eq})_2} = \frac{2KA}{\ell} + \frac{KA}{\ell}$$

$$\Rightarrow \quad \frac{3KA}{\ell}$$

$$\Rightarrow \quad \frac{dQ}{dt} = \frac{(T_1 - T_2)}{R_{eq.}}$$

Illustration 54

Find K_{eff} of the setup ?



Solution:

$R_{eq.} = R_1 + R_2$

$$\frac{3\ell_1}{K_{eff}A} = \frac{2\ell_1}{KA} + \frac{\ell_1}{3KA}$$

Get $K_{eff.}$

Illustration 55

Calculate the quantity of heat conducted through $2m^2$ of a brick wall $12cm$ thick in 1 hour if the temperature on one side is $8°C$ and on the other side is $28°C$.
(Thermal conductivity of brick = $0.13\ W\,m^{-1}\,K^{-1}$)

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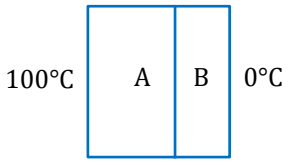
Temperature gradient = $\frac{28-8}{12\times10^{-2}}\ K\,m^{-1}$ and $t = 3600\ s$

$\therefore \quad Q = kAt \times$ Temperature gradient $= 0.13 \times 2 \times 3600 \times \frac{28-8}{12\times10^{-2}} = 156000\ J$

Illustration 56

Two parallel plates A and B are joined together to form a compound plate (figure). The thicknesses of the plates are 4.0 cm and 2.5 cm respectively and the area of cross-section is 100 cm^2 for each plate. The thermal conductivities are $K_A = 200\ W/m\text{-}°C$ for the plate A and $K_B = 400\ W/m\text{-}°C$ for the plate B. The outer surface of the plate A is maintained at $100°C$ and the outer surface of the plate B is maintained at $0°C$. Find:

(a) the rate of heat flow through any cross-section,
(b) the temperature at the interface and
(c) the equivalent thermal conductivity of the compound plate.



Solution:

(a) Let the temperature of the interface be θ.
The area of cross-section of each pate is $A = 100\ cm^2 = 0.01m^2$.
The thicknesses are $x_A = 0.04\ m$ and $x_B = 0.025\ m$
The thermal resistance of the plate A is $R_1 = \frac{1}{K_A}\frac{x_A}{A}$

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and that of the plate B is $R_2 = \frac{1}{K_B}\frac{x_B}{A}$

The equivalent thermal resistance is

$$R = R_1 + R_2 = \frac{1}{A}\left(\frac{x_A}{K_A} + \frac{x_B}{K_B}\right) \quad \text{...(i)}$$

Thus, $\frac{\Delta Q}{\Delta t} = \frac{\theta_1 - \theta_2}{R} = \frac{A(\theta_1 - \theta_2)}{x_A/K_A + x_B/K_B}$

$$= \frac{(0.01 m^2)(100°C)}{(0.04m)/(200 W/m-°C) + (0.025m)/(400 W/m-°C)} = 3810\ W.$$

(b) We have $\frac{\Delta Q}{\Delta t} = \frac{A(\theta - \theta_1)}{x_B/K_B}$

or, $3810\ W = \frac{(0.01 m^2)(\theta - °C)}{(0.025m)/(400 W/w-°C)}$

or, $\theta = 25°C$

(c) If K is the equivalent thermal conductivity of the compound plate, its thermal resistance is

$$R = \frac{1}{A}\frac{x_A + x_B}{K}$$

Comparing with (i),

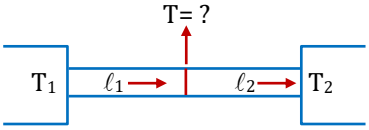
$$\frac{x_A + x_B}{K} = \frac{x_A}{K_A} + \frac{x_B}{K_B}$$

or, $K = \frac{x_A + x_B}{x_A/K_A + x_B/K_B} = 248\ W/m-°C$

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Junction:

(eg.) $\frac{T_1 - T}{R_1} = \frac{T - T_2}{R_2} \Rightarrow \frac{T - T_1}{R_1} + \frac{T - T_1}{R_2} = 0$

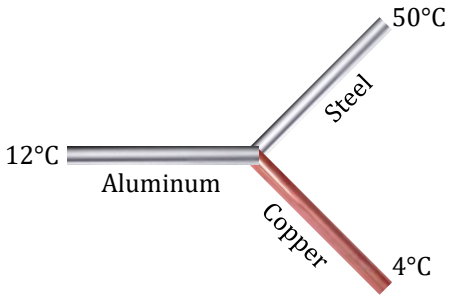


If there is no source or sink at junction then sum of outgoing $\frac{dQ}{dt} = 0$

Illustration 57

Three identical rods of length $1m$ each, having cross-section area of $1cm^2$ each and made of Aluminium, copper and steel respectively are maintained at temperatures of $12°C$, $4°C$ and $50°C$ respectively at their separate ends. Find the temperature of their common junction.

[$K_{Cu} = 400\ W/m\text{-}K$, $K_{Al} = 200\ W/m\text{-}K$, $K_{steel} = 50\ W/m\text{-}K$]

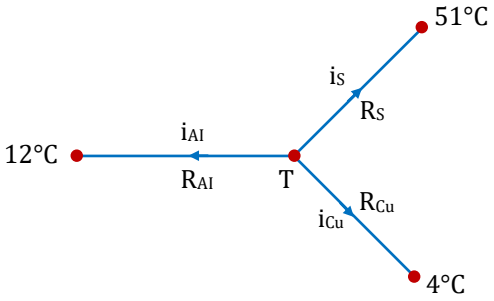


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Solution:

$$R_{Al} = \frac{L}{KA} = \frac{1}{200 \times 10^{-4}} = \frac{10^4}{200}$$

Similarly, $R_{steel} = \frac{10^4}{50}$ and $R_{copper} = \frac{10^4}{400}$



Let temperature of common junction = T

Then from Kirchhoff's current laws, $i_{Al} + i_{steel} + i_{Cu} = 0$

$$\Rightarrow \quad \frac{T-12}{R_{Al}} + \frac{T-51}{R_{steel}} + \frac{T-4}{R_{Cu}} = 0$$

$$\Rightarrow \quad (T-12)\,200 + (T-50)\,50 + (T-4)\,400 = 0$$

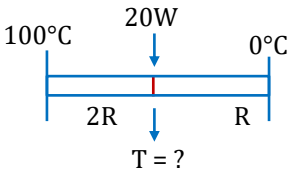
$$\Rightarrow \quad 4(T-12) + (T-50) + 8\,(T-4) = 0$$

$$\Rightarrow \quad 13T = 48 + 50 + 32 = 130$$

$$\Rightarrow \quad T = 10°C$$

Illustration 58

Find T.



Solution:

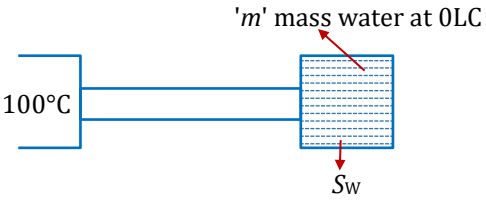
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$$\frac{T-100}{2R} + \frac{T-0}{R} = 20 \Rightarrow \frac{T-100+2T}{2R} = 20 \Rightarrow 3T = 100 + 40R$$

$$T = \frac{100+40R}{3}$$

Illustration 59

Find time after which temperature of water become 50°C from 0°C ?



Solution:

Here $\frac{dQ}{dt}$ is not constant it is variable.

Now let at a time t their be a temperature T.

$$\frac{dQ}{dt} = \frac{kA}{L}(100-T) = ms\frac{dT}{dt}$$

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$$\frac{kA}{Lms}\int_0^t dt = \int_0^{50}\frac{dT}{(100-T)}$$

$$\frac{kA}{Lms}t = -\ell n[100-50] + Ln[100-0]$$

$$t = \frac{Lms\ \ell n2}{kA}$$

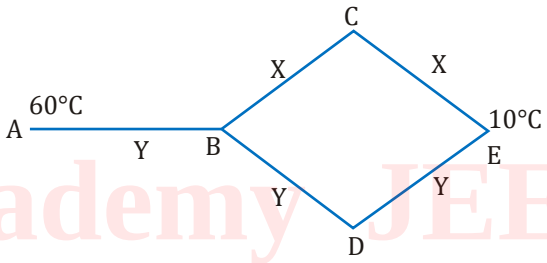
Illustration 60

Two rods of material X and three rods of material Y are connected as shown in figure. All the rods are identical in length and cross–sectional area. If the end A is maintained at $60°C$ and the junction E at $10°C$, calculate the temp. of the junctions B, C, D. The thermal conductivity of X is 0.92 CGS units and that of Y is 0.46 CGS units.

Solution:

$$R_X \propto \frac{1}{K_X},\ R_Y \propto \frac{1}{K_Y}$$

$$\Rightarrow \quad \frac{R_X}{R_Y} = \frac{K_Y}{K_X} = \frac{0.46}{0.92} = \frac{1}{2}$$



Let, $R_X = R \quad \therefore \ R_Y = 2R$

The total resistance $\Sigma R = R_Y +$ effective resistance in the bridge

$$\Sigma R = 2R + \frac{2R\times 4R}{2R+4R} = 2R + \frac{4}{3}R = \frac{10}{3}R$$

and $\quad \Delta\theta = I\times R$

Further, $I_{BCE}\,(2R) = I_{BDE}(4R)$ and $I_{BCE} + I_{BDE} = I$; $I_{BCE} = \frac{2}{3}I$ and $I_{BDE} = \frac{1}{3}I$

For A and B $\quad \theta_A - \theta_B = 60^0 - \theta_B \Rightarrow 60 - \theta_B = 2R\times I$ ...(i)

For B and C $\quad \theta_B - \theta_C = \frac{2}{3}(I\times R)$ ...(ii)

$$\theta_C - \theta_E = \frac{2}{3}\times R\times I$$

For A and E $\quad \theta_A - \theta_E = 60-10 = 50 \Rightarrow \frac{10}{3}(R\times I) = 50$ ...(iii)

$$\therefore \ R\times I = 15$$

$$\therefore \ \theta_A - \theta_B - 2\times 15 = 30,\ \theta_B = 60-30 \ = 30°C,\ \theta_B - \theta_C = \left(\frac{2}{3}\right)\times 15 = 10$$

$$\therefore \ \theta_C = 30-10 = 20°C \text{ Obviously, } \theta_C = \theta_D = 20°C$$

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Thermal Radiation:

The process of the transfer of heat from one place to another place without heating the intervening medium is called radiation. When a body is heated and placed in vacuum, it loses heat even when there is no medium surrounding it. The heat can not go out from the body by the process of conduction or convection since both of these process require the presence of a material medium between source and surrounding objects. The process by which heat is lost in this case is called radiation. This does not require the presence of any material medium.

It is by radiation that the heat from the Sun reaches the Earth. Radiation has the following properties:

(a) Radiant energy travels in straight lines and when some object is placed in the path, it's shadow is formed at the direction.

(b) It can travel through vacuum.

(c) Intensity of radiation follows the law of inverse square.

(d) Thermal radiation can be polarised in the same way as light by transmission through a nicol prism.

All these and many other properties establish that heat radiation has nearly all the properties possessed by light and these are also electromagnetic waves with the only difference of wavelength or frequency. The wavelength of heat radiation is larger than that of visible light.

Radiation:

Radiation is the process in which energy is transferred by means of electromagnetic waves.

All bodies continuously radiate energy in the form of electromagnetic waves. Even an ice cube radiates energy, although so little of it is in the form of visible light that an ice cube cannot be seen in the dark.

The surface of an object plays a significant role in determining how much radiant energy the object will absorb or emit.

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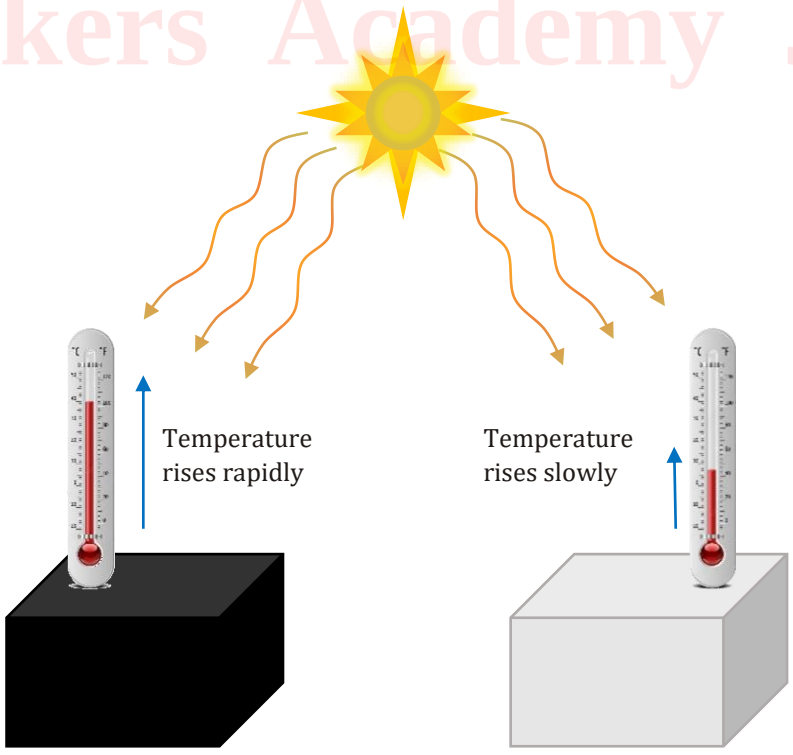


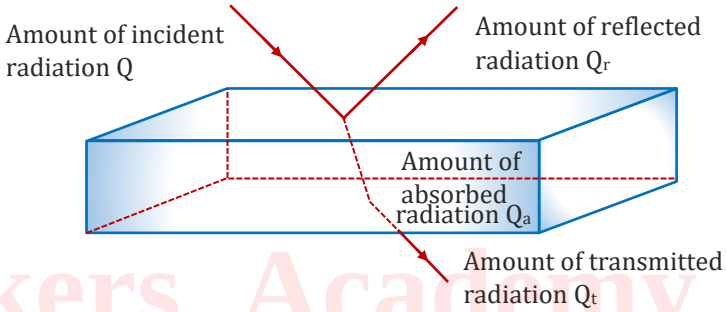
Figure: The temperature of the block coated with lampblack rises faster than the temperature of black coated absorbs radiant energy from the sun at the greater rate

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The two blocks in sunlight in figure, for example, are identical, except that one has a rough surface coated with lamblack (a fine black soot), while the other has a highly polished silver surface. As the thermometers indicate, the temperature of the black block rises at a much faster rate than that of the silver block. This is because lampblack absorbs about 97% of the incident radiant energy, while the silvery surface absorbs only about 10%. The remaining part of the incident energy is reflected in each case. We observe the lampblack as black in color because it reflects so little of the light falling on it, while the silvery surface looks like a mirror because it reflects so little of the light falling on it, while the silvery surface looks like a mirror because it reflects so much light. Since the color black is associated with nearly complete absorption of visible light, the term perfect blackbody or, simply, blackbody is used when referring to an object that absorbs all the electromagnetic waves falling on it.

Spectral, Emissive, Absorptive & Transmittive Power of a Given Body Surface:

Due to incident radiations on the surface of a body following phenomena occur by which the radiation is divided into three parts. (a) Reflection (b) Absorption (c) Transmission



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- **From Energy Conservation:**

$$Q = Q_r + Q_a + Q_t \Rightarrow \frac{Q_r}{Q} + \frac{Q_a}{Q} + \frac{Q_t}{Q} = 1 \quad \Rightarrow \quad r + a + t = 1$$

Reflection Coefficient $r = \frac{Q_r}{Q}$, Absorptive Coefficient $a = \frac{Q_a}{Q}$, Transmittive Coefficient $t = \frac{Q_t}{Q}$

$r = 1$ and $a = 0$, $t = 0$ $\Rightarrow$ Perfect reflector

$a = 1$ and $r = 0$, $t = 0$ $\Rightarrow$ Ideal absorber (ideal black body)

$t = 1$ and $a = 0$, $r = 0$ $\Rightarrow$ Perfect transmitter

Reflection power $(r) = \left[\frac{Q_r}{Q} \times 100\right]\%$, Absorption power $(a) = \left[\frac{Q_a}{Q} \times 100\right]\%$

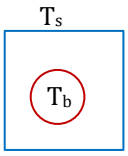
Transmission power $(t) = \left[\frac{Q_t}{Q} \times 100\right]\%$

Prevost's Theory of Heat Energy Exchange:

- All bodies absorb as well as emit radiation at all temperature.
- Rate of emission of radiation depends on nature and temperature of the surface of body.

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- Rate of absorbing radiation depends on nature of surface of the body and temperature of the surrounding.



$T_b > T_s$ then rate of emission > Rate of absorbing and $T_b \downarrow$

$T_b < T_s$ then rate of emission < Rate of absorbing and $T_b \uparrow$

$T_b = T_s$ then rate of emission = Rate of absorbing and T_b no change

- According to Prevost at every possible temperature (Not absolute temperature) there is a continuous heat energy exchange between a body and its surrounding and this exchange carry on for infinite time.
- The relation between temperature difference of body with its surrounding decides whether the body experience cooling effect or heating effect.

When a cold body is placed in the hot surrounding : The body radiates less energy and absorbs more energy from the surrounding, therefore the temperature of body increases.

When a hot body placed in cooler surrounding : The body radiates more energy and absorb less energy from the surroundings. Therefore, temperature of body decreases.

When the temperature of a body is equal to the temperature of the surrounding:

The energy radiated per unit time by the body is equal to the energy absorbed per unit time by the body, therefore its temperature remains constant.

- At absolute zero temperature (0 kelvin) all atoms of a given substance remains in ground state, so, at this temperature emission of radiation from any substance is impossible, so Prevost's heat energy exchange theory does not applied at this temperature, so it is called **limited temperature** of prevosts theory.
- With the help of Prevost's theory rate of cooling of any body w.r.t. its surroundings can be worked out (applied to Stefen Boltzman law, Newton's law of cooling.)
- **Absorptive Power or Absorptive Coefficient 'a' :** The ratio of amount of radiation absorbed by a surface (Q_a) to the amount of radiation incident (Q) upon it is defined as the coefficient of absorption $a = \frac{Q_a}{Q}$. It is unitless.

  For black body $a = 1$; $0 \le a \le 1$
- **Emissive Power (E):**

  The amount of heat radiation emitted by unit area of the surface in one second at a particular temperature or It is radiation intensity at a particular temperature.

  SI UNIT : J/m^2s

  $E = \frac{\Delta Q}{\Delta t A}$; unit is W/m^2

Spectral Emissive power (E_λ):

Emissive power per unit wavelength range at wavelength λ is known as spectral emissive power, E_λ. If E is the total emissive power and E_λ is spectral emissive power, they are related as follows,

$$E = \int_0^\infty E_\lambda \, d\lambda \text{ and } \frac{dE}{d\lambda} = E_\lambda$$

Emissivity: $e = \frac{\text{Emissive power of a body at temperature T}}{\text{Emissive power of a black body at same temperature T}} = \frac{E}{E_0}$.

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Kirchhoff's Law:

At a given temperature for all bodies the ratio of their spectral emissive power (e_λ) to spectral absorptive power (a_λ) is constant and this constant is equal to spectral emissive power (E_λ) of the ideal black body at same temperature

$$\frac{e_\lambda}{a_\lambda} = E_\lambda = \text{Constant} ; \left[\frac{e_\lambda}{a_\lambda}\right]_1 = \left[\frac{e_\lambda}{a_\lambda}\right]_2 = \text{Constant} \quad e_\lambda \propto a_\lambda$$

Good absorbers are good emitters and bad absorbers are bad emitters

- For a constant temperature the spectral emissive power of an ideal black body is a constant parameter
- The practical confirmation of Kirchhoff's law carried out by Rishi apparatus and the main base of this apparatus is a Lessilie container.

- The main conclusion predicted from Kirchhof's law can be expressed as
  Good absorber $\rightleftharpoons$ Good emitter
  Bad absorber $\rightleftharpoons$ Bad emitter

Kirchhoff's Law : Absorptivity of a surface is equal to its emissivity.
($e = a$) for a surface
"Good absorbers are good emitters."

Applications of Kirchhoff Law:

- **In Deserts Days are Hot and Nights Cold:**
  Sand is rough and black, so it is a good absorber and hence in deserts, days (When radiation from Sun is incident on sand) will be very hot. Now in accordance with Kirchhoff's Law, good absorber is a good emitter.
  So, nights (when sand emits radiation) will be cold.

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Stefan's Law:

The amount of radiation emitted per second per unit area by a black body is directly proportional to the fourth power of its absolute temperature.

Amount of radiation emitted $E \propto T^4$

where T = temperature of ideal black body (in K)

$E = \sigma T^4$

This law is true for only ideal black body

SI Unit : E = watt/m^2 σ = Stefen's constant = 5.67 $x10^{-8}$ watt /$m^2 K^4$

Dimensions of σ : $M^1 L^0 T^{-3} \theta^{-4}$

Total radiation energy emitted out by surface of area A in time t : If T is constant.

Ideal black body $Q_{IBB} = \sigma A T^4 t$ and for any other body $Q_{GB} = e_r \sigma A T^4 t$

$e \rightarrow$ emissivity of surface. It is property of surface

$\rightarrow 0 \le e \le 1$

$\rightarrow e = 1$ for black body

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Stefan's Radiation Law:

An idealized body that absorbs all the radiation incident upon it is called a blackbody. A blackbody absorbs not only all visible light, but infrared, ultraviolet, and all other wavelengths of electromagnetic radiation. It turns out that a good absorber is also a good emitter of radiation. A blackbody emits more radiant power per unit surface area than any real object at the same temperature. The rate at which a blackbody emits radiation per unit surface area is proportional to the fourth power of the absolute temperature.

$P = \sigma A T^4$

In equation, A is the surface area and T is the surface temperature of the blackbody in kelvins. Since Stefan's law involves the absolute temperature and not a temperature difference, $^\circ C$ cannot be substituted. The universal constant σ (Greek letter sigma) is called Stefan's constant:

$\sigma = 5.670 \times 10^{-8}\ W/(m^2.K^4)$

The fourth-power temperature dependence implies that the power emitted is extremely sensitive to temperature changes. If the absolute temperature of a body doubles, the energy emitted increases by a factor of $2^4 = 16$.

Since real bodies are not perfect absorbers and therefore emit less than a blackbody, we define the emissivity (e) as the ratio of the emitted power of the body to that of a blackbody at the same temperature. Then Stefan's law becomes.

$P = e\sigma A T^4$

The emissivity ranges from 0 to 1 ; $e = 1$ for a perfect radiator and absorber (a blackbody) and $e = 0$ for a perfect reflector. The emissivity for polished aluminium, an excellent reflector, is about 0.05 ; for soot (carbon black) it is about 0.95.

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Illustration 61

A black body is at $27^\circ C$. It surface area is equal to $\frac{1}{5.67}$. Find power emitted by it ?

Solution:

$P = e\sigma A T^4$

$(1)\frac{1}{5.67}\times 5.67\times 10^{-8}\times (300)^4$

$\Rightarrow \quad 10^{-8} \times 81 \times 10^8 \Rightarrow \ 81$

Illustration 62

Calculate the temperature at which a perfect black body radiates at the rate of 5.67 $W\ cm^{-2}$. Stefan's constant is $5.67 \times 10^{-8}\ J\ s^{-1}\ m^{-2}\ K^{-4}$.

Solution:

Given $E = 5.67\ W\ cm^{-2} = 5.67 \times 10^{+4}\ W\ m^{-2}$, $\sigma = 5.67 \times 10^{-8}\ J\ s^{-1}\ m^{-2}\ K^{-4}$

Using, $E = \sigma T^4$; $\quad T^4 = \frac{E}{\sigma}$

or $\quad T = \left[\frac{E}{\sigma}\right]^{\frac{1}{4}} = \left[\frac{5.67 \times 10^{+4}}{5.67 \times 10^{-8}}\right]^{\frac{1}{4}} = (10^{12})^{1/4} = 1000\ K$

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Illustration 63

If temperature of ideal black body is increased by 50%, what will be percentage increase in quantity of radiations emitted from its surface.

Solution:

$E \propto T^4$ and

$\therefore \quad E' \propto (1.5)^4 T^4 \propto \left[\frac{15}{10}\right]^4 T^4 \propto \left[\frac{3}{2}\right]^4 T^4 \propto \frac{81}{16} T^4 \approx 5T^4$

$$\frac{E'-E}{E} \times 100\% = \left[\frac{5T^4 - T^4}{T^4}\right] \times 100\% = 400\ \%$$

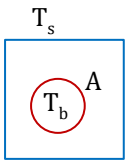
Rate of Emission of Radiation:

Net Rate of Heat Loss Through Radiation:

Rate of emission = $eA\sigma T_b^4$

Rate of absorption = $eA\sigma T_s^4$

Net rate of heat loss = $eA\sigma\left(T_b^4 - T_s^4\right)$



$T_b > T_s$ $\quad \boxed{\frac{dQ}{dt} = eA\sigma\left(T_b^4 - T_s^4\right) = -ms\frac{dT_b}{dt}}$

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Illustration 64

A body of $e = \frac{1}{3}$ at $127°C$ is kept in a room at $27°C$. Calculate initial rate of heat loss if surface area of body is 10 cm^2 (Take $\sigma = 6 \times 10^{-8}$)

Solution:

$$\frac{dQ}{dt} = \frac{1}{3} \times 10 \times 10^{-4} \times 6 \times 10^{-8}\left[(400)^4 - (300)^4\right] = \frac{1}{3} \times 6 \times 10^{-12} \times 10\left[10^8(256-81)\right]$$

$\Rightarrow \quad 2 \times 10^{-3} \times 175 \quad \Rightarrow \quad 350 \times 10^{-3} \quad \Rightarrow \quad 0.35$ Watt

Rate of Cooling Means Rate of Loss of Temperature:

Rate of cooling = $\left|\frac{dT_b}{dt}\right| = \frac{eA\sigma\left(T_b^4 - T_s^4\right)}{ms}$

If in time dt the net heat energy loss for ideal black body is dQ and because of this its temperature falls by $d\theta$

Rate of loss of heat $R_H = \frac{dQ}{dt} = e\sigma A(T^4 - T_0^4)$

It is also equal to emitted power or radiation emitted per second.

Rate of fall in temperature (Rate of cooling) $R_F = \frac{d\theta}{dt} = \frac{\sigma A}{ms}(T^4 - T_0^4)\left[\because \frac{dQ}{dt} = ms\frac{d\theta}{dt}\right]$

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Illustration 65

Calculate the energy radiated per second from the filament of an incandescent lamp at 2000K, if the surface area is $5.0 \times 10^{-5}\ m^2$ and its relative emittance is 0.85 and $\sigma = 5.7 \times 10^{-8}\ W\,m^2\,K^{-4}$.

Solution:

We know, $E = Ae_r\sigma T^4$

Here $A = 5.0 \times 10^{-5}\ m^2$, $\sigma = 5.7 \times 10^{-8}\ W\,m^{-2}\,K^{-4}$, $e = 0.85$, $T = 2000\ K$

$E = 5.0 \times 10^{-5}\ m^2 \times 0.85 \times 5.7 \times 10^{-8}\ W\,m^{-2}\,K^{-4} \times (2000)^4\ K^4$

$E = 38.76\ W = 38.76\ J\,s^{-1}$

Illustration 66

Two bodies A or B have same rate of cooling $\frac{m_A}{m_B}=\frac{3}{1},\frac{s_A}{s_B}=\frac{1}{2}$. Find ratio of heat loss of A and B ?

Solution:

Same rate of cooling

$$\left(\frac{eA\sigma\left(T_b^4-T_s^4\right)}{m_As_A}\right)_A=\left(\frac{eA\sigma\left(T_b^4-T_s^4\right)}{m_Bs_B}\right)_B$$

$$\left(\frac{\text{Heat loss}}{3\times1}\right)_A=\left(\frac{\text{Heat loss}}{1\times2}\right)_B$$

$2(\text{Heat loss})_A = 3(\text{Heat loss})_B$

$$\frac{(\text{Heat loss})_A}{(\text{Heat loss})_B}=\frac{3}{2}$$

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Illustration 67

Two spheres of same material and temperature are kept in same surrounding. $r_2 = 2r_1$. Find rate of cooling of second : first ?

Solution:

$$\frac{(\text{Rate of cooling})_1}{(\text{Rate of cooling})_2}=\frac{\frac{A_1}{m_1}}{\frac{A_2}{m_2}} \Rightarrow \frac{A_1}{m_1}\times\frac{m_2}{A_2}$$

$$\Rightarrow \frac{\pi\times(r_1)^2}{\rho\times\frac{4}{3}\times\pi\times(r_1)^3}\times\frac{\rho\times\frac{4}{3}\times\pi\times(r_2)^3}{\pi\times(r_2)^2} \Rightarrow \frac{(r_1)^2\times(2r_1)^3}{(r_1)^3\times(2r_1)^2} \Rightarrow \frac{8r_1^5}{4r_1^5}=\frac{2}{1}$$

$$\therefore \frac{(\text{Rate of cooling})_1}{(\text{Rate of cooling})_2}=2$$

Newton's Law of Cooling:

Condition of Newton's Law of Cooling:

(a) The difference in temperature between the body and surroundings must be small

(b) The loss of heat from the body should be by radiation only.

(c) The temperature of surroundings must remain constant during the cooling of the body.

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For small temperature difference between a body and its surrounding, the rate of cooling of the body is directly proportional to the temperature difference and the surface area exposed.

$\frac{d\theta}{dt} \propto (\theta - \theta_0)$, where θ and θ_0 are temperature corresponding to object and surroundings.

From above expression, $\frac{d\theta}{dt} = -k(\theta - \theta_0)$ ...(i)

This expression represents Newton's law of cooling. It can be derived directly from Stefan's law, which gives,

$$k = \frac{4e\sigma\theta_0^3}{ms} A$$

Proof:

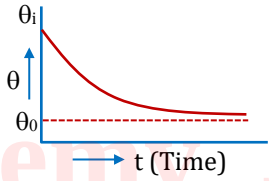
$$\frac{dQ}{dt} = eA\sigma\left(\theta^4 - \theta_0\right) = eA\sigma\left[\left(\theta_0^4 + \Delta\theta\right)^4 - \theta_0^4\right] = eA\sigma\theta_0^4\left[1 + \frac{4\Delta\theta}{\theta_0} - 1\right]$$

$$\frac{dQ}{dt} = 4eA\sigma\theta_0^3\left(\theta - \theta_0\right)$$

$$\frac{dQ}{dt} = -ms\frac{d\theta}{dt}$$

$$-ms\frac{d\theta}{dt} = 4\sigma eA\theta_0^3\Delta\theta$$

Now $\frac{d\theta}{dt} = -k[\theta - \theta_0] \Rightarrow \int_{\theta_i}^{\theta_f} \frac{d\theta}{(\theta - \theta_0)} = \int_0^t -kdt$



where θ_i = initial temperature of object and
θ_f = final temperature of object.

$\Rightarrow \quad \ell n\frac{(\theta_f - \theta_0)}{(\theta_i - \theta_0)} = -kt \quad \Rightarrow \quad (\theta_f - \theta_0) = (\theta_i - \theta_0)\, e^{-kt}$

$\Rightarrow \quad \theta_f = \theta_0 + (\theta_i - \theta_0)\, e^{-kt}$ ...(ii)

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Approximate Method for Applying Newton's Law of Cooling:

Sometime when we need only approximate values from Newton's law, we can assume a constant rate of cooling, which is equal to the rate of cooling corresponding to the average temperature of the body during the interval.

$$\left\langle \frac{d\theta}{dt} \right\rangle = -k(\langle\theta\rangle - \theta_0) \quad \text{...(iii)}$$

If θ_i and θ_f be initial and final temperature of the body then,

$$\langle\theta\rangle = \frac{\theta_i + \theta_f}{2} \quad \text{...(iv)}$$

Remember equation (iv) is only an approximation and equation (i) must be used for exact values.

Illustration 68

A body at temperature 40°*C* is kept in a surrounding of constant temperature 20°*C*. It is observed that its temperature falls to 35°*C* in 10 minutes. Find how much more time will it take for the body to attain a temperature of 30°*C*.

Solution:

From equation (ii),

$\Delta\theta_f = \Delta\theta_i\, e^{-kt}$

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For the interval in which temperature falls from 40 to 35°C.

$$(35 - 20) = (40 - 20)\, e^{-k.\,10}$$

$$\Rightarrow \quad e^{-10\,k} = \frac{3}{4} \qquad \Rightarrow \quad k = \frac{\ell n \frac{4}{3}}{10}$$

For the next interval

$$(30 - 20) = (35 - 20)e^{-kt}$$

$$\Rightarrow \quad e^{-kt} = \frac{2}{3} \qquad \Rightarrow \quad kt = \ell n \frac{3}{2}$$

$$\Rightarrow \quad \frac{\left(\ell n \frac{4}{3}\right)t}{10} = \ell n \frac{3}{2} \qquad \Rightarrow \quad t = 10\, \frac{\left(\ell n \frac{3}{2}\right)}{\left(\ell n \frac{4}{3}\right)} \text{minute} = 14.096 \text{ min}$$

Aliter:

(By approximate method) for the interval in which temperature falls from 40 to 35°C

$$<\theta> = \frac{40+35}{2} = 37.5°C$$

From equation (iii) $\left\langle \frac{d\theta}{dt} \right\rangle = -k(<\theta> - \theta_0)$

$$\Rightarrow \quad \frac{(35°C - 40°C)}{10(\text{min})} = -K(37.5°C - 20°C)$$

$$\Rightarrow \quad K = \frac{1}{35}(\text{min}^{-1})$$

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For the interval in which temperature falls from 35°C to 30°C

$$<\theta> = \frac{35+30}{2} = 32.5°C$$

From equation (14.4)

$$\frac{(30°C - 35°C)}{t} = -(32.5°C - 20°C)\, K$$

$$\Rightarrow \quad \text{Required time, } t = \frac{5}{12.5} \times 35\text{min} = 14 \text{ min}$$

Illustration 69

A heater is connected to 200 watt power supply and kept in surrounding at 20°C. Temperature of heater becomes steady at 40°C.

(a) Find rate of heat absorption by heater at 35°C

(b) If rate of increase of temperature was 1°C per second when heater was at 35° ? Find heat capacity of heater ?

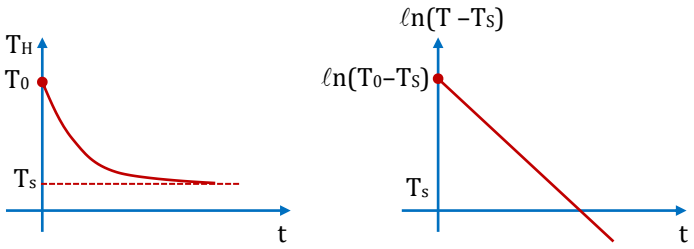
(c) If heat capacity is 50 Joule per kelvin? Find rate of increase of temperature of heater when its temperature was 38°C ?

(d) After steady state, heater is switched off. Find rate of heat loss and rate of cooling just after switching off the heater?

(e) Draw the graph of temperature of heater after switching off?

(f) Find the time taken to cool from 40°C to 25°C ?

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(g) Find time taken by heater to cool from 39.5°C to 38.5°C ?

Solution:

(a) $\frac{dQ}{dt}=K\left(T_H-T_S\right)$

$200 = K(40 - 20)$

$K = 10$

$\left(\frac{dQ}{dt}\right)_{T_H=35^\circ}=10(35^\circ-20^\circ) \quad \Rightarrow \ 150$

$\therefore$ Rate of absorption = 200 – 150 = 50 watt

(b) $\frac{dQ}{dt}=150$

$\therefore$ Rate of absorption = 50 watt

$(50)=ms\left(\frac{dT}{dt}\right)$

$ms = C = 50\ J/Kelvin$

(c) Rate of emission = 10(38 – 20) = 180

Rate of absorption = 200 – 180 = 20 watt

$20=ms\left(\frac{dT}{dt}\right)$

$20=50\frac{dT}{dt} \quad \Rightarrow \quad \frac{dT}{dt}=\frac{2}{5}=0.4\%$

(d) $\frac{dQ}{dt}=10(40-20)=20watt$ (Rate of heat loss)

$\frac{dT}{dt}=\frac{\frac{dQ}{dt}}{ms} \quad \Rightarrow \quad \frac{200}{50} \quad \Rightarrow 4°C/sec.$

(e) $\frac{dQ}{dt}=K(T-T_s)=ms\frac{dT}{dt}$

$\int_{40}^{T}\frac{dT}{T-T_s}=\frac{-K}{ms}\int_{40}^{T}dt \quad \Rightarrow \quad \left[\ell n\left(T-T_s\right)\right]_{T_0}^{T}=\frac{-K}{ms}t$

$\ell n\left(T_0-T_s\right)-\frac{K}{ms} \quad \Rightarrow \quad \frac{T-T_s}{T_0-T_s}=e^{\frac{-Kt}{ms}}$

$T=T_s+\left(T_0-T_s\right)e^{\frac{-Kt}{ms}}$

(f) $T=T_s+\left(T_0-T_s\right)e^{\frac{-Kt}{ms}}$

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$$25=20+(40-20)e^{\frac{-10t}{50}}$$

$$\frac{5}{20}=e^{\frac{-t}{5}} \qquad \Rightarrow \quad e^{\frac{-t}{5}}=\frac{1}{4}$$

$$t = 5\,\ell n4$$

(g) We approximate this as linear variation

$$\frac{39.5+38.5}{2}=39°C$$

$$\frac{d\theta}{dt}=10(39-20)=190$$

$\therefore$ $\Delta\theta = 190 \times t = ms\Delta T$

$190 \times t = 50(39.5 - 38.5)$

$$t=\frac{50}{190}\text{sec.}$$

Illustration 70

If a liquid takes 30 seconds in cooling from 80°C to 70°C and 70 seconds in cooling 60°C to 50°C, then find the room temperature.

Solution:

$$\frac{\theta_1-\theta_2}{t} = K\left(\frac{\theta_1+\theta_2}{2}-\theta_0\right)$$

In first case, $\frac{80-70}{30} = K\left(\frac{80+70}{2}-\theta_0\right)$; $\frac{1}{3} = K(75-\theta_0)$ ...(i)

In second case, $\frac{60-50}{70} = K\left(\frac{60+50}{2}-\theta_0\right)$; $\frac{1}{7} = K(55-\theta_0)$ ...(ii)

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Equation (i) divide by equation (ii)

$$\frac{7}{3}=\frac{(75-\theta_0)}{(55-\theta_0)}$$

$\Rightarrow$ $385 - 7\theta_0 = 225 - 3\theta_0$

$\Rightarrow$ $\theta_0=\frac{160}{4} = 40°C$

Solar Constant 's':

The Sun emits radiant energy continuously in space of which an insignificant part reaches the Earth. The solar radiant energy received per unit area per unit time by a black surface held at right angles to the Sun's rays and placed at the mean distance of the Earth (in the absence of atmosphere) is called solar constant. The solar constant S is taken to be 1340 watts/m^2 or 1.937 Cal/cm^2–$minute$

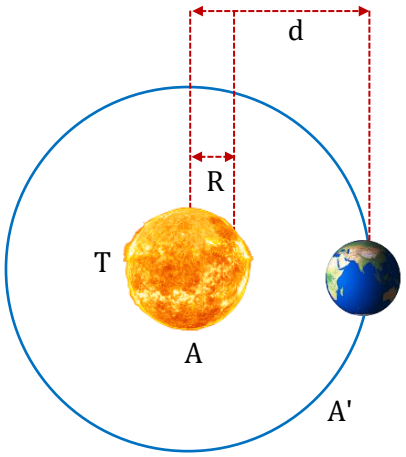
Temperature of the Sun:

Let R be the radius of the Sun and $'d'$ be the radius of Earth's orbit around the Sun. Let E be the energy emitted by the Sun per second per unit area. The total energy emitted by the Sun in one second = $E.A = E \times 4\pi R^2$. (This energy is falling on a sphere of radius equal to the radius of the Earth's orbit around the Sun i.e., on a sphere of surface area $4\pi d^2$).

So, the energy falling per unit area of Earth = $\frac{4\pi R^2 \times E}{4\pi d^2}=\frac{E\,R^2}{d^2}$

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$R = 7\times 10^{8} m$, $d = 1.5 \times 10^{11} m$, $s = 5.7 \times 10^{-8}\ W m^{-2} K^{-4}$



Solar constant,

$$S = \frac{E R^2}{d^2}$$

By Stefan's Law,

$$E = \sigma T^4$$

$$S = \frac{\sigma T^4 R^2}{d^2}$$

$$\Rightarrow \quad T = \left[\frac{S\times d^2}{\sigma\times R^2}\right]^{\frac{1}{4}} = \left[\frac{1340\times(1.5\times10^{11})^2}{5.7\times10^{-8}\times(7\times10^{8})^2}\right]^{\frac{1}{4}} = 5732\ K$$

Spectral Emissive Power (E_λ):

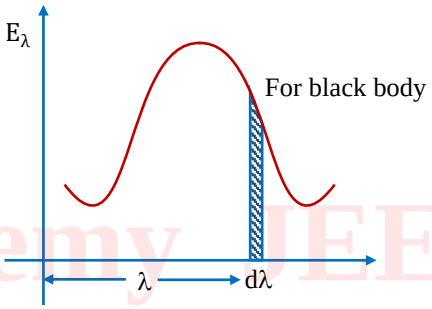
'E_λ' is emissive power per unit wavelength near a given wavelength (λ)

$$E_\lambda = \frac{dE}{d\lambda}$$

$E_\lambda d\lambda$ = emissive power of radiation having wavelength from λ to $\lambda + d\lambda$

$\int_{\lambda_1}^{\lambda_2} E_\lambda d\lambda$ = emissive power of radiation having wavelength from λ_1 to λ_2

$\int_0^\infty E_\lambda d\lambda$ = total emissive power

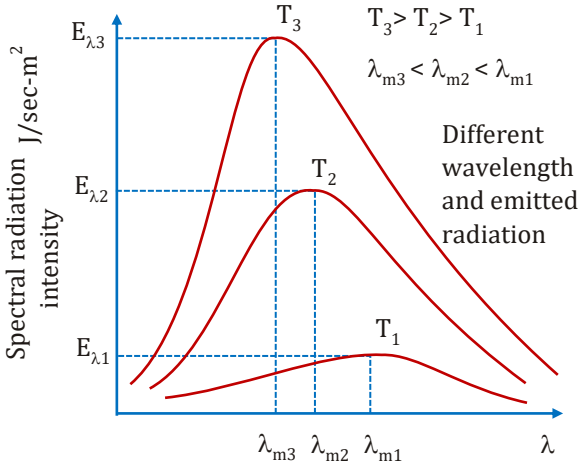


Total area of graph = Total emissive power = $e\sigma T^4$ (here e = emissivity of surface)

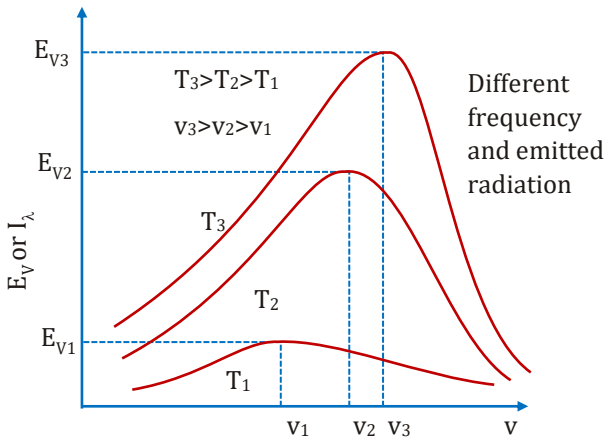
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Spectral Energy Distribution Curve of Black Body Radiations:

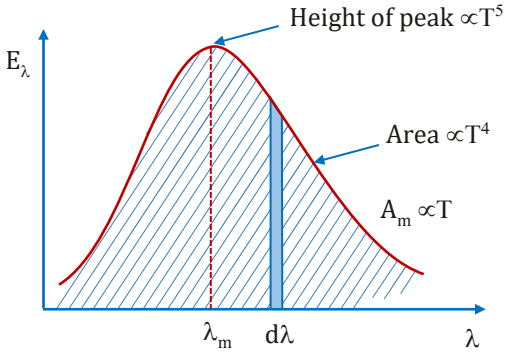
Practically given by : Lumers and Pringshem



Mathematically given by : Plank



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Spectral energy distribution curve ($E_\lambda - \lambda$)

(i) $\lambda_m \propto \frac{1}{T}$ (ii) $E_{\lambda_{max}} \propto T^5$ (iii) Area $\int_0^\infty E_\lambda d\lambda = E = \sigma T^4$ $\frac{A_1}{A_2} = \left[\frac{T_1}{T_2}\right]^4$

- Spectral energy distribution curves are continuous. At any temperature in between, possible wavelength (0 – ∞) radiation emitted but for different wavelength quantity of radiations are different.
- As the wave length increases, the amount of radiation emitted first increase, becomes maximum and then decreases.
- At a particular temperature the area enclosed between the spectral energy curve shows the spectral emissive power of the body.

$$\text{Area} = \int_0^\infty E_\lambda d\lambda = E = \sigma T^4$$

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Wein's Displacement Law:

The wavelength corresponding to maximum emission of radiation decrease with increasing temperature $\left[\lambda_m \propto \frac{1}{T}\right]$. This is known as Wein's displacement law.

$\lambda_m T = b$; where b Wein's constant = 2.89 x 10^{-3} mK. $\Rightarrow$ $\lambda_{m_1} T_1 = \lambda_{m_2} T_2 = \lambda_{m_3} T_3 = b$

Dimensions of b : = $M^0 L^1 T^0 \theta^1$

Relation between frequency and temperature $\nu_m = \frac{c}{b} T$

Illustration 71 0

The temperature of furnace is 2000°C, in its spectrum the maximum intensity is obtained at about 4000Å, If the maximum intensity is at 2000Å calculate the temperature of the furnace in °C.

Solution:

By using $\lambda_m T = b$, 4000 (2000+273) = 2000(T) $\Rightarrow$ T = 4546K

The temperature of furnace = 4546 – 273 = 4273 °C

Illustration 72

Body A has $\lambda_{m_A} = 500Å$ and body B has $\lambda_{m_B} = 1000Å$.

Find: (a) $\frac{T_A}{T_B}$ (b) $\frac{\text{Emissive power of A}}{\text{Emissive power of B}}$

Solution:

(a) $\frac{T_A}{T_B} = \frac{\lambda_{m_B}}{\lambda_{m_A}} = \frac{1000Å}{500Å} = \frac{2}{1}$ (b) $\frac{\text{Emissive power of } A}{\text{Emissive power of } B} = \frac{\sigma T_A^4}{\sigma T_B^4} = \left(\frac{2}{1}\right)^4 = \frac{16}{1}$

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KINETIC THEORY

Notations and Ideal Gas Concept

Kinetic Energy of Gas

All molecules of gas are continuously performing motion in random direction or say zig-zag direction. Apart from translational motion these molecules also perform rotational as well as vibrational motion. Thus, these molcules have kinetic energy associated with these motions.

Kinetic energy of a gas is defined as some of kinetic energies of all molecules.

It is dependent on absolute temperature. Increasing temperature increases kinetic energy of gas.

Potential Energy

Intermolecular attraction force exists between molecules and thus every molecules have some potential energy.

Greater Intermolecular force (IMF), lesser will be potential energy and greater will be magnitude of potential energy.

Orders

(i) IMF

Solid > Liquid > Gas (Real) > Ideal Gas (IMF = 0)

(ii) Potential Energy

Solid < Liquid < Gas (Real) < Ideal Gas (IMF = 0)

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Internal Energy of Gas

Internal energy of gas is defined as some of potential energy and kinetic energy at a temperature.

At a given temperature for solid, liquid and gas:

(i) Internal kinetic energy : Same for all

(ii) Internal potential Energy : Maximum for ideal gas ($PE = 0$) and Minimum for solids ($PE = -ve$)

(iii) Internal Energy : Maximum for Ideal gas and Minimum for solid

Notation

Moles of gas $\mu = \frac{M}{M_W} = \frac{N}{N_0}$

Avogadro constant $N_0 = 6.023 \times 10^{23} \ \frac{molecules}{mole}$

Universal gas constant $R = 8.31 \ \frac{J}{\text{mole-}K} \approx 2 \frac{\text{cal}}{\text{mole-}K}$

Specific gas constant $r = \frac{R}{M_W}$

Boltzmann constant $k = \frac{R}{N_0} = 1.38 \times 10^{-23} \ \frac{J}{\text{kelvin}}$

Mass of a gas molecule = m

Mass of gas = M

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Molecular weight = M_w

No. of gas molecules = N

Molecular density $n = \frac{N}{V}$

Gas density $\rho = \frac{M}{V} = mn$

Postulates of Kinetic Theory of Gases

The Kinetic Theory of Gases

Rudolph Claussius (1822–88) and James Clark Maxwell (1831–75) developed the kinetic theory of gases in order to explain gas laws in terms of the motion of the gas molecules. The theory is based on following assumptions regarding the motion of molecules and the nature of the gases.

Basic Postulates of Kinetic Theory of Gases

- Every gas consists of extremely small particles known as molecules. The molecules of a given gas are all identical but are different than those of another gas.
- The size is negligible in comparision to inter molecular distance (10^{-9} m).
- The gas molecules keep colliding among themselves as well as with the walls of containing vessel. These collision are perfectly elastic. (i.e. the total energy before collision = total energy after the collision)
- No attractive or repulsive force acts between gas molecules.
- Gravitational attraction among the molecules is ineffective due to extremely small masses and very high speed of molecules.
- Molecules constantly collide with the walls of container due to which their momentum changes. This change in momentum is transferred to the walls of the container. Consequently pressure is exerted by gas molecules on the walls of container.

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Ideal Gas

A gas which follows all gas laws and gas equation at every possible temperature and pressure is known as ideal or perfect gas.

Equation of State for Ideal Gas

$PV = \mu RT$ (μ = number of moles of gas)

$\Rightarrow PV = \frac{M}{M_w}RT$

$\Rightarrow PV = \left[\frac{mN}{mN_0}\right]RT$

$\Rightarrow PV = \left[\frac{R}{N_0}\right]NT$

$\Rightarrow PV = NkT \quad \left(\text{Boltzmann constant}, k = \frac{R}{N_0}\right)$

$P = \frac{\rho RT}{M_W} \quad \left(\rho = \frac{M}{V}\right)$

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Properties of Ideal Gas

- Ideal gas molecules can do only translational motion, so their kinetic energy is only translational kinetic energy.
- Ideal gas can not be liquified because *IMF* is zero.
- Specific heat of ideal gas is constant quantity and it does not change with temperature.
- All real gases behave as an ideal gas at high temperature and low pressure and low density.
- Gas molecules have point mass and negligible volume and velocity is very high (10^7 *cm/s*). That's why there is no effect of gravity on them.

NTP & STP

Properties	N.T.P. (Normal Temperature and Pressure)	S.T.P. (Standard Temperature and Pressure)
Temperature	0° C = 273.15 *K*	0.01° *C* = 273.16K
Pressure	1 atm	1 atm
Volume (1 mole)	22.4 litres	22.4 litres

1 atm = 1.01325×10^5 pascal
1 litre = 10^{-3} m^3 = 10^3 cm^3 = $10^3 cc$ = $10^3 ml$

Illustration 1

Calculate volume of 6 moles of CO_2 gas at NTP?

Solution:

Volume of 1 mole CO_2 gas at NTP = 22.4 litres
Volume of 6 mole CO_2 gas at NTP = 6 × 22.4 litres = 134.4 litres

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Illustration 2

Calculate pressure exerted by 3 moles of an ideal gas kept in a vessel of volume 6 litres at 27°*C*?

Solution:

Given:

V = 6 litres
T = 300 *K*
μ = 3 moles
$PV = \mu RT$

$$\Rightarrow \quad P = \frac{\mu RT}{V}$$

$$\Rightarrow \quad P = \frac{3 \times 0.0821 \times 300}{6}$$

$$\Rightarrow \quad P = 12.315 \text{ atm}$$

Illustration 3

A vessel of volume 8.3×10^{-3} m^3 contains an ideal gas at temperature 27°*C* and pressure 200 *kPa*. The gas is allowed to leak till the pressure falls to 100 *kPa* and temperature increases to 327°*C*. What is the amount of gas in moles will be leaked out?

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Solution:

Initial molar amount

$$\mu_1 = \frac{PV}{RT} = \frac{200\times10^3\times8.3\times10^{-3}}{8.3\times300} = \frac{2}{3}\text{ mole}$$

$$\Rightarrow \quad \frac{P_1}{P_2} = \frac{\mu_1 T_1}{\mu_2 T_2} \quad \Rightarrow \quad \frac{200}{100} = \frac{\frac{2}{3}\times300}{\mu_2\times600}$$

$$\Rightarrow \quad \mu_2 = \frac{1}{6}\text{ mole} \quad \Rightarrow \quad \text{leaked out moles} = \mu_1 - \mu_2 = \frac{2}{3} - \frac{1}{6} = \frac{1}{2}\text{ mole}$$

Charles' Law and Boyle's Law

Gas Laws

An ideal gas is described by following terms or say following properties Temperature (T), Pressure(P), Volume of the gas(V) and Number of moles(μ). Relations between these properties for an ideal gas have been established and are know as gas laws.

(1) Charle's Law
(2) Boyle's Law
(3) Gay-Lussac's Law
(4) Avogadro's Law
(5) Dalton's Partial Pressure Law

Charle's Law

According to this law, for a given mass of an ideal gas at constant pressure; volume of a gas is directly proportional to its absolute temperature,

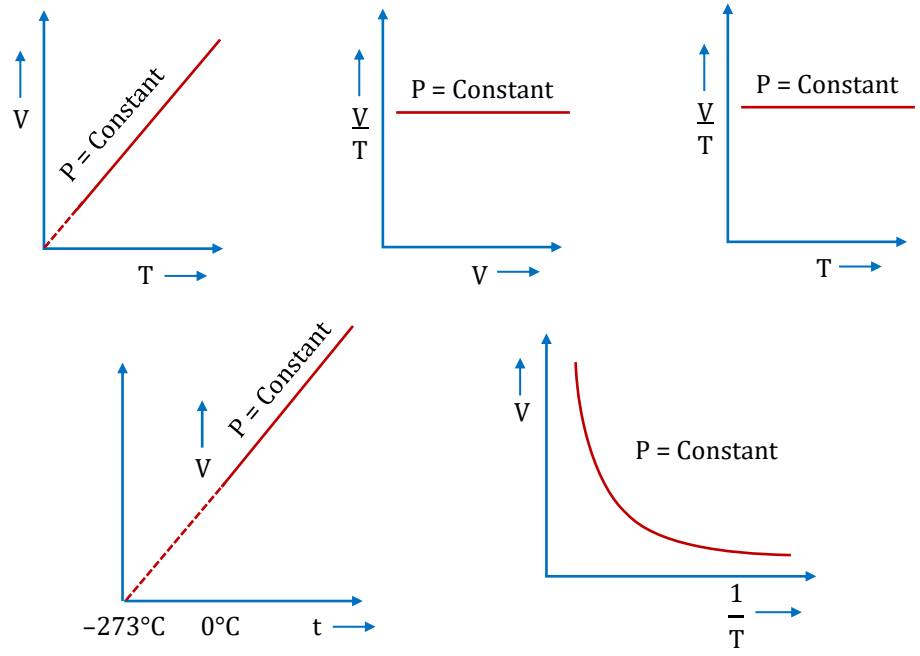
Mathematically

$V \propto T$ (m and P are Constant)

$$\frac{V_1}{V_2} = \frac{T_1}{T_2}$$

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Graph for Charle's Law



Here t represents temperature in °C and T represents in K i.e. absolute temperature.

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Boyle's Law

According to this law, for a given mass of an ideal gas at constant temperature; the volume of a gas is inversely proportional to its pressure,

Mathematically

$$V \propto \frac{1}{P} \qquad (m \text{ and } T \text{ are Constant})$$

$$P_1V_1 = P_2V_2$$

Graphs for Boyle's Law

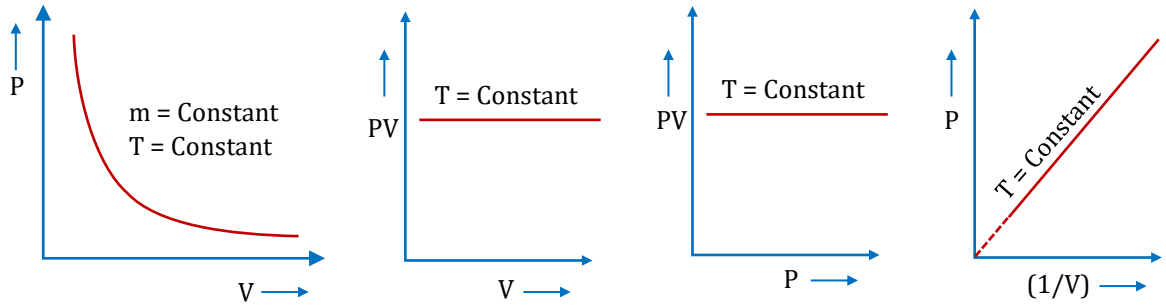
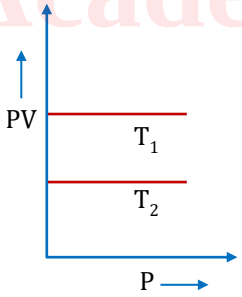


Illustration 4

Find relation between T_1 and T_2 for the given curve.

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Solution:

From Boyle's Law $T_1 > T_2$

Gay-Lussac's Law, Avogadro's Law and Dalton's Law

Gay–Lussac's Law

According to this law, for a given mass of an ideal gas; at constant volume, pressure of a gas is directly proportional to its absolute temperature,

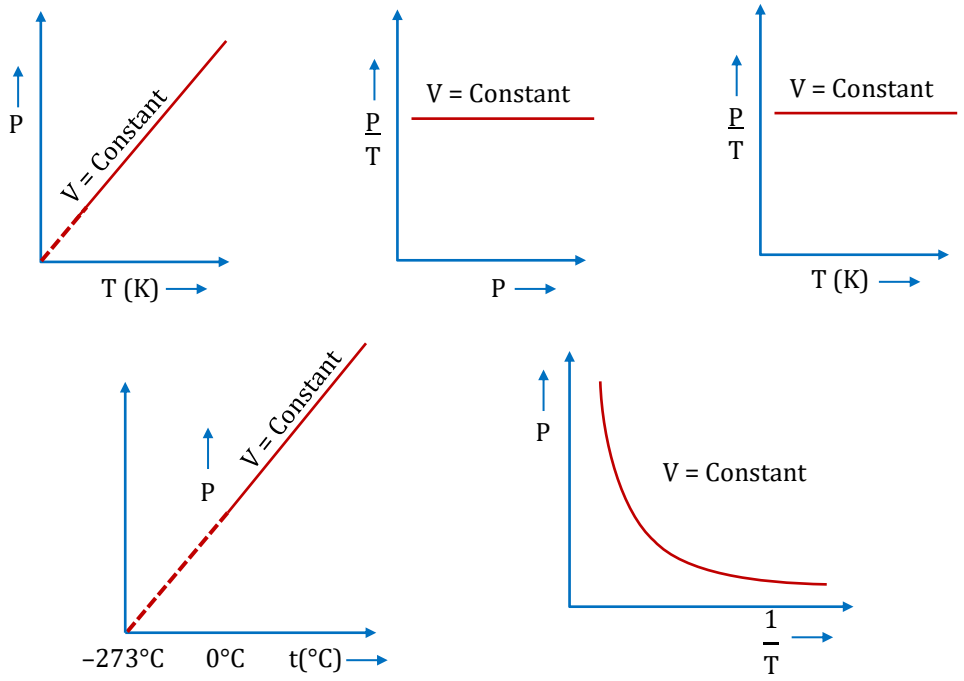
Mathematically

$$P \propto T \quad (m \text{ and } V \text{ are Constant})$$

$$\frac{P_1}{P_2} = \frac{T_1}{T_2}$$

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Graph for Gay–Lussac's Law



Here t represents temperature in $°C$ and T represents in K i.e. absolute temperature.

Avogadro's Law

According to this law, at same temperature and pressure, equal volume of all gases contain equal number of molecules, i.e., $N_1 = N_2$ if P, V and T are same.

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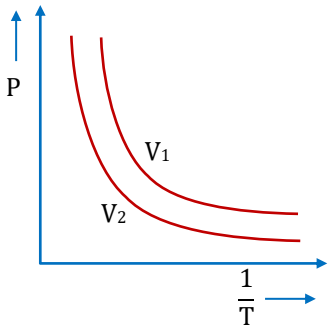
Dalton's Partial Pressure Mixture Law

According to this law, the pressure exerted by mixture of non-reactive gases is equal to the sum of partial pressure of each component gases present in the mixture.

$$P = P_1 + P_2 + P_3 \ldots.$$

Illustration 5

Find relation between V_1 and V_2 for given curve.



Solution:

From Gay–Lussac's Law, $V_1 < V_2$

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Illustration 6

At a particular temperature, N_2 exerts a pressure of 60 bar and O_2 exerts a pressure of 25 bar on the wall of closed vessel. Find pressure exerted by them on the wall if mixed together.

Solution:

From Dalton's Partial Pressure Mixture Law, $P_{net} = P_1 + P_2$

where $P_1 \rightarrow$ pressure exerted by N_2 alone = 60 bar

$P_2 \rightarrow$ pressure exerted by O_2 alone = 25 bar

$\therefore$ $P_{net} = 60 + 25 = 85$ bar

Illustration 7

At a particular temperature and pressure, 4 moles of N_2 occupy a volume of 30 litres. Then find volume occupied by 16 moles of O_2 at same temperature and pressure.

Solution:

From Avogadro's Law

$$V \propto N$$

$$\frac{V_1}{N_1} = \frac{V_2}{N_2}$$

$$\Rightarrow \quad \frac{30}{4} = \frac{V_2}{16}$$

$$\Rightarrow \quad V_2 = 120 \text{ litres}$$

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Different Types of Speeds of Gas Molecules

Average velocity

Because molecules are in random motion in all possible direction with all possible velocity. Therefore, the average velocity of the gas molecules in container is zero.

$$<\vec{v}> = \frac{\vec{v}_1 + \vec{v}_2 + ...\vec{v}_N}{N} = 0$$

Average speed or Mean speed (v_{avg})

By Maxwell's velocity distribution law v_M or $<|\vec{v}|> \equiv v_{mean} = v_{avg}$

$$<|\vec{v}|> \equiv v_{mean} \equiv v_{avg} = \frac{|\vec{v}_1| + |\vec{v}_2| + ...|\vec{v}_n|}{N}$$

Most probable speed (v_{mp})

At a given temperature, the speed to which maximum number of molecules belongs is called as most probable speed (v_{mp})

RMS Speed (v_{rms})

RMS Speed (root mean squared speed) is equal to square root of the sum of square of speed of each molecule of a gas.

$$v_{rms} = \sqrt{\frac{v_1^2 + v_2^2 + v_3^2 .....}{N}}$$

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Illustration 8

4 gas molecules have speeds $3u$, $5u$, $2u$ and $6u$. Find their rms speed.

Solution:

$$v_{rms}=\sqrt{\frac{(3u)^2+(5u)^2+(2u)^2+(6u)^2}{4}}=\sqrt{\frac{(9+25+4+36)u^2}{4}}=\sqrt{\frac{37}{2}}u$$

Illustration 9

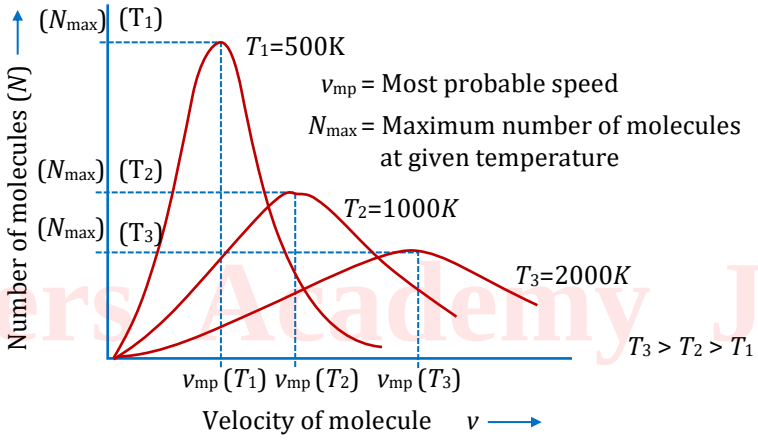
For above question find average speed.

Solution:

$$v_{avg}=\frac{3u+5u+2u+6u}{4}=4u$$

Maxwell's law of distribution of velocities

Maxwell had drawn a curve between number of molecules, moving with a particular velocity, and velocity of molecules known as velocity distribution curve shown here.



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According to this law

- The velocities of molecules of a gas are in between zero and infinity (0 – ∞).
- The area of the graph with velocity axis gives as total number of molecules available in the container which remains constant always.
- With the increase in the temperature, the most probable velocity increases but corresponding number of molecules decreases.
- The number of molecules within certain velocity range is constant although the velocity of molecule changes continuously at particular temperature.
- According to Maxwell

  $v_{rms} > v_{mean} > v_{mp}$

Illustration 10

If for a gas, their most probable speed at temperature T_1 is v, at temperature T_2 is $\frac{3v}{2}$ and at temperature T_3 is $\frac{2v}{3}$. Find order of temperature.

Solution:

As we know most probable speed increases with increase in temperature. Hence $T_3 < T_1 < T_2$.

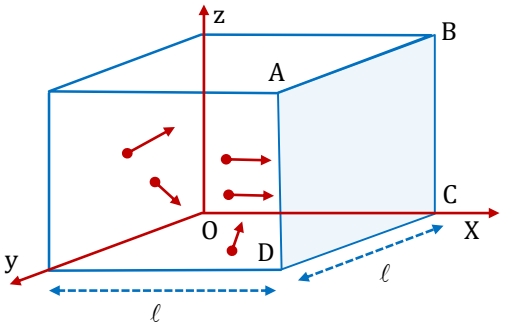
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Expression of Pressure

Expression for Pressure of An Ideal gas

Consider an ideal gas enclosed in a cubical vessel of length ℓ.

Suppose there are 'N' molecules in a gas which are moving with velocities $\vec{v}_1, \vec{v}_2 ..... \vec{v}_N$.



If we consider any single molecule then its instantaneous velocity $\vec{v}$ can be expressed as $\vec{v} = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$

Due to random motion of the molecule $v_x = v_y = v_z$; $|\vec{v}| = v_x\sqrt{3} = v_y\sqrt{3} = v_z\sqrt{3} = \sqrt{v_x^2 + v_y^2 + v_z^2}$

Suppose a molecule of mass m is moving with a velocity v_x towards the face ABCD. It strikes the face of the cubical vessel and returns back to strike the opposite face.

Change in momentum of the molecule per collision $\Delta p = -mv_x - mv_x = -2mv_x$

Momentum transferred to the wall of the vessel per molecule per collision $\Delta p = 2mv_x$

The distance travelled by the molecule in going to face ABCD and coming back is 2ℓ.

So, the time between two successive collision is $\Delta t = \frac{2\ell}{v_x}$

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Number of collision per sec per molecule is $f_c = \frac{v_x}{2\ell} = \frac{\text{molecule velocity}}{\text{mean free path}}$; $f_c = \frac{v_{rms}}{\lambda_m}$ or $f_c = \frac{v_m}{\lambda_m}$

Hence momentum transferred to the wall per second by the molecule is equal to force applied on the wall therefore, force

$$F = (2mV_x)\frac{v_x}{2\ell} = \frac{mv_x^2}{\ell} = \frac{mv^2}{3\ell} \quad \left[\text{As } V = \sqrt{3}V_x\right]$$

Pressure exerted by gas molecule $P = \frac{F}{A} = \frac{1}{3}\frac{mv^2}{\ell \times A}$

$$\Rightarrow \quad P = \frac{1}{3}\frac{mv^2}{V} \qquad \left[\because A \times \ell = V\right]$$

Pressure exerted by gas $P = \sum\frac{1}{3}\frac{mv^2}{V} = \sum\frac{1}{3}\frac{mv^2}{V} \times \frac{N}{N} = \frac{1}{3}\frac{mN}{V}\frac{\sum v^2}{N} = \frac{1}{3}\frac{mN}{V}v_{rms}^2$

$$v_{rms}^2 = \frac{3PV}{M} = \frac{3\mu RT}{\mu M_w}$$

$$\Rightarrow \quad v_{rms} = \sqrt{\frac{3RT}{M_w}} \ , \ P = \frac{1}{3}\frac{M}{V}v_{rms}^2 = \frac{1}{3}\rho v_{rms}^2$$

- Average number of molecules for each wall = $\frac{N}{6}$.

  No. of molecules along each axis = $\frac{N}{3}$ $(N_x = N_y = N_z)$

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- $\overline{v}_x^2 = \overline{v}_y^2 = \overline{v}_z^2 = \frac{v_{rms}^2}{3}$ Root mean square velocity along any axis for gas molecule is

$(v_{rms})_X = (v_{rms})_y = (v_{rms})_z = \frac{v_{rms}}{\sqrt{3}}$

All gas laws and gas equation can be obtained by expression of pressure of gas (except Joule's law)

Translational Kinetic Energy

$$E_T = \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 + \frac{1}{2}mv_3^2 + ...$$

$$E_T = \frac{1}{2}m[v_1^2 + v_2^2 + v_3^2 + ...] \times \frac{N}{N}$$

$$E_T = \frac{1}{2}MV_{rms}^2$$

$$\begin{cases} P = \frac{1}{2}\rho(V_{rms})^2 \\ (V_{rms})^2 = \frac{3P}{\rho} \end{cases}$$

$$E_T = \frac{1}{2}M\left(\frac{3P}{\rho}\right)$$

$$\begin{cases} \rho = \frac{M}{Volume} \\ \frac{M}{\rho} = Volume \end{cases}$$

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$$E_T = \frac{3}{2}PV$$

$\frac{E_T}{V} = E_v$ = Energy density

$$E_V = \frac{3}{2}P$$

Expression of RMS Speed

$\Rightarrow$ $P = \frac{1}{3}\rho(V_{rms})^2$

$\Rightarrow$ $V_{rms} = \sqrt{\frac{3P}{\rho}}$

$$\frac{P}{\rho} = \frac{RT}{M_W} = \frac{K_bT}{m}$$

$\Rightarrow$ $P = \frac{1}{3}\rho(V_{rms})^2$

$\Rightarrow$ $V_{rms} = \sqrt{\frac{3P}{\rho}}$

$$\frac{P}{\rho} = \frac{RT}{M_W} = \frac{k_bT}{m}$$

$$V_{rms} = \sqrt{\frac{3P}{\rho}} = \sqrt{\frac{3RT}{M_w}} = \sqrt{\frac{3k_bT}{m}}$$

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If gas is same (M_w= const.) and temp. is changed

$$V_{rms} \alpha \sqrt{T}$$

If temperature is constant and gas is changed

$$V_{rms} \propto \frac{1}{\sqrt{M_w}}$$

Effect of pressure at constant temp.:

$$V_{rms} \propto \frac{1}{\sqrt{M_w}}$$

$$P = \frac{\rho RT}{M_w}$$

$\because$ T & M_w = constant

$\therefore$ $\frac{P}{\rho} = const.$

So no effect on V_{rms}

Note:

If only temp. is changed

$$V_{rms} = \sqrt{\frac{3RT}{M_W}}$$

If only pressure is changed

$$V_{rms} = \sqrt{\frac{3P}{\rho}}$$

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If both pressure & temp. is changed

$$V_{rms} = \sqrt{\frac{3RT}{M_W}}$$

Most probable speed

$$v_{mp} = \sqrt{\frac{2P}{\rho}} = \sqrt{\frac{2RT}{M_w}} = \sqrt{\frac{2k_bT}{m}}$$

Average speed

$$v_{av} = \sqrt{\frac{8P}{\pi\rho}} = \sqrt{\frac{8RT}{\pi M_w}} = \sqrt{\frac{8k_bT}{\pi m}}$$

Comparision

$\Rightarrow$ $$v_{rms} = \sqrt{\frac{3P}{\rho}} = \sqrt{\frac{3RT}{M_w}} = \sqrt{\frac{3k_bT}{m}}$$

$\Rightarrow$ $$v_{av} = \sqrt{\frac{8P}{\pi\rho}} = \sqrt{\frac{8RT}{\pi M_w}} = \sqrt{\frac{8k_bT}{\pi m}}$$

$\Rightarrow$ $$v_{mp} = \sqrt{\frac{2P}{\rho}} = \sqrt{\frac{2RT}{M_w}} = \sqrt{\frac{2k_bT}{m}}$$

$$V_{rms} > V_{Mean} > V_{MP}$$

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Escape Velocity

The minimum velocity required for which molecule or object to escape out from earth surface.

$$V_{es} = \sqrt{\frac{2GM}{R}}$$

$(V_{es})_{earth} = 11.2 km/sec$

Relation of escape Velocity with rms

Condition for atmosphere to be present at the surface of any planet:

$V_{rms} < V_{es} \rightarrow$ Atmosphere present

$V_{rms} \geq V_{es} \rightarrow$ No atmosphere

$$\sqrt{\frac{3RT}{M_W}} \geq V_{es}$$

$$T \geq \frac{(V_{es})^2 M_W}{3R}$$

$$T_{min} = \frac{(V_{es})^2 M_w}{3R}$$

$$T_{min} = \frac{Mw}{2} \times 10^7 K \quad \rightarrow \quad \text{here } M_w \text{ is in } kg$$

$$T_{min} = \frac{Mw}{2} \times 10^4 K \quad \rightarrow \quad \text{here } M_w \text{ is in gram}$$

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Note:

If we continuously increase the temperature at $10^4 K$ hydrogen gas molecules will leave the earth surface first.

Moon has no atmosphere because there rms speed of gas molecules is greater then the escape velocity.

Illustration 11

The rms velocity of gas molecules of a given amount of a gas at $27°C$ and 1.0×10^5 $N\ m^{-2}$ pressure is 100 $m\ sec^{-1}$. If temperature and pressure are respectively $527°C$ and 2.5×10^5 $N\ m^{-2}$, the rms velocity will be:

Solution:

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

For a given gas, $v_{rms} \propto \sqrt{T}$

$$\frac{v_{rms_1}}{\sqrt{T_1}} = \frac{v_{rms_2}}{\sqrt{T_2}}$$

$$\Rightarrow \quad \frac{100}{\sqrt{300}} = \frac{v_{rms_2}}{\sqrt{800}}$$

$$\Rightarrow \quad v_{rms_2} = 163 m/s$$

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Illustration 12

The rms velocity of H_2 is 8×10^3 m/s. What will be the rms velocity of O_2 molecules at the same temperature:

Solution:

$$v_{r.m.s.} \propto \sqrt{\frac{1}{M_w}}$$

$$\Rightarrow \quad \frac{(v_{r.m.s.})_{O_2}}{(v_{r.m.s.})_{H_2}} = \sqrt{\frac{2}{32}} = \frac{1}{4}$$

$$\Rightarrow \quad (v_{r.m.s.})_{O_2} = \frac{1}{4} \times 8 \times 10^3 = 2 \times 10^3 \ m/s$$

Illustration 13

At a constant temperature if pressure of a gas is tripled. Find fractional change in most probable speed.

Solution:

$$v_{mp} = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2P}{\rho}}$$

At constant temperature $\frac{P}{\rho}$ is constant hence v_{mp} does not change by change pressure.

Illustration 14

Calculate the minimum temperature at which N_2 molecules escape from the earth.

Solution:

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$$v_{rms} = \sqrt{\frac{3RT}{M}} \geq 11.2 \times 10^3 m/s$$

$$T \geq \left(11.2 \times 10^3\right)^2 \times \frac{M}{3R}$$

For minimum temperature

$$T = \left(11.2 \times 10^3\right)^2 \times \frac{M}{3R}$$

On solving we get $T = 14 \times 10^4 K$

Degrees of Freedom (f)

The number of independent ways in which a molecule or an atom can exhibit motion or have energy is called its degrees of freedom.

The number of independent coordinates required to specify the dynamic state of a system is called its degrees of freedom.

Types of Degrees of freedom

The degrees of freedom are of three types:

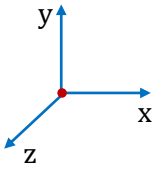
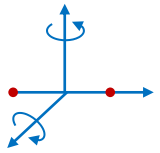
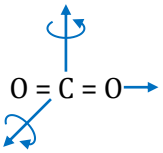
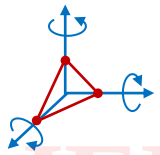
(a) **Translational Degree of freedom :** Number of independent ways in which a gas molecule may perform translatory motion. There are maximum three degree of freedom corresponding to translational motion.

(b) **Rotational Degree of freedom :** Number of independent ways in which a gas molecule may perform rotational motion. The number of degrees of freedom in this case depends on the structure of the molecule.

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(c) **Vibrational Degree of freedom :** Number of independent ways in which a gas molecule may vibrate. It is exhibited at high temperature only.

Degrees of freedom for different gases according to atomicity of gas at low temperature

Atomicity of gas	Translational	Rotational	Total	
Monoatomic **Ex.** Ar, Ne, Ideal gas etc	3	0	3	
Diatomic **Ex.** O_2, Cl_2, N_2 etc.	3	2	5	
Triatomic (linear) **Ex.** CO_2, C_2H_2	3	2	5	
Triatomic (Non–linear) or Polyatomic **Ex.** H_2O, NH_3, CH_4	3	3	6	

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Note:

Degrees of freedom depends on atomicity, structure & temperature of gases.

Total number of degree of freedom = $3N$

Where N = Atomicity of gas molecule

For a gas molecule

Vibrational DoF = Total DoF – (Translational + Rotational) DoF

Example: Diatomic gas molecule like O_2, Cl_2, N_2 (Atomicity $N = 2$)

Translational DoF = 3

Rotational DoF = 2

Vibrational DoF = 1

Total DoF = 3 + 2 + 1 = 6 = 3×2

Example: Triatomic linear gas molecule like CO_2 (Atomicity $N = 3$)

Translational DoF = 3

Rotational DoF = 3

Vibrational DoF = 3

Total DoF = 3 + 3 + 3 = 9 = 3×3

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Law of equipartition of energy

Maxwell's law of equipartition of energy

The total kinetic energy of a gas molecules is equally distributed among its all degree of freedom and the energy associated with each degree of freedom at absolute temperature T is $\frac{1}{2}kT$

For one molecule of gas

Energy related with each degree of freedom = $\frac{1}{2}kT$

Energy related with all degree of freedom = $\frac{f}{2}kT$

$\because \quad \overline{v}_x^2 = \overline{v}_y^2 = \overline{v}_z^2 = \frac{v_{rms}^2}{3}$

$\Rightarrow \quad \frac{1}{2}mv_{rms}^2 = \frac{3}{2}kT$

So energy related with one degree of freedom $= \frac{1}{2}m\frac{v_{rms}^2}{3} = \frac{3}{2}\frac{kT}{3} = \frac{1}{2}kT$

Gas	f	Translational	Rotational	Total
Monoatomic	3	$\frac{3}{2}k_bT$	0	$\frac{3}{2}k_bT$
Diatomic	5 (3+2)	$\frac{3}{2}k_bT$	$\frac{2}{2}k_bT$	$\frac{5}{2}k_bT$
Triatomic Or Polyatomic (Linear)	5 (3+2)	$\frac{3}{2}k_bT$	$\frac{2}{2}k_bT$	$\frac{5}{2}k_bT$
Triatomic Or Polyatomic (Non-Linear)	6 (3+3)	$\frac{3}{2}k_bT$	$\frac{3}{2}k_bT$	$\frac{6}{2}k_bT$

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Note:

For ideal gas if atomicity is not known then take $f = 3$.

Illustration 15

Find rotational energy of 4 moles of N_2 at 127°C.

Solution:

$T = 400\ K$

$\mu = 4$

Rotational energy of N_2 (Diatomic gas) $= \frac{2}{2}\mu RT$

$= 4 \times 8.314 \times 400$

$= 13302\ J$

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Illustration 16

At $127°C$ temperature, the total energy of an ideal gas is E_1. If the temperature is increased to $527°C$, then total energy would be

Solution:

Total energy $E \propto T$

$$\frac{E_1}{T_1} = \frac{E_2}{T_2}$$

$$\Rightarrow \quad \frac{E_1}{400} = \frac{E_2}{800}$$

$$\Rightarrow \quad E_2 = 2E_1$$

Illustration 17

7 moles of triatomic gas is at 400 K then what fraction of total energy is rotational energy ?

Solution:

$$\text{Kinetic energy} = \frac{3}{2}\mu RT$$

$$\text{Rotational energy} = \frac{2}{2}\mu RT$$

$$\text{Total energy} = \frac{3}{2}\mu RT + \frac{2}{2}\mu RT = \frac{5}{2}\mu RT$$

$$\text{Fraction} = \frac{\frac{2}{2}\mu RT}{\frac{5}{2}\mu RT} = \frac{2}{5}$$

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Molar Specific Heat of gas and Mayer's law

Specific Heat

The amount of energy required to raise the temperature of unit mass of a substance by $1°C$ (or $1K$) is called its specific heat. It is represented by s or c.

If the temperature of a substance of mass m changes from T to $T + dT$ when it exchanges an amount of heat dQ with its surroundings then its specific heat is $c = \frac{1}{m}\frac{dQ}{dT}$

Specific heats can be molar or gram.

(i) Gram specific heat

The amount of energy required to raise the temperature of 1 gram of a substance by $1°C$ (or $1K$) is called its specific heat.

If Q amount of heat is required to raise the temperature of mass m (in gram) of a substance by temperature ΔT, then molar specific heat of the gas is given by,

$$c_{gram} = \frac{1}{m}\frac{Q}{\Delta T}$$

Unit : Joule/$kg-°C$, cal/$g-°C$

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(ii) Molar specific heat

The amount of energy required to raise the temperature of 1 mole of a substance by $1°C$ (or $1K$) is called its specific heat.

If Q amount of heat is required to raise the temperature of μ moles of a substance by temperature ΔT, then molar specific heat of the gas is given by,

$$c_{molar} = \frac{1}{\mu}\frac{Q}{\Delta T}$$

Unit : Joule/mol–°C, cal/mol –°C

Note:

- Value of specific heats of gas can vary from zero (0) to infinity.
- Generally two types of specific heats are defined for a gas –
  (a) Specific heat at constant volume (c_V)
  (b) Specific heat at constant pressure (c_P)
- There are many possible processes to give heat to a gas.
  A specific heat can be associated to each such process which depends on the nature of process.

Molar Specific of gas for different processes

(i) Isochoric Process

V = Constant

$$c_V = \frac{Q}{\mu\Delta T}$$

(ii) Isobaric Process

P = Constant

$$c_P = \frac{Q}{\mu\Delta T}$$

(iii) Isothermal Process

$$c = \frac{Q}{\mu\Delta T}$$

$\because$ T = Constant

$\therefore$ $\Delta T = 0$

$c_{isothermal} = \infty$

(iv) Adiabatic Process

$$c = \frac{Q}{\mu\Delta T}$$

$\because$ $Q = 0$

$c_{adiabatic} = 0$

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Mayer's Law

For an ideal gas

$$c_V = \frac{f}{2}R$$

$$c_V = \left(\frac{f}{2}+1\right)R$$

$c_P - c_V = R$

This is known as Mayer's law.

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If gram specific heats are given:

$c_{molar} = M_w\,(c_{gram})$

Where M_w is molar mass of the gas

So, $(c_p)_{molar} = M_w(c_p)_{gram}$

$(c_V)_{molar} = M_w(c_V)_{gram}$

By Mayer's Law,

$c_P - c_V = R$

$\Rightarrow \quad M_w(c_p)_{gram} - M_w(c_V)_{gram} = R$

$\Rightarrow \quad (c_p)_{gram} - (c_V)_{gram} = \dfrac{R}{M_w}$

Gas	f	c_V	c_P	γ
Monoatomic	3	$\frac{3}{2}R$	$\frac{5}{2}R$	$\frac{5}{3} \approx 1.67$
Diatomic	5	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{5} \approx 1.4$
Triatomic or Polyatomic (Linear)	5	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{5} \approx 1.4$
Triatomic or Polyatomic (Non-Linear)	6	$\frac{6}{2}R$	$\frac{8}{2}R$	$\frac{4}{3} \approx 1.33$

Molar specific heat for mixture of gases

Suppose we have two ideal gases of moles μ_1 & μ_2 with specific heats c_1 & c_2 respectively. Now if they are mixed and then heated till their temperature changes by ΔT.

If $Q_1 \rightarrow$ heat required to change the temperature of ideal gas A by ΔT

$Q_2 \rightarrow$ heat required to change the temperature of ideal gas B by ΔT

$Q_{mix} \rightarrow$ total heat required to change the temperature of mixture by ΔT

then from energy conservation

$Q_{mix} = Q_1 + Q_2$

$\Rightarrow \quad \mu_{mix}\, c_{mix}\, \Delta T = \mu_1 c_1 \Delta T + \mu_2 c_2 \Delta T$

$\Rightarrow \quad c_{mix} = \dfrac{\mu_1 c_1 + \mu_2 c_2}{\mu_1 + \mu_2} \qquad (\because\ \mu_{mix} = \mu_1 + \mu_2)$

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If c_v of individual gases are given then

$$(c_V)_{mix} = \frac{\mu_1 c_{V_1} + \mu_2 c_{V_2}}{\mu_1 + \mu_2}$$

If c_P of individual gases are given then

$$(c_P)_{mix} = \frac{\mu_1 c_{P_1} + \mu_2 c_{P_2}}{\mu_1 + \mu_2}$$

$$\gamma_{mix} = \frac{(c_P)_{mix}}{(c_V)_{mix}} = \frac{\mu_1 c_{P_1} + \mu_2 c_{P_2}}{\mu_1 c_{V_1} + \mu_2 c_{V_2}}$$

Illustration 18

3 moles of monoatomic gas is heated at constant pressure from $27°C$ to $227°C$. Find amount of heat required (in terms of R).

Solution:

At constant pressure molar specific of monoatomic gas $c = c_P = \dfrac{5}{2}R$

$\Delta T = 200\ K$

Heat required

$$Q = \mu c_P \Delta T = 3 \times \frac{5}{2} R \times 200 = 1500R$$

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Illustration 19

If 4 moles of diatomic gas is heated at constant volume by giving $300R$ heat. Then calculate change in temperature of the gas.

Solution:

At constant volume

$$c_V = \frac{5}{2}R \qquad \text{(for diatomic gas)}$$

$$\mu c_V \Delta T = Q$$

$$\Rightarrow \quad 4 \times \frac{5}{2} R \times \Delta T = 300R$$

$$\Rightarrow \quad \Delta T = 30\ K$$

Illustration 20

If 4 moles of a monoatomic gas and 2 moles of a diatomic gas are mixed then find value of γ for the mixture.

Solution:

Monoatomic gas:

$$\mu_1 = 4,\ c_{V_1} = \frac{3}{2}R,\ c_{P_1} = \frac{5}{2}R$$

Diatomic gas:

$$\mu_2 = 2,\ c_{V_2} = \frac{5}{2}R,\ c_{P_2} = \frac{7}{2}R$$

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$$\gamma_{mix} = \frac{\mu_1 c_{P_1} + \mu_2 c_{P_2}}{\mu_1 c_{V_1} + \mu_2 c_{V_2}} = \frac{4 \times \frac{5}{2}R + 2 \times \frac{7}{2}R}{4 \times \frac{3}{2}R + 2 \times \frac{5}{2}R} = 1.54$$

Mixture of Gases

Case – I

Here two gases kept in two different vessels are mixed in another vessel.

Here following result may or may not be possible.

$$P_1 + P_2 \neq P_3$$

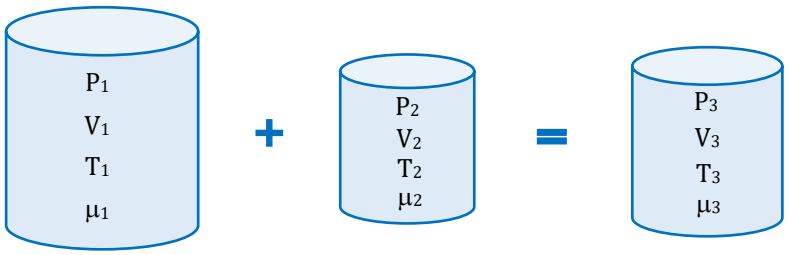
$$V_1 + V_2 \neq V_3$$

$$T_1 + T_2 \neq T_3$$

But we can surely say that

$$\mu_1 + \mu_2 = \mu_3$$

[From mass conservation]



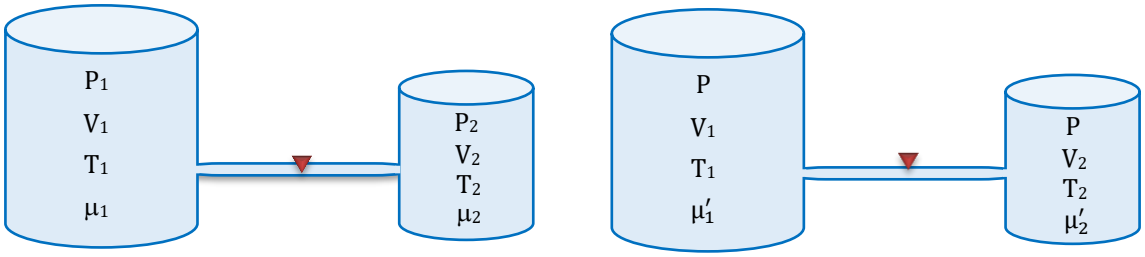
$$\Rightarrow \quad \frac{P_1V_1}{RT_1} + \frac{P_2V_2}{RT_2} = \frac{P_3V_3}{RT_3}$$

$$\Rightarrow \quad \frac{P_1V_1}{T_1} + \frac{P_2V_2}{T_2} = \frac{P_3V_3}{T_3}$$

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Case – II

Here two gases kept in two different vessels connected via a tube as shown. Cross-section area of this tube is very small so its volume can be neglected w.r.t. volume of vessel.



Initially the valve is closed which does not let mixing of gases but after opening of the valve gases will mix until pressure of both vessels become equal.

Also, we can say from mass conservation $\mu_1 + \mu_2 = \mu_1' + \mu_2'$

$$\Rightarrow \quad \frac{P_1V_1}{RT_1} + \frac{P_2V_2}{RT_2} = \frac{PV_1}{RT_1'} + \frac{PV_2}{RT_2'} \qquad \Rightarrow \quad \frac{P_1V_1}{T_1} + \frac{P_2V_2}{T_2} = \frac{PV_1}{T_1'} + \frac{PV_2}{T_2'}$$

Illustration 21

Two gases kept in two identical vessel of volume 30 litres at same temperature 300 K and same pressure 20 atm are mixed in a vessel having same volume V. The resulting pressure of the mixture will be:

Solution:

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$$\frac{P_1V_1}{T_1} + \frac{P_2V_2}{T_2} = \frac{P_3V_3}{T_3}$$

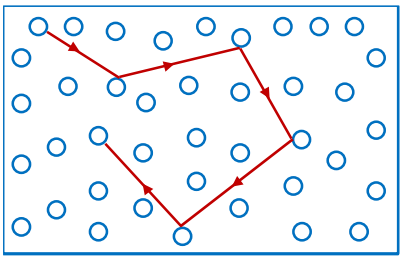
$$\Rightarrow \quad \frac{P_1}{T_1} + \frac{P_2}{T_2} = \frac{P_3}{T_3} \qquad \Rightarrow \quad \frac{20}{300} + \frac{20}{300} = \frac{P}{300} \qquad \Rightarrow \quad P = 40 \text{ atm}$$

Mean Free Path and Real Gases

Mean Free Path

A molecule in its path undergoes a number of collisions so the path traversed by it is not a straight line but somewhat zigzag. Between two successive collisions a molecule travels in a straight line with uniform velocity. As motion is random thus the distance travelled by molecule between two successive collisions is not always equal.

The average distance travelled by a molecule between two successive collisions is called as mean free path (λ_m) of the molecule.



$$\lambda_m = \frac{1}{\sqrt{2}\pi d^2 n}$$

Here $n = \frac{N}{V}$ = No. of molecules per unit volume

d = Diameter of a molecule

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Real Gas

Practically, no gas is ideal at all. They show some deviation from ideal gas behaviour and are known as Real gas.

Main cause of deviation is two postulates that have been assumed in Kinetic Theory of Gases:-

(i) Interaction force between gas molecules are considered negligible

(ii) Size of Gas molecules is considered negligible

But practically they are not negligible.

Compressibility Factor

It is used to express the extent of deviation of a real gas from ideal gas behaviour.

It is denoted by Z.

For a real gas, unlike ideal gas

$$PV_{real} \neq \mu RT$$

Compressibility factor of a real gas is given by

$$Z = \frac{PV_{real}}{\mu RT}$$

Where $P \rightarrow$ Pressure of real gas

$\mu \rightarrow$ no. of moles of real gas

$T \rightarrow$ Temperature of real gas

$V_{real} \rightarrow$ Volume of real gas

For an ideal gas of same mole μ at same temperature T and pressure P, we can write

$$PV_{ideal} = \mu RT$$

Then $$Z = \frac{PV_{real}}{PV_{ideal}} = \frac{V_{real}}{V_{ideal}}$$

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Now we will have three different cases of Z

Case-I $Z = 1$

$V_{real} = V_{ideal}$

Thus real gas behaves as ideal gas.

Case-II $Z > 1$

$V_{real} > V_{ideal}$

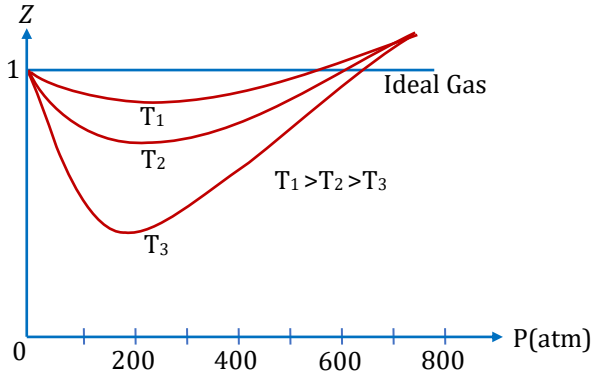
Thus real gas expand with respect to ideal gas and hence we can say positive deviation.

Case-III $Z < 1$

$V_{real} < V_{ideal}$

Thus real gas compress with respect to ideal gas and hence we can say negative deviation.

Compressibility Factor Curve



Thus, we can see from the curve that real gas approaches ideal gas behaviour at **Low Pressure** and **High Temperature**.

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THERMODYNAMICS

Introduction

Branch of physics which deals with the inter–conversion between heat energy and any other form of energy is known as thermodynamics. In this branch of physics, we deal with the processes involving heat, work and internal energy. In this branch of science, the conversion of heat into mechanical work and vice versa is studied.

Thermodynamical System

The system which can be represented in of pressure (P), volume (V) and temperature (T), is known thermodynamic system. A specified portion of matter consisting of one or more substances on which the effects of thermodynamic variables such as temperature, volume and pressure are to be studied, is called a system. e.g. A gas enclosed in a cylinder fitted with a piston is a system.

Heterogeneous System

A system which is not uniform throughout is said to be heterogeneous. e.g. A system consisting of two or more immiscible liquids.

Homogeneous System

A system is said to be homogeneous if it is completely uniform throughout. e.g. Pure solid or liquid.

Surroundings

Anything outside the system, which exchanges energy with the system and which tends to change the properties of the system is called its surroundings.

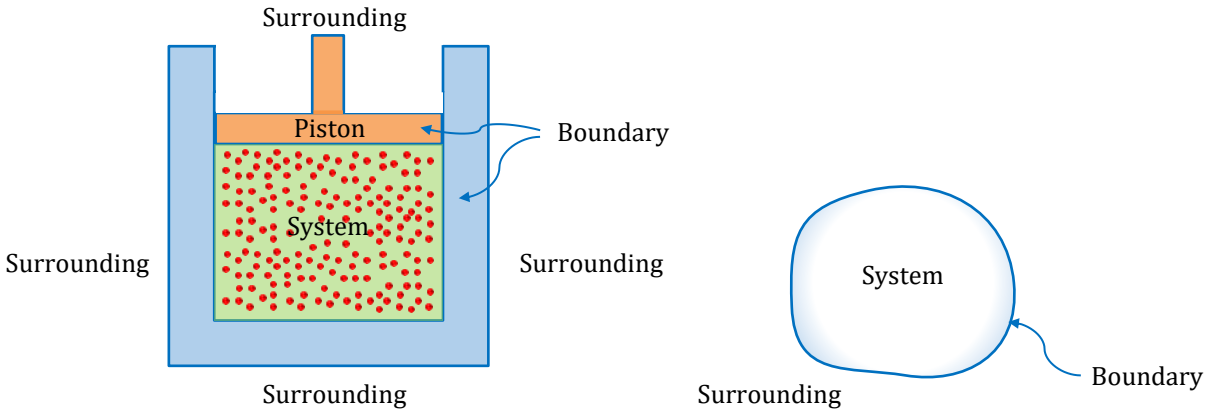
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Universe

The system and its surroundings are together known as the universe.

Boundary

Real or imaginary wall that separates the system from the surroundings



Types of Systems

(1) **Open System :** Any interaction is possible.

(2) **Closed System :** Impenetrable by matter, but other kinds of interactions can occur.

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(3) **Semipermeable :** Penetrable by some species, but not by others; rest interactions possible.
(4) **Insulated System :** Thermal interactions are not possible, but nonthermal interactions can occur.
(5) **Rigid System :** Boundary cannot be mechanically deformed.
(6) **Isolated System :** No interactions can occur.



Closed system



Open system



Isolated System

- **Intensive**
  Independent of the size (or mass) of the system. Example: - pressure, temperature etc.
- **Extensive**
  depends on the size (or mass) of the system. Example: - volume, number of moles etc.

Thermodynamic variables of the system:

(i) Composition (m) (ii) Temperature (T) (iii) Volume (V)
(iv) Pressure (P) (v) Mass

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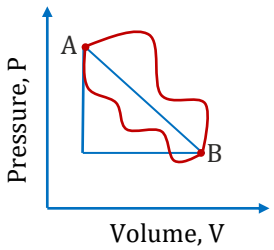
Thermodynamic state

The state of a system can be described completely by composition, temperature, volume and pressure.

If a system is homogeneous and has definite mass and composition, then the state of the system can be described by the remaining three variables namely temperature, pressure and volume. These variables are interrelated by equation $PV = \mu RT$. The thermodynamic state of the system is its condition as identified by two independent thermodynamic variables (P, V or P, T or V, T).

Process

When a system changes from one equilibrium state to another the path of successive states through which the system passes is called a process.



- **Cyclic process**
  Cyclic process is a thermodynamic process in which the system returns to its initial stage after undergoing a series of changes.
- **Non–cyclic process**
  Non–cyclic process is a process in which the system does not return to its initial stage.
- **Quasi–static or equilibrium process**
  Quasi–static is a thermodynamic process which proceeds extremely slowly such that at every instant of time, the temperature and pressure are the same in all parts of the system.

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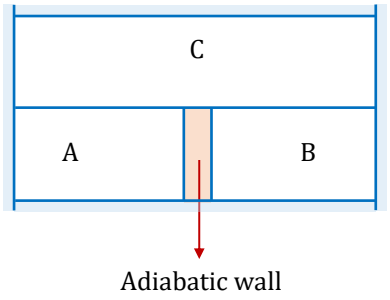
- **Reversible and Irreversible processes**

  A reversible process is one in which the changes in heat and work of direct process from initial to a final state are exactly retraced in opposite sense in the reverse process and the system and surroundings are left in their initial states. The reversibility is an ideal concept and can not be realized in practice.

  The process which is not reversible is the irreversible process. In nature the processes are irreversible.

Zeroth and First Law of Thermodynamics

If objects A and B are separately in thermal equilibrium with a third object C (say thermometer), then objects A and B are in thermal equilibrium with each other. Zeroth law of thermodynamics introduce the concept of temperature. Two objects (or systems) are said to be in thermal equilibrium if their temperatures are the same.



- If two systems are separately in a thermal equilibrium with the third system then they must be in thermal equilibrium with each other.
- ZLOT defines temperature only.
- In measuring the temperature of a body, it is important that the thermometer be in the thermal equilibrium with the body whose temperature is to be measured.

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Heat and Work done in Thermodynamic Process

Work and Heat:

$W \propto Q$ (Heat)

$W = JQ$

Here J = conversion factor

= Joule's constant (mechanical equivalent of heat)

$J = 4.18 \frac{joule}{calorie}$ (Dimensionless)

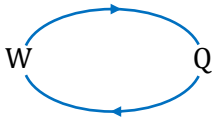
Note

- 1 cal = 4.18 J ≃ 4.2
- 1 joule = 0.24 cal = $\frac{1}{4.2}$ cal
- 1 cal > 1 joule

Special Point:

(a) Work and heat both are the form of energy. Work is ordered energy where as heat is disordered form of energy.

(b) Conversion of mechanical work into heat is completely possible but its reverse is not true.



100% ✓

100% ×

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Two Types of Conversions:

(a) Mechanical work → Heat

$$PE: \frac{mgh}{J} = msd\theta(cal)\left\{\frac{1}{J} = 0.24\right\}$$

$$KE: \frac{1}{2}\frac{mv^2}{J} = msd\theta(cal)$$

Rotational kinetic energy = $\frac{1}{2}\frac{I\omega^2}{J} = msd\theta(cal)$

(b) Heat → mechanical work

Illustration 22

A block of 100 *gm* is sliding on a rough horizontal surface. If speed of block reduces from 10 *m*/sec to 5 *m*/sec then find heat generated in this process.

Solution:

Heat generated = change in *KE*

$\Rightarrow$ Heat generated = $\frac{1}{2}m\left[v_1^2 - v_2^2\right] = \frac{1}{2}\times 100\times 10^{-3}\left[(10)^2 - (5)^2\right] = \frac{75}{2}\times 10^{-1} = 3.75$ joule

Illustration 23

A bullet of mass 84 *gm* is moving with a velocity of 200 *m/s*. It collides with wall by which one fourth of the kinetic energy is converted into heat. Find the heat developed by bullet in calorie.

Solution:

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$$Q = \frac{1}{4}(W) = \frac{1}{4}\left(\frac{1}{2}\frac{mv^2}{J}\right) = \frac{1}{4}\times\frac{1}{2}\times\frac{84}{1000}\times\frac{200\times 200}{4.2} = 100cal.$$

Illustration 24

A man gains 100 kcal of heat by eating bread. His efficiency is 25% and mass is 70 *kg*. Calculate the height to which he can climb.

Solution:

$$\frac{25}{100}\times Q = W$$

$$\frac{25}{100}\times 100\times 10^3 = \frac{Mgh}{J}$$

$$25\times 10^3 = \frac{70\times 10\times h}{4.2}$$

h = 150 meter

Internal Energy

Internal energy of a system is the energy possessed by the system due to molecular motion and molecular configuration. The energy due to molecular motion is called internal kinetic energy (U_k) and that due to molecular configuration is called internal potential energy (U_p).

$dU = dU_k + dU_p$

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If there no intermolecular forces, then $dU_p = 0$ and $dU = dU_k = m\, c_v\, dT$ [ideal gas]

c_v = Specific heat at constant volume and dT = Infinitesimal change in temperature

m = Mass of system M = Molecular weight

Molar heat capacity $C_v = Mc_v$ For μ–moles of ideal gas = $dU = \mu C_v dT = \frac{m}{M} C_v dT$

Internal energy in the absence of inter–molecular forces is simply the function of temperature and state only, it is independent of path followed. $\Delta U = U_f - U_i$

U_i = Internal energies in initial state and U_f = Internal energies in final state

Properties of Internal Energy

1. It depends mainly on states of matter and temperature of system.
2. For ideal gas it a function of temperature, identity and moles.
3. For real gas it is a function of T and V or T and P as well as identity and moles.
4. Absolute value of internal energy is impossible to determine, so it is generally expressed as a difference between the two states of system, or by applying zero value to a reference state.
5. Internal energy is an Exact quantity or a state function.
6. Internal energy of a system is the energy possessed by the system due to molecular motion and molecular configuration.
7. The energy due to molecular motion is called internal kinetic energy (U_k) and that due to molecular configuration is called internal potential energy (U_p).

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Sign Convention used in Physics Thermodynamics

FLOT $\Delta Q = \Delta U + \Delta W$ or $[Q = \Delta U + W]$

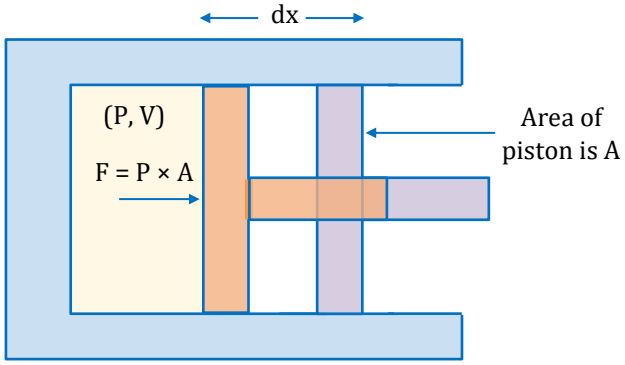
Rules:

1. Q/ΔQ =
   - +ve ⇒ heat given to system / heat absorbed by system.
   - –ve ⇒ heat rejected / released by system or heat taken out from system
2. ΔU/dU =
   - +ve ⇒ U ↑ (internal energy) T ↑ (increases)
   - –ve ⇒ U ↓ (internal energy) T ↓ (decreases)
3. W/ΔW =
   - V ↑ (expansion) dV = +ve ⇒ ΔW = PdV = +ve (work is done by system)
   - V ↓ (compression) ; dV = –ve ⇒ ΔW = PdV = –ve (work is done on system)

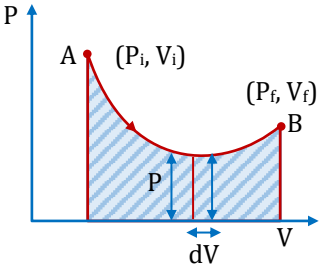
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Work done by thermodynamic system

One of the simple example of a thermodynamic system is a gas in a cylinder with a movable piston.



- If the gas expands against the piston Gas exerts a force on the piston and displace it through a distance and does work on the piston.
- If the piston compresses the gas When piston moved inward, work is done on the gas.
- The work associated with volume changes
  If pressure of gas on the piston = P.
  Then the force on the piston due to gas is $F = PA$
- When the piston is pushed outward an infinitesimal distance dx, the work done by the gas is $dW = F \times dx = PA\,dx$
  The change in volume of the gas is dV = Adx,



$\therefore \quad dW = PdV$

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For a finite change in volume from Vi to Vf, this equation is then integrated between V_i to V_f to find the net work done = $W = \int_{V_i}^{V_f} dW = \int_{V_i}^{V_f} PdV$

Hence the work done by a gas is equal to the area under P–V graph.

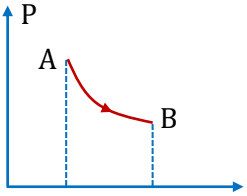
Work Done in Various Thermodynamic Process:

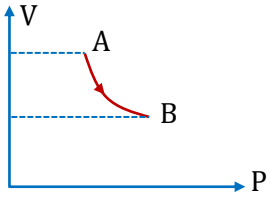
(1) Mathematical method $W = \int_{V_i}^{V_f} PdV$

Process	W
Cyclic	Area of the closed curve
Isochoric	0
Isothermal	$\mu RT \log_e\left[\frac{V_f}{V_i}\right] = \mu RT \log_e\left[\frac{P_i}{P_f}\right]$
Adiabatic	$\frac{\mu R\left(T_f - T_i\right)}{1-\gamma}$
Isobaric	$P\,(V_f - V_i) = \mu R(T_f - T_i)$
Polytropic process	$\frac{\mu R\left(T_f - T_i\right)}{1-x}$

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(2) Graphical method ΔW = Area enclosed between PV curve and Volume axis.
Example: -



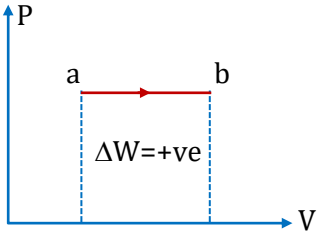


Work done $\Rightarrow W_{A\to B}$ = Area shaded.

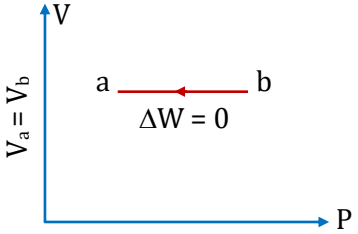
Sign concept: -

- Expansion work $\rightarrow$ +ve (work by system)/volume of a gas increases
- Compression work $\rightarrow$ –ve (work on system)/ volume of a gas decreases

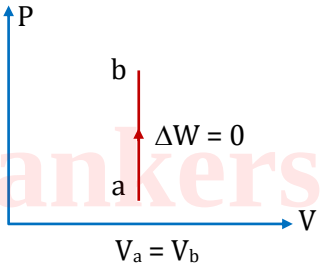
(i)



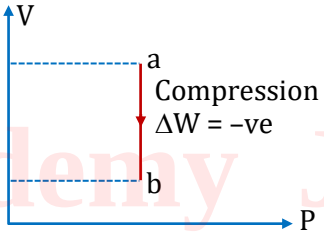
(ii)



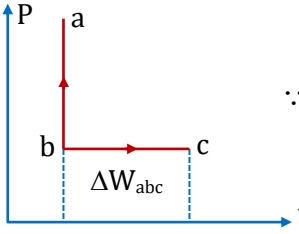
(iii)



(iv)



(v)

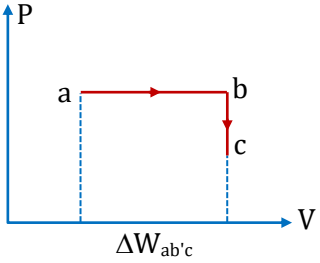


$W_{ba} = 0$

$\because V_a = V_b$

W_{bc} = +ve

(vi)



W_{ab} = +ve

$W_{bc} = 0$

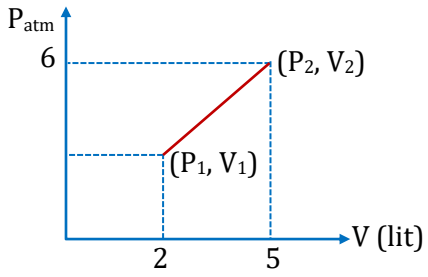
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Note:

Area enclosed between PV curve and volume axis.

Illustration 25

A system changes from the state (P_1,V_1) to (P_2,V_2) as shown in figure. What is the work done by the system?



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Solution:

W = Area between curve line and volume axis.

$$W = \frac{1}{2}\times\left[(2+6)\times10^{5}\right]\times\left(3\times10^{-3}\right)$$

W = 1200 joule

Note:

(a) work and heat are both path functions, it depends on path

(b) internal energy depends on initial and final state, it is a state function.

- **Order of work:**

  $\Delta W_C > \Delta W_b > \Delta W_a$

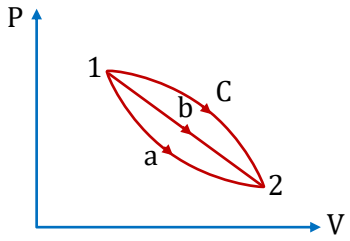
- **Order of internal energy:**

  $dU_a = dU_b = dU_c$ (constant)

- **Order of heat:**

  $\Delta Q_c > \Delta Q_b > \Delta Q_a$

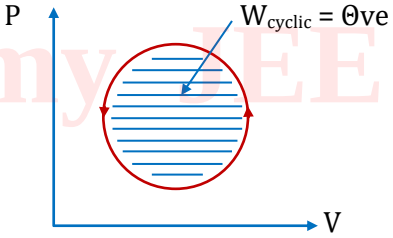
  $\therefore\ Q = W + \Delta U$



Cyclic Process:

1. Work done in cyclic process:

(a) ΔW = Area enclosed between closed loop



(b) In P-V graph

Direction → C.W. ⇒ ΔW = +ve

Direction → ACW ⇒ ΔW = –ve

(c) In V-P graph

Direction → C.W. : ΔW = –ve

Direction → ACW ⇒ ΔW = +ve

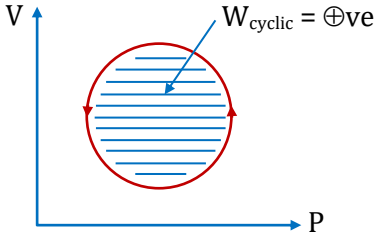
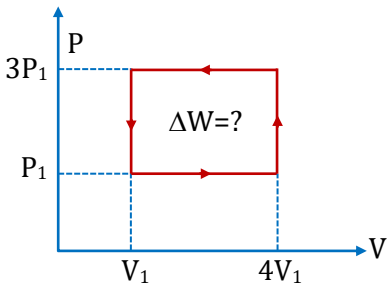


Illustration 26



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Solution:

1. Area = $2P_1 \times 3V_1 = 6P_1V_1$

   Work = $-6P_1V_1$ Joule [$\because$ Cycle is ACW]

2. In cyclic process: $dU = 0$

3. Heat: $\Delta Q = \Delta W$ [By FLOT $\Delta Q = dU + \Delta W$]

Illustration 27

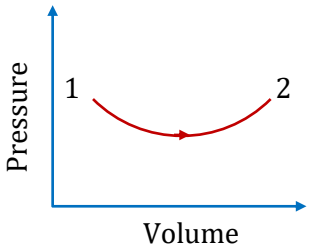
Consider the process on a system shown in figure. During the process, the work done by the system.

(A) Continuously increases

(B) Continuously decreases

(C) First increases then decreases

(D) First decreases then increases



Solution:

$W = \int PdV$

As volume is continuously increasing. So, work done will keep on increasing.

Illustration 28

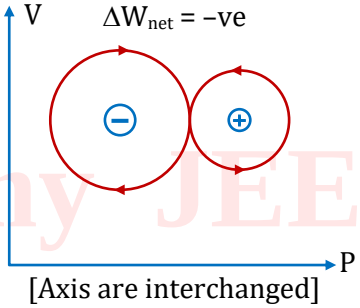
The net work done in the given process is

(A) Positive

(B) Negative

(C) zero

(D) None of these



Solution:

V–P graph

$W_1 = -ve$ (CW)

$W_2 = +ve$ (ACW)

$W_{net} = -ve$

First law of thermodynamics

If some quantity of heat is supplied to a system capable of doing external work, then the quantity of heat absorbed by the system is equal to the sum of the increase in the internal energy of the system and the external work done by the system.

$$\boxed{\Delta Q = \Delta U + \Delta W} \quad \text{or} \quad \boxed{Q = W + \Delta U}$$

- This law is applicable to every process in nature.
- Thermodynamic force is non-conservative force.
- The first law of thermodynamics introduces the concept of internal energy.
- The first law of thermodynamics is based on the law of conservation of energy.
- ΔQ, ΔU and ΔW must be expressed in the same units (either in units of work or in units of heat).
- This law is applicable to all the three phases of matter, i.e., solid, liquid and gas.
- dU is a characteristic of the state of a system, it may be any type of internal energy–translational kinetic energy, vibrational, rotational kinetic energy, binding energy etc.

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Illustration 29

In a thermodynamic process, work done on the system is 100 J and heat given to system is 500 cal. Then calculate change in internal energy of system.

Solution:

As from FLOT : $\Delta Q = \Delta U + \Delta W$

$\Rightarrow$ $(500 \times 4.2) = \Delta U - 100$

$\Delta U = 2100 + 100 = 2200\ J = 524$ cal ΔU is positive so temperature will increase

Illustration 30

A gas under constant pressure of $4.5\times10^5\ Pa$ when subjected to 800 kJ of heat, changes the volume from $0.5m^3$ to $2.0\ m^3$. Find the change in internal energy of the gas?

Solution:

$\Delta U = \Delta Q - \Delta W = \Delta Q - P\Delta V$

$\Delta U = 800 \times 10^3 - 4.5 \times 10^5\ (2 - 0.5) = 1000(800 - 675)$

$\Rightarrow$ $\Delta U = 125\ kJ$

Illustration 31

If Q amount of heat is given to a diatomic ideal gas in process in which the gas perform a work $\frac{2Q}{3}$ its surrounding. Find the molar heat capacity (in terms of R) for the process.

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Solution:

According to FLOT

$Q = W + \Delta U$

$$Q = \frac{2}{3}Q + \mu C_v \Delta T$$

$$\frac{Q}{3} = \mu \times \frac{5}{2} R\Delta T$$

$$\mu C\Delta T = \frac{15}{2}\mu R\Delta T$$

$$C = \frac{15}{2}R$$

Illustration 32

The pressure of one mole monoatomic gas increases linearly from $4 \times 10^5\ Nm^{-2}$ to $8 \times 10^{+5}\ Nm^{-2}$ when its volume increases from $0.2m^3$ to $0.5\ m^3$. Calculate.

(i) Work done by the gas, (ii) Increase in the internal energy,

(iii) Amount of heat supplied, (iv) Molar heat capacity of the gas $R = 8.31\ J\ mol^{-1}\ K^{-1}$

Solution:

$P_1 = 4 \times 10^5\ Nm^{-2}$

$P_2 = 8 \times 10^{+5}\ Nm^{-2}$, $V_1 = 0.2\ m^3$, $V_2 = 0.5\ m^3$

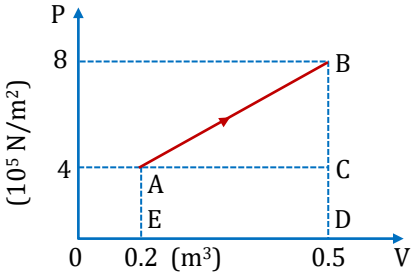
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(i) Work done by the gas

= Area under P–V graph (Area ABCDEA)

$$(AE + BD) \times AC = \frac{1}{2}(4 \times 10^5 + 8 \times 10^5) \times (0.5 - 0.2)$$

$$= \frac{1}{2} \times 12 \times 10^5 \times 0.3 = 1.8 \times 10^5\ J$$



(ii) Increase in internal energy

$$\Delta U = C_V(T_2 - T_1) = \frac{C_V}{R}R(T_2 - T_1) = \frac{C_V}{R}(P_2V_2 - P_1V_1)$$

For monoatomic gas $C_V = \frac{3}{2}R$

$$\therefore \quad \Delta U = \frac{3}{2}[(8 \times 10^5 \times 0.5) - (4 \times 10^5 \times 0.2)] = \frac{3}{2}[4 \times 10^5 - 0.8 \times 10^5] = 4.8 \times 10^5\ J$$

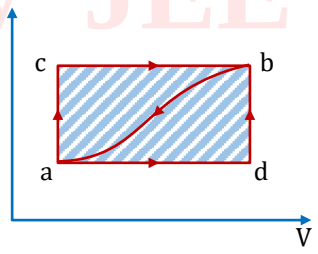
(iii) $Q = \Delta U + W = 4.8 \times 10^5 + 1.8 \times 10^5 = 6.6 \times 10^5\ J$

(iv) $$C = \frac{Q}{\mu\Delta T} = \frac{QR}{\mu R\Delta T} = \frac{QR}{(P_2V_2 - P_1V_1)} = \frac{6.6 \times 10^5 \times 8.31}{1 \times 3.2 \times 10^5} \ 17.14\ J/\text{mol}\ K$$

Illustration 33

As shown in figure when a system is taken from state a to state b, along the path $a \to c \to b$, 60 J of heat flow into the system, and 30 J of work is done:

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(i) How much heat flows into the system along the path $a \to d \to b$ if the work is 10 J.

(ii) When the system is returned from b to a along the curved path, the work done by the system is –20 J. Does the system absorb or liberate heat, and how much?

(iii) If, $U_a = 0$ and $U_d = 22\ J$, find the heat absorbed in the process $a \to d$ and $d \to b$.

Solution:

For the path acb $\Delta U = Q - W = 60 - 30 = 30\ J$ or $U_b - U_a = 30\ J$

(i) Along the path adb $Q = \Delta U + W = 30 + 10 = 40\ J$

(ii) Along the curved path ba $Q = (U_a - U_b) + W = (-30) + (-20) = -50\ J$, heat liberates from system

(iii) $Q_{ad} = U_d - U_a + W_{ad}$

but $W_{ad} = W_{adb} - W_{db} = 10 - 0 = 10$ Hence $Q_{ad} = 22 - 0 + 10 = 32\ J$

and $Q_{db} = U_b - U_d + W_{db} = 30 - 22 + 0 = 8\ J$

Illustration 34

1 kg of water at 373 K is converted into steam at same temperature. Volume of 1 cm^3 of water becomes 1671 cm^3 on boiling. What is the change in the internal energy of the system if the latent heat of vaporisation of water is 5.4×10^5 cal kg^{-1}?

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Solution:

Volume of 1 kg of water = 1000 cm^3 = 10^{-3} m^3, Volume of 1 kg of steam = 10^3 × 1671 cm^3 = 1.671 m^3 Change in volume ΔV = (1.671 – 10^{-3}) m^3 = 1.670 m^3, Pressure, P = 1 atm. = 1.01 × 10^5 $N\,m^{-2}$ In expansion work done,

$W = P\Delta V = 1.01 \times 10^5 \times 1.67\ J = \frac{1.686 \times 15^5}{4.2}$ cal = 4.015 × 10^4 cal

But $\Delta U = Q - W$ (first law of thermodynamics)

or ΔU = (5.4 × 10^5 – 0.4015 × 10^5) cal = 4.9985 × 10^5 cal

Isochoric Process

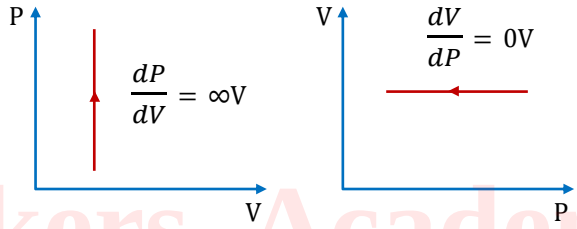
Isometric or Isochoric Process

Isochoric process is a thermodynamic process that takes place at constant volume of the system, but pressure and temperature varies for change in state of the system.

Equation of state P/T = constant (P and T are variable, V is constant)

Work done In this process volume remains constant so $dV = 0 \Rightarrow W = \int_{V_i}^{V_f} PdV = 0$

Form of first Law $Q = \Delta U$



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It means whole of the heat supplied is utilized for change in internal energy of the system. $Q = \Delta U = \mu C_v \Delta T$

Slope of the *P–V* curve $\frac{dP}{dV} = \infty$

Specific heat or molar heat capacity at constant volume (C_V)

The quantity of heat required to raise the temperature of 1-gram mole of gas through 1 °C at constant volume is equal to the specific heat at constant volume.

- A gas enclosed in a cylinder having rigid walls and a fixed piston. When heat is added to the gas, there would be no change in the volume of the gas.
- When a substance melts, the change in volume is negligibly small. So, this may be regarded as a nearly isochoric process.
- Heating process in pressure cooker is an example of isometric process.

Isobaric Process

Isobaric process is a thermodynamic process that takes place at constant pressure, but volume and temperature varies for change in state of the system.

- **Equation of state** V/T = Constant or $V \propto T$
- **Work done** In this process pressure remains constant $\therefore\ dP = 0$

  Work done $W = \int_{V_i}^{V_f} PdV = P(V_f - V_i)$

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$\Rightarrow$ Fraction of Heat given which –

(i) is converted into internal energy $\boxed{\frac{\Delta U}{\Delta Q}=\frac{1}{\gamma}}$

(ii) does work against external pressure $\boxed{\frac{\Delta W}{\Delta Q}=1-\frac{1}{\gamma}}$

- **Form of first Law**

  $Q = \Delta U + P\,(Vf - Vi)$

  $C_p\,(T_f - T_i) = \mu C_v\,(T_f - T_i) + P\,(V_f - V_i)$

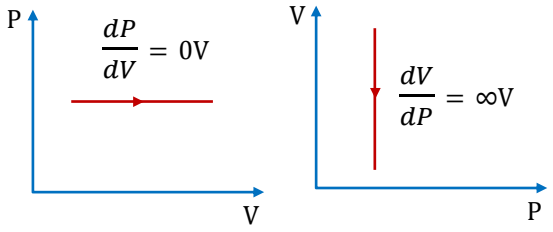
  It is clear that heat supplied to the system is utilized for:

  (i) Increasing internal energy and

  (ii) Work done against the surrounding atmosphere.

- **Slope of the PV curve:**

$$\left(\frac{dP}{dV}\right)_{isobaric}=0$$



Isothermal Process

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In this process pressure and volume of system change but temperature remains constant. In an isothermal process, the exchange of heat between the system and the surroundings is allowed. Isothermal process is carried out by either supplying heat to the substance or by extracting heat from it. A process has to be extremely slow to be isothermal.

- **Equation of state:** $P\,V$ = constant (μRT) (T is constant)
- **Work Done**

  Consider μ moles of an ideal gas, enclosed in a cylinder, at absolute temperature T, fitted with a frictionless piston. Suppose that gas undergoes an isothermal expansion from the initial state (P_1, V_1) to the final state

  (P_2, V_2) Work done: $W=\int_{V_1}^{V_2} P dV \qquad \therefore PV=\mu RT$ then $P=\frac{\mu RT}{V}$

  $$\therefore\ W=\int_{V_1}^{V_2}\frac{\mu RT}{V}dV=\mu RT\int_{V_1}^{V_2}\frac{dV}{V}=\mu RT\left[\log_e V\right]_{V_1}^{V_2}=\mu RT\left[\log_e V_2-\log_e V_1\right]=\mu RT\log_e\left[\frac{V_2}{V_1}\right]$$

  $$\Rightarrow W = 2.303\,\mu RT\log_{10}\left[\frac{P_1}{P_2}\right] \qquad \left[\because P_1V_1=P_2V_2\right]$$

- **Form of First Law**

  There is no change in temperature and internal energy of the system depends on temperature only

  So $\Delta U = 0$, $Q = 2.303\,\mu RT\log_{10}\left[\frac{V_2}{V_1}\right]$

  It is clear that whole of the heat energy supplied to the system is utilized by the system in doing external work. There is no change in the internal energy of the system.

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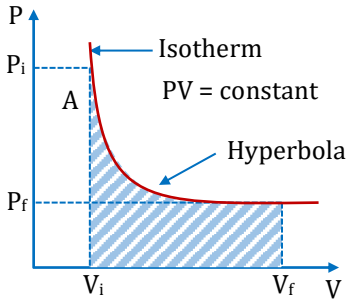
- **Slope of the isothermal curve**

  For an isothermal process, PV = constant

  Differentiating, $PdV + VdP = 0$

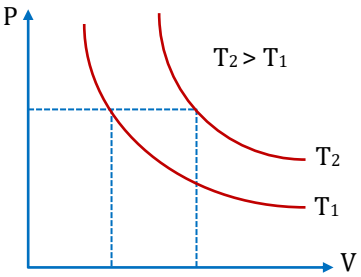
  $\Rightarrow \quad VdP = -PdV \quad \Rightarrow \quad \frac{dP}{dV} = -\frac{P}{V}$

  Slope of isothermal curve, $\left[\frac{dP}{dV}\right]_{isothermal} = -\frac{P}{V}$



For a given system:

- The product of the pressure and volume of a given mass of a perfect gas remains constant in an isothermal process.
- Boyle's law is obeyed in an isothermal process.
- A graph between pressure and volume of a given mass of a gas at constant temperature is known as isotherm or isothermal curve of the gas.
- Two isotherms for a given gas at two different temperatures T_1 and T_2 are shown in figure.



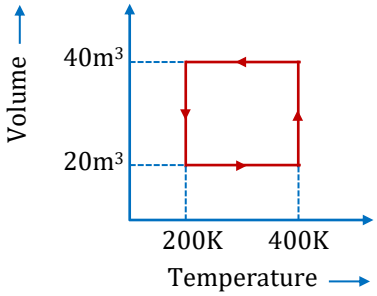
At const P ; V ∝ T so $T_2 > T_1$

- The curves drawn for the same gas at different temperatures are mutually parallel and do not cut each other.
- If two isotherms intersect each other at a single point we get same value of P and V at intersection point. $PV = \mu RT_1$ for temperature T_1 and $PV = \mu RT_2$ for temperature T_2.
  It means $T_1 = T_2$ which is not possible.
- An ideal gas enclosed in a conducting cylinder fitted with a conducting piston.
- Let the gas be allowed to expand very–very slowly. This shall cause a very slow cooling of the gas, but heat will be conducted into the cylinder from the surrounding. Hence the temperature of the gas remains constant. If the gas is compressed very–very slowly, heat will be produced, but this heat will be conducted to the surroundings and the temperature of the gas shall remain constant.
- The temperature of a substance remains constant during melting. So, the melting process is an isothermal process.
- Boiling is an isothermal process, when a liquid boils, its temperature remains constant.
- If sudden changes are executed in a vessel of infinite conductivity then they will be isothermal.

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Illustration 35

Following figure is given for 1 mole gas.



Then find out net work done in the cyclic process.

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Solution:

Work done = $W_{AB} + W_{BC} + W_{CD} + W_{DA}$

$= 0 + 2.303\mu RT_B \log_{10}\left(\frac{V_C}{V_B}\right) + 0 + 2.303\mu RT_D \log_{10}\left(\frac{V_A}{V_D}\right)$

$= 2.303 \times \mu RT\left[T_B \log_{10}\left(\frac{V_C}{V_B}\right) + T_D \log\left(\frac{V_A}{V_D}\right)\right]$

$= 2.303 \times 1 \times 8.314\left[400\log_{10}\left(\frac{40}{20}\right) + 200\log\left(\frac{20}{40}\right)\right]$

$= 2.303 \times 8.314\,[400\log_{10} 2 - 200\log_{10} 2]$

$= 2.303 \times 8.314 \times 200 \times \log_{10}2$

$= 2.303 \times 8.314 \times 200 \times 0.3010$

$= 1152.65\ J$

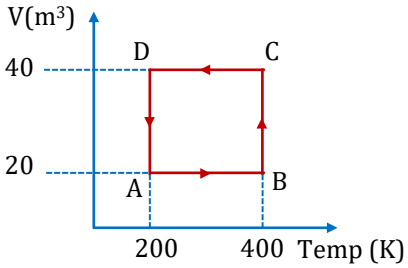
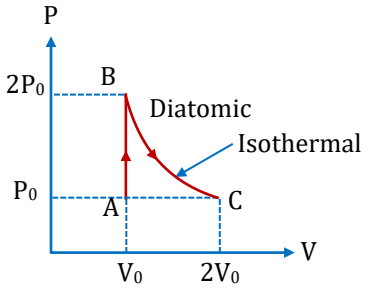


Illustration 36

P-V curve of a diatomic gas is shown in the figure.

Find the total heat given to the gas in the process *AB* and *BC*.



Solution:

From first law of thermodynamics

$\Delta Q_{ABC} = \Delta U_{ABC} + W_{ABC}$

$W_{ABC} = W_{AB} + W_{BC} = 0 + nRT_B \ln\frac{V_C}{V_B} = nRT_B \ln\frac{2V_0}{V_0}$

$= nRT_B \ln 2 = 2P_0V_0 \ln 2$

$\Delta U = nC_V\Delta T = \frac{5}{2}(2P_0V_0 - P_0V_0)$

$\Rightarrow \quad \Delta Q_{ABC} = \frac{5}{2}P_0V_0 + 2P_0V_0 \ln 2$

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Adiabatic Process

It is that thermodynamic process in which pressure, volume and temperature of the system do change but there is no exchange of heat between the system and the surroundings.

A sudden and quick process will be adiabatic since there is no sufficient time available for exchange of heat so process adiabatic.

Equation of state : $PV = \mu RT$

Equation for adiabatic process PV^γ = constant, $TV^{\gamma-1}$ = constant, $T^\gamma p^{1-\gamma}$ = constant

Work done

Let initial state of system is (P_1, V_1, T_1) and after adiabatic change final state of system is (P_2, V_2, T_2) then we can write $P_1V_1^\gamma = P_2V_2^\gamma = K$ (here K is const.)

So $W = \int_{V_1}^{V_2} PdV = K\int_{V_1}^{V_2} V^{-\gamma}dV = K\left(\frac{V^{-\gamma+1}}{-\gamma+1}\right)_{V_1}^{V_2} = \frac{K}{(-\gamma+1)}\left[V_2^{-\gamma+1} - V_1^{-\gamma+1}\right]\ (\because K = P_1V_1^\gamma = P_2V_2^\gamma)$

$\Rightarrow \quad W = \frac{1}{(\gamma-1)}\left[P_1V_1^\gamma V_1^{-\gamma}.V_1 - P_2V_2^\gamma V_2^{-\gamma}.V_2\right] = \frac{1}{(\gamma-1)}\left[P_1V_1 - P_2V_2\right] \Rightarrow W = \frac{\mu R}{(\gamma-1)}(T_1 - T_2) \quad (\because PV = \mu RT)$

Form of first law : $\Delta U = -\Delta W$

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It means the work done by an ideal gas during adiabatic expansion (or compression) is on the cost of change in internal energy proportional to the fall (or rise) in the temperature of the gas.

If the gas expands adiabatically, work is done by the gas. So, W_{adia} is positive.

The gas cools during adiabatic expansion and $T_1 > T_2$.

If the gas is compressed adiabatically, work is done on the gas. So, W_{adia} is negative.

The gas heats up during adiabatic compression and $T_1 < T_2$.

Slope of the adiabatic curve

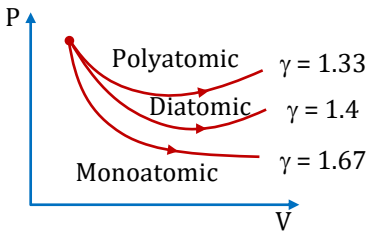
For an adiabatic process, PV^{γ} = constant

Differentiating, $\gamma PV^{\gamma-1}\,dV + V^{\gamma}\,dP = 0$

$$\Rightarrow \quad V^{\gamma}\,dP = -\gamma PV^{\gamma-1}\,dV \Rightarrow \frac{dP}{dV} = -\frac{\gamma PV^{\gamma-1}}{V^{\gamma}} = -\gamma\frac{P}{V} = \gamma\left(-\frac{P}{V}\right)$$

Slope of adiabatic curve, $\left[\frac{dP}{dV}\right]_{adiabatic} = -\frac{\gamma P}{V}$

(Adiabatic expansion of gas)



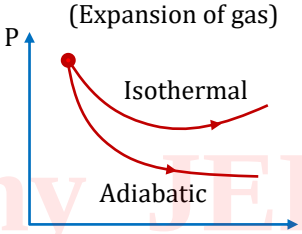
Magnitude of slope of adiabatic curve is greater than the slope of isotherm

$$\left|\frac{dP}{dV}\right|_{adiabatic} = \gamma\frac{P}{V} = \gamma\left|\frac{dP}{dV}\right|_{isothermal}$$

$$\Rightarrow \quad \frac{\text{Slope of adiabatic changes}}{\text{Slope of isothermal changes}} = \gamma$$

Since γ is always greater than one so an adiabatic curve is steeper than an isothermal curve

(Expansion of gas)



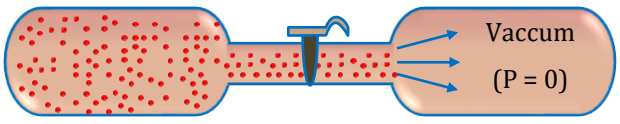
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Examples of adiabatic process

- A gas enclosed in a thermally insulated cylinder fitted with a non–conducting piston. If the gas is compressed suddenly by moving the piston downwards, some heat is produced. This heat cannot escape from the cylinder. Consequently, there will be an increase in the temperature of the gas.
- If a gas is suddenly expanded by moving the piston outwards, there will be a decrease in the temperature of the gas.
- Bursting of a cycle tube.
- Propagation of sound waves in a gaseous medium.
- In diesel engines burning of diesel without spark plug is done due to adiabatic compression of diesel vapour and air mixture

Free Expansion

Take a thermally insulated bottle with ideal gas at some temperature T_1 and, by means of a pipe with a stopcock, connect this to another insulated bottle which is evacuated. If we suddenly open the stopcock, the gas will rush from the first bottle into the second until the pressures are equalized.



Free expansion of a gas

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Experimentally, we find that this process of free expansion does not change the temperature of the gas – when the gas attains equilibrium and stops flowing, the final temperature of both bottles are equal to the initial temperature T_1. This process is called a free expansion.

The change in the internal energy of the gas can be calculated by applying the first law of thermodynamics to the free–expansion process.

The process is adiabatic because of the insulation, So $Q = 0$.

No part of the surroundings moves so the system does no work on its surroundings.

For ideal gas $(\gamma W)_{ext.}$ = Work done against external atmosphere $= P\,dV = 0$

$(\because P = 0) \Rightarrow$ because vacuum

$(\Delta W)_{int.}$ = Work done against internal molecular forces = 0

$\Delta Q = \Delta U + \Delta W \Rightarrow 0 = \Delta U + 0$

The internal energy does not change $dU = 0$ So U and T are constant.

The initial and final states of this gas have the same internal energy.

Which implies that the internal energy of an ideal gas does not depend on the volume at all.

The free–expansion process has led us to the following conclusion:

The internal energy $U(T)$ of an ideal gas depends only on the temperature.

Illustration 37

Temperature and pressure of air in tyre tube is $27°C$ and 8 atm respectively. If tube suddenly burst then calculate final temperature of air. [$\gamma = 1.5$)

Solution:

$$P_1^{1-\gamma}T_1^{\gamma} = P_2^{1-\gamma}T_2^{\gamma}$$

$$(8)^{1-\gamma}(300)^{\gamma} = (1)^{1-\gamma}(T_2)\gamma$$

$T_2 = 150\ K$

$T_2 = 150 - 273 = -123°C$

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Illustration 38

Find the order of magnitude of slope of PV curve in different process.

Solution:

Slope IC > AD > IT > IB

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$\infty \quad \frac{\gamma P}{V} \quad \frac{P}{V} \quad 0$$

IC [Isochoric process] or IM [Iso volumetric process] or Isometric process

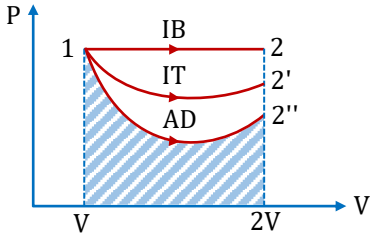
Illustration 39

A gas is expanded to double of its initial volume through three different process (IB, IT, AD) then find

(a) Order of work done Ans. $IB > IT > AD$

(b) Order of final pressure Ans. $IB > IT > AD$

(c) Order of final temperature Ans. $IB > IT > AD$



Solution:

(a) $IB > IT > AD$ (b) $IB > IT > AD$ (c) $IB > IT > AD$

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Illustration 40

A gas is compressed to half of its initial volume along three different process, then

(a) Order of work done $AD > IT > IB$

(b) Order of final pressure $AD > IT > IB$

(c) Order of temperature $AD > IT > IB$

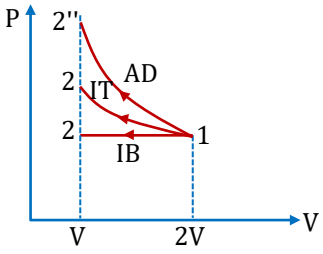


Illustration 41

An ideal gas $\left(\gamma = \frac{5}{3}\right)$ is adiabatically compressed from to 80 cm^3 to 10 cm^3. If the initial pressure is P then find the final pressure?

Solution:

$\therefore \quad P_1V_1^{\gamma} = P_2V_2^{\gamma}$

$$\frac{P_1}{P_2} = \left(\frac{V_2}{V_1}\right)^{\gamma}$$

$$\frac{P}{P_2} = \left(\frac{10}{80}\right)^{\frac{5}{3}}$$

$P_2 = 32P$

Polytropic Process

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General expression for C (C_P or C_V) in the process PV^x = constant $C = \frac{R}{\gamma - 1} + \frac{R}{\gamma - 1}$

For isobaric process P = constant so $x = 0$ $\therefore C = C_P = \frac{R}{\gamma - 1} + R = C_V + R$

For isothermal process, PV = constant so $x = 1$ $\therefore C = \infty$

For adiabatic process PV^{γ} = constant so $x = g$ $\therefore C = 0$

Values of f, U, C_V, C_P and γ for different gases are shown in table below.

Atomicity of gas	f	C_v	C_P	γ
Monoatomic	3	$\frac{3}{2}R$	$\frac{5}{2}R$	$\frac{5}{3} = 1.67$
Diatomic, Triatomic and Triatomic linear (at normal temperature)	5	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{5} = 1.4$
Poly atomic Triangular Non-linear	6	$\frac{6}{2}R = 3R$	$\frac{8}{2}R = 4R$	$\frac{4}{3} = 1.33$

Illustration 42

An ideal gas is taken through a process in which pressure and volume vary as $P = kV^2$. Show that the molar heat capacity of the gas for the process is given by $C = C_v + \frac{R}{3}$

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Solution:

$PV^{-2} = K$

As $C = C_v + \frac{R}{(1-x)}$

$x = -2 = C_V + \frac{R}{1-(-2)} \Rightarrow C = C_v + \frac{R}{3}$

Illustration 43

Match the column-I with column-II

	Column-I		Column-II
(A)	Adiabatic expansion	(p)	$Q = 0$
(B)	Adiabatic free expansion	(q)	$\Delta U = 0$
(C)	Isobaric expansion	(r)	$W = 0$
(D)	Isothermal expansion	(s)	$Q = W + \Delta U$
(E)	Isochoric cooling		

Solution:

(A) – p,s, (B) – p,q,r,s, (C) – s , (D) – q,s, (E) – r,s

Second Law of Thermodynamics

Draw back of FLOT:

- No information about what part of heat converted into mechanical-work.
- Does not give proper direction of heat flow

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Statement:

1. **Kelvin plank statement :** It state that in a cyclic process total heat can not be converted into mechanical work.
2. **Claussius statement :** it is impossible to have net heat flow from low temperature body to high temperature body.

Heat Engines

Efficiency of heat engine

$$\eta = \frac{\text{output}}{\text{input}}$$

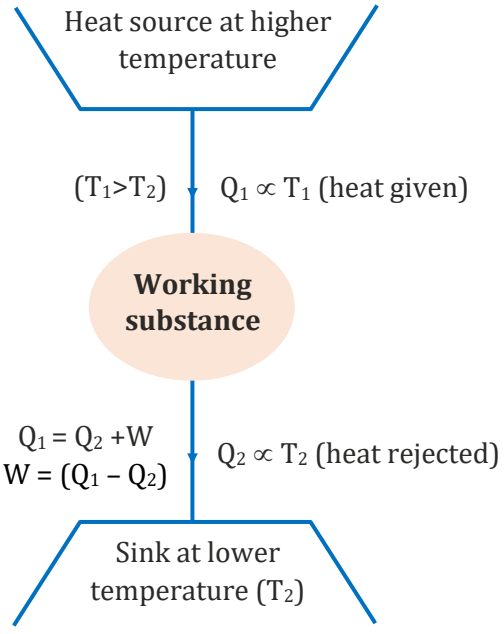
$$\eta = \frac{\text{work}}{\text{heat given}}$$

$$\eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1}$$

Maximum efficiency

$$\eta = \left(\frac{T_1 - T_2}{T_1}\right) \times 100\%$$

$$\eta = \left(1 - \frac{T_2}{T_1}\right) \times 100\%$$



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Note:

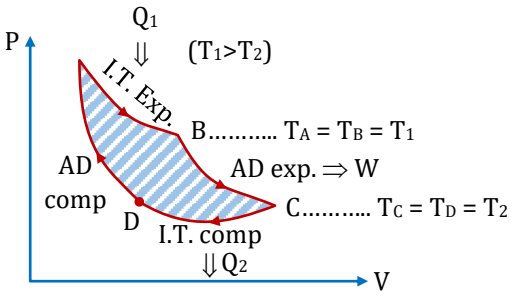
(1) If $T_1 = \infty\ K$ or $T_2 = 0\ K$ then $\eta = 100\%$ (That is practically impossible).

(2) Efficiency depends on temperature of source ($T_1 K$) and temperature of sink ($T_2\ K$).

Carnot Cycle

Enclosed area represent work done by the engine.

1. It is a hypothetical engine because working substance is an ideal gas.
2. It's efficiency is maximum but not 100%.
3. This cycle process have two isothermal and two adiabatic process.



Efficiency of Carnot heat engine:

$$\eta = \frac{\text{output}}{\text{input}} = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = \frac{T_1 - T_2}{T_1}$$

$$\eta\% = \left[1 - \left(\frac{T_2}{T_1}\right)\right] \times 100\%$$

Illustration 44

If a carnot engine works at a temperature between 300 K to 400 K, then find the efficiency of that engine.

Solution:

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$$\eta\% = \left(\frac{T_1 - T_2}{T_1}\right) = \frac{400 - 300}{400}$$

$$= \frac{100}{400} = \frac{1}{4} \text{ or } 25\%$$

Illustration 45

A diatomic ideal gas is used in a Carnot engine as the working substance. if during the adiabatic expansion part of the cycle the volume of the gas increases from V to 32 V, the efficiency of the engine is

Solution:

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$$

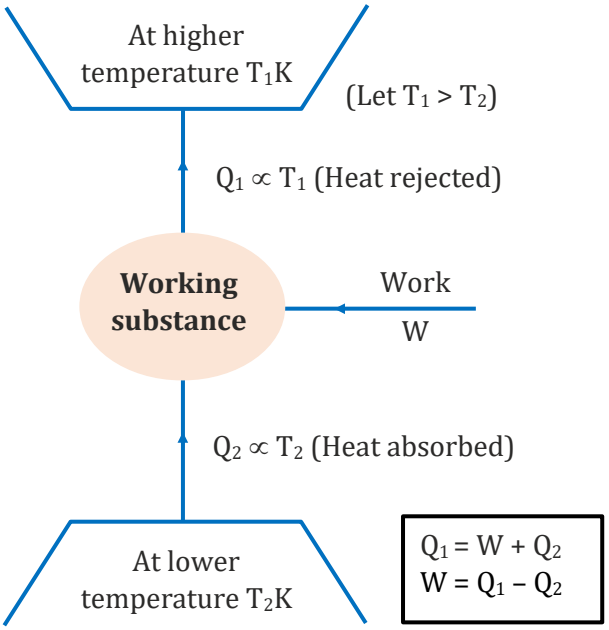
$$\Rightarrow \quad \frac{T_2}{T_1} = \left(\frac{1}{32}\right)^{\frac{7}{5}-1} = \frac{1}{4}$$

$$\text{Efficiency } \eta = \left(1 - \frac{T_2}{T_1}\right)$$

$$= 1 - \frac{1}{4} = 0.75$$

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Refrigerators



1. In Carnot refrigerator heat is absorbed from a cold body & some external work is to be done on refrigerator and the total heat is given out at higher temperature.

- Refrigerator is just opposite to heat engine so (COP) of refrigerator is

$$COP\ (\beta) = \frac{\text{Heat absorbed}}{\text{Work}} = \frac{Q_2}{W} = \frac{Q_2}{(Q_1 - Q_2)}$$

$$\boxed{COP(\beta) = \left(\frac{T_2}{T_1 - T_2}\right)}$$

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Relation between (η and β) $\boxed{\eta = \frac{1}{1+\beta}}$ or $\boxed{\beta = \frac{1}{\eta} - 1}$

Illustration 46

Calculate *COP* of a Carnot refrigerator working between 30°*C* to 40°*C*.

Solution:

$$\beta = \frac{T_2}{T_1 - T_2} = \frac{303}{10} = 30.3$$

Illustration 47

A Carnot engine takes 103 kilocalories of heat from a reservoir at 627°*C* and exhausts the remaining heat to a sink at 27°*C*. What will be the efficiency of the engine?

Solution:

$Q_1 = 10^3 \times 10^3$ cal $= (10^6 \times 4.2\ J)$

$T_1 = 900\ K$

$T_2 = 300\ K$

$$\eta = \frac{T_1 - T_2}{T_1} \times 100 = \frac{600}{900} \times 100 = 67\%$$

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Illustration 48

In the above problem, what will be the work performed by the engine?

Solution:

Work performed by the engine

$$W = \eta Q_1 = \frac{2}{3} \times 10^6 \times 4.2 \quad \text{or} \quad W = 2.8 \times 10^6 \text{ joule}$$

Illustration 49

A Carnot engine absorbs an amount Q of heat from a reservoir at an absolute temperature T and rejects heat to a sink at a temperature of $T/3$. The amount of heat rejected is: -

(A) $Q/4$ (B) $Q/3$ (C) $Q/2$ (D) $2Q/3$

Solution:

$$\because \quad \eta = 1 - \frac{T/3}{T} = \frac{W}{Q_1}$$

Where Q_1 = heat absorbed, Q_2 = heat rejected

$$\Rightarrow \quad 1 - \frac{T/3}{T} = \frac{W}{Q_1}$$

$$\Rightarrow \quad \frac{2}{3} = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1}$$

$$\Rightarrow \quad \frac{2}{3} = 1 - \frac{Q_2}{Q_1}$$

$$\Rightarrow \quad \frac{Q_2}{Q_1} = \frac{1}{3}$$

$$\Rightarrow \quad \frac{Q_2}{3} = \frac{Q}{3}$$

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Illustration 50

A Carnot engine having an efficiency of 1/10 as heat engine, is used as a refrigerator. If the work done on the system is 10 J, the amount of energy absorbed from the reservoir at lower temperature is: -

(A) 90 J (B) 99 J (C) 100 J (D) 1 J

Solution:

$$\eta = \frac{1}{10}$$

$$\beta = 1 - \frac{1}{\eta}$$

$$\beta = 9 = \frac{Q_2}{W}$$

$$Q_2 = 9 \times 10$$

$Q_2 = 90 J$ Heat absorbed at lower temp.

$Q_1 = 90 + 10 = 100 J$ Heat released at higher temp.

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